Chair of Spacecraft Systems
TUM School of Engineering and Design
Technical University of Munich

# Effect of Central Nodes on Task Dissemination in Distributed Satellite Systems

## Wildfire Follow-Up as a Case Study for Architecture Trade-Space Analysis

**Maneesh Manjunath**

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**Examiner:**
Prof. Dr. Alessandro Golkar

**Advisor:**
M.Sc. Vincenzo Messina

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I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.



Munich, 14 October 2026 Maneesh Manjunath

# Zusammenfassung

Eine schnelle Nachbeobachtung von Waldbränden durch ein verteiltes Satellitensystem erfordert mehr als den geometrischen Zugang zum betroffenen Gebiet. Eine nach der Erstentdeckung erzeugte Aufgabe muss einen geeigneten Satelliten vor dessen nächstem nutzbaren Überflug erreichen, obwohl Kontakte nur zeitweise bestehen und die Verbindungskapazität begrenzt ist. Die Zuweisung einer Central-Node-Relaisrolle an ausgewählte Raumfahrzeuge kann zusätzliche Verteilungsmöglichkeiten schaffen, zugleich aber den Datenverkehr erhöhen und ungünstig mit begrenzten Weiterleitungsregeln zusammenwirken. Diese Arbeit untersucht diesen Architekturkonflikt mit einem durchgängigen Ablauf, der Fernerkundungsdetektionen, Aufgabenverteilung, späteren Zielzugang, Robustheitsprüfung und multikriterielle Entscheidungsunterstützung verbindet.

Aktive-Feuer-Detektionen des Sentinel-3-SLSTR-Instruments werden in nachvollziehbare Aufgaben mit Position, Freigabezeit, Priorität, Frist und NachrichtengröSSe umgewandelt. Ein diskretes Ereignismodell überträgt jede Aufgabe über zeitabhängige Boden-Weltraum- und Inter-Satelliten-Kontakte. Eine Nachbeobachtung wird nur gewertet, wenn der beobachtende Satellit die vollständige Aufgabe vor dem maSSgeblichen Zugang empfangen hat. Central Nodes verändern Relaisberechtigung, Weiterleitungspräferenz und Kommunikationsreichweite, während Umlaufbahn, Nutzlast, Rechenleistung und Komponentenzuverlässigkeit unverändert bleiben. Die abschlieSSend verifizierte Simulationskampagne bewertet Walker-Konstellationen mit 60, 120 und 180 Satelliten hinsichtlich Bahnebenen, Höhe, Inklination, Central-Node-Anteil, Routing-Regel, PaketgröSSe und Störungsbedingungen. Kalifornien und Indien werden auf übereinstimmenden Architekturen mit 60 und 120 Satelliten verglichen. Ein geprüfter indischer Block mit 282 Fällen und 180 Satelliten ermöglicht 216 übereinstimmende Vergleiche zwischen Flotten mit 120 und 180 Satelliten.

Die Nachverarbeitung trennt Beobachtungserfüllung von vollständiger Aufgabenverteilung, Abdeckung nützlicher Empfänger, Abschlusszeit, Kommunikationsaufwand und Robustheit, bevor Pareto- und Rangakzeptanzanalysen durchgeführt werden. Der übereinstimmende regionale Vergleich zeigt, dass die Zusammensetzung der Aufgabenlast sowohl Kennwerte als auch Architekturpräferenzen verändert: Die indische Aufgabenlast erreicht eine um 4,66 Prozentpunkte höhere Beobachtungserfüllung als Kalifornien, jedoch eine um 2,98 Punkte geringere strikte Vervollständigung, eine um 5,95 Punkte geringere Abdeckung nützlicher Empfänger und eine ungefähr 175 Minuten längere ungünstigste Verteilungszeit. Eine Vervierfachung der AufgabengröSSe verringert die strikte Vervollständigung in Kalifornien um 14,94 und in Indien um 37,02 Prozentpunkte. Im übereinstimmenden indischen Flottenvergleich verbessert der Ausbau von 120 auf 180 Satelliten die Beobachtungserfüllung um 1,23 Punkte und die strikte Vervollständigung um 3,99 Punkte und verkürzt die ungünstigste Verteilungszeit um 44,68 Minuten. Dafür entstehen pro Fall 25.827 Mbit zusätzlicher Aufgabenverkehr. Hohe Central-Node-Anteile helfen bei unbeschränktem Routing, können jedoch bei aktiven Kopien- und Sprunggrenzen schlecht abschneiden. Der Ausfall der Bodenstation ist die stärkste einzelne Störung; gröSSere Pakete in Verbindung mit Satellitenausfällen verursachen den gröSSten kombinierten Verlust.

Das zentrale Ergebnis besteht darin, dass geometrische Abdeckung, Verteilungsleistung und Ressourcenbedarf weder durch eine einzelne Kennzahl noch durch eine allgemein optimale Architektur dargestellt werden können. Der Missionsnutzen hängt davon ab, ob die Information vor der nutzbaren Beobachtungsmöglichkeit eintrifft und welcher Datenverkehr dafür erforderlich ist. Der Beitrag ist daher keine einzelne empfohlene Konstellation, sondern ein nachvollziehbares Architekturportfolio, das die Zielkonflikte zwischen Aufgabenlast, Zeitverhalten, Routing-Grenzen, Kommunikationsaufwand und Robustheit sichtbar macht.

# Abstract

Rapid wildfire follow-up by a distributed satellite system requires more than geometric access to the affected area. A task produced after an initial detection must reach a suitable satellite before its next useful overflight, despite intermittent contacts and finite link capacity. Assigning selected spacecraft a Central Node relay role can create additional dissemination opportunities, but it can also increase traffic and interact unfavourably with bounded forwarding. This thesis investigates that architecture trade-off through an end-to-end workflow that connects remote-sensing detections, task dissemination, later target access, robustness testing, and multi-objective decision support.

Sentinel-3 SLSTR active-fire detections are converted into traceable tasks containing a location, release time, priority, deadline, and message size. A discrete-event model propagates each task through time-varying ground-to-space and inter-satellite contacts. A follow-up observation is counted only when the observing satellite has received the complete task before the relevant access. Central Nodes change relay eligibility, forwarding preference, and communication range, while orbit, payload, processing capability, and component reliability remain unchanged. The final verified simulation campaign evaluates Walker constellations with 60, 120, and 180 satellites across plane count, altitude, inclination, Central Node fraction, routing policy, packet size, and disturbance conditions. California and India are compared on matched 60- and 120-satellite architectures. A verified 282-case India block at 180 satellites provides 216 matched comparisons between the 120- and 180-satellite fleets.

Post-processing separates observation fulfilment from complete dissemination, useful-recipient coverage, completion time, communication traffic, and robustness before Pareto and rank-acceptability analysis. The matched regional comparison demonstrates that workload composition changes both metric values and architecture preference: the India workload achieves 4.66 percentage points higher observation fulfilment than California, but 2.98 points lower strict completion, 5.95 points lower useful-recipient coverage, and approximately 175 minutes longer worst-case dissemination. Quadrupling task size reduces strict completion by 14.94 points in California and 37.02 points in India. Within the matched India fleet comparison, increasing the constellation from 120 to 180 satellites improves observation fulfilment by 1.23 points and strict completion by 3.99 points and reduces worst-case dissemination time by 44.68 minutes, at the cost of 25,827 Mbit more task traffic per case. High Central Node fractions help under unlimited routing but can perform poorly when copy and hop limits are active. Ground-station outage is the strongest single disturbance, while larger packets combined with satellite failure produce the greatest compound loss.

The central finding is that geometric coverage, dissemination performance, and resource demand cannot be represented by one metric or one universally optimal architecture. Mission value depends on whether information arrives before the useful observation opportunity and on the traffic required to achieve that outcome. The resulting contribution is therefore not a single recommended constellation, but a traceable architecture portfolio that exposes the trade-offs among workload, timing, routing limits, communication burden, and robustness.

# Acknowledgements

Sincere thanks are extended to Vincenzo Messina for his supervision throughout this thesis. The discussions were particularly valuable when the work moved from running architecture cases to deciding which comparisons were scientifically meaningful. He provided the original communication-model basis, supported the integration with TAT-C, and helped define the questions used to evaluate the final campaigns.

The Chair of Spacecraft Systems at the Technical University of Munich is also thanked for providing the academic and computing environment used for this work. Gratitude is likewise extended to family and friends for their patience and support during the long simulation and writing phases.

OpenAI's ChatGPT, including the Codex environment, was used as an assistive tool during this thesis. Its use included support with Python code development and debugging, preparation of simulation and post-processing workflows, generation and refinement of plots and tables, language correction, and restructuring of draft sections in response to supervisory feedback. The research questions, modelling assumptions, input data, simulation design, selection of comparisons, validation decisions, and scientific interpretation remained the responsibility of the author. All AI-assisted code, calculations, figures, and text were reviewed against the simulation outputs, source literature, and supervisory feedback before inclusion. ChatGPT was not treated as a scientific source and is not used to substantiate technical claims.

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# Acronyms

**API** application programming interface.

**CDF** cumulative distribution function.

**CN** central node.

**CSV** comma-separated values.

**DBSCAN** Density-Based Spatial Clustering of Applications with Noise.

**DSS** distributed satellite system.

**EO** Earth observation.

**FIRMS** Fire Information for Resource Management System.

**FRP** fire radiative power.

**GS** ground station.

**HPC** high-performance computing.

**ISL** inter-satellite link.

**MW** megawatt.

**PBL** planetary boundary layer.

**SLSTR** Sea and Land Surface Temperature Radiometer.

**TAT** task-allocation time.

**TAT-C** Trade-space Analysis Tool for Constellations.

**UTC** Coordinated Universal Time.

# 1 Introduction

## 1.1 Work Developed in This Thesis

This thesis develops a complete path from a wildfire detection to an architecture decision for a distributed satellite system. Sentinel-3 active-fire products are converted into time-stamped follow-up tasks; a temporal simulator then evaluates whether each task can be transmitted through the satellite network and received before a useful target access. The architecture study varies Walker geometry, fleet size, Central Node participation, forwarding policy, packet size, and component failures. The resulting outputs are checked for completeness and input consistency before mission performance, timing, traffic, robustness, Pareto efficiency, and decision sensitivity are calculated.

Figure 1.1 locates the contribution within the operational chain. The implemented work covers task construction, satellite acquisition, inter-satellite dissemination, ground-station delivery, campaign execution, and post-processing. Central Nodes affect the dissemination stage by providing an additional relay role; they do not command the constellation and do not use different sensors.

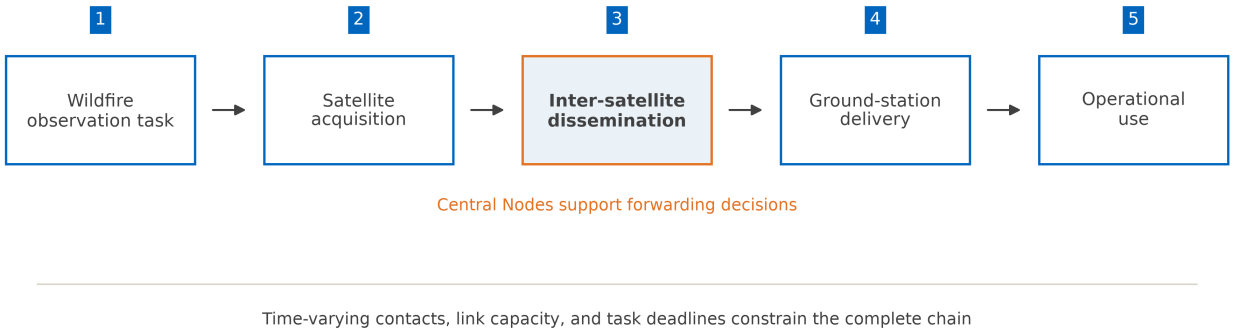



**Figure 1.1** Operational chain evaluated in the thesis. Central Nodes support the dissemination stage, while time-varying contacts, link capacity, and task deadlines constrain the complete chain.

The completed work extends an existing communication and TAT-C-based orbital model supplied by Vincenzo Messina. The thesis contribution consists of wildfire-task processing, simulator integration, Central Node placement, campaign execution and restart controls, validation, post-processing, statistical analysis, figures, and the final architecture trade-space.

## 1.2 Motivation

Coverage is necessary for a rapid-response Earth-observation mission, but it is not sufficient. A satellite can pass over a target within minutes and still miss the opportunity if the task arrives after the pass. The opposite can also happen: a request can spread through the network while no informed satellite obtains a useful access before the deadline. Information delivery and orbital access are therefore treated as two connected stages of one operational problem.

Wildfire follow-up makes this timing problem concrete. Each source product becomes a request with a position, release time, priority, deadline, and message size. The request must cross intermittent ground-to-space and space-to-space contacts, and a later access counts only if the observing satellite has already received the complete request. Failure can therefore occur through late communication or through the absence of a suitable post-reception access.

A Central Node is a satellite with an additional relay role. In the implemented model, that role changes link eligibility, the order in which forwarding candidates are considered, and the commu-

nication range assumed for a link with a Central Node endpoint. Nothing else about the spacecraft changes: it has the same orbit, sensor, processor, and reliability model. This is close to the information-broker role used in federated satellite concepts [11, 20], while observation authority remains distributed.

The relevant design question is not a yes-or-no question about Central Nodes. With only a few, useful temporal paths may still be missing. With many, the system may create extra traffic or depend too heavily on one forwarding mechanism. The answer may also change with geometry, workload, packet size, or routing limits. Performance and burden are therefore compared across the full Central Node fraction sweep rather than selecting a fraction from one metric.

## 1.3 Problem Statement

The adapted problem contains many tasks released at different times, four priority classes, finite packets, different deadlines, and two event windows with very different loads. The engineering problem is:

Determine how Walker-constellation geometry, routing policy, and the fraction of Central Nodes affect deadline-constrained task dissemination and post-reception wildfire observation, while retaining the communication burden and response to failures needed for an architecture decision.

Two uses of the word robustness are separated throughout the report. Physical robustness is the retained mission performance after a node, link, ground-station, or capacity disturbance. Decision stability is the persistence of an architecture ranking when the objective weights change. A design can be stable under a weighting model and still perform poorly after a physical failure.

## 1.4 Aim and Objectives

The aim is to construct a traceable performance-resource-robustness trade-space for communication-constrained wildfire follow-up in a distributed satellite system. Four objectives support that aim:

1. Convert Sentinel-3 active-fire detections into a reproducible task table with location, release time, priority, deadline, packet size, and source provenance.
2. Simulate reception, finite-capacity forwarding, and post-reception observation over a controlled Walker-constellation and Central Node design grid.
3. Compare mission performance, communication burden, and failure response without treating one metric or an unadjusted correlation as a complete architecture decision.
4. Identify Pareto-efficient and decision-stable candidates while stating the assumptions under which the shortlist is valid.

Task preparation, simulation, and decision support remain separate objectives. This separation shows whether a changed conclusion originates in the task definition, the simulated architecture, or the preference model used for ranking.

## 1.5 Research Questions

The report answers three research questions:

**RQ1.** How do Central Node fraction and forwarding policy change timely task dissemination and follow-up observation in the implemented distributed satellite system?

**RQ2.** How do workload, packet size, constellation geometry, and injected failures change performance and the relative ranking of architectures?

**RQ3.** Which architectures provide defensible trade-offs among observation fulfilment, deadline-complete dissemination, communication burden, resource proxies, and robustness?

Task generation forms part of the experimental method rather than a research question in its own right. Observation-radius sensitivity, alternative placement rules, and individual Central Node components are not assigned numerical conclusions because dedicated experiments for those questions were not executed.

## 1.6 Expected Relationships

Three expectations guide the analysis. First, an initial set of preferred relays should open useful temporal paths, while later additions should offer less once most relevant paths are already available [1, 3, 4]. Second, a lower delay may require more transfers and more differentiated nodes; delay alone is therefore not enough for selection. Third, the preferred architecture should change with workload and with the value assigned to capacity, deadlines, and resource burden. These are expectations derived from the literature and model structure, not simulation findings.

## 1.7 Scope and Boundaries

The simulated mission begins when a processed wildfire task is released. It ends when task reception and a possible follow-up observation have been evaluated. Thermal measurement, onboard fire detection, cloud screening, image production, image downlink, and delivery of an alert to civil-protection authorities are outside the model.

The Central Node fraction is the primary treatment. Satellite count, plane count, altitude, and inclination form the architecture context. Packet-size multipliers, routing policy, node and link failures, ground-station outage, capacity degradation, and objective weights are tested sensitivities. The following assumptions remain fixed in the completed campaign:

- one deterministic balanced Central Node placement rule;
- one bundled Central Node treatment combining eligibility, preference, and a link-budget multiplier;
- a 700 km observation-radius proxy;
- one Munich ground station in the baseline;
- two selected event windows rather than a statistical sample of global fire conditions; and
- engineering resource proxies rather than monetary lifecycle cost.

The final verified simulation campaign is analysed as one body of evidence. Its matched California and India blocks cover 60- and 120-satellite architectures using the intended 32-task and 773-task inputs. The campaign also contains 282 verified India cases at 180 satellites under the same 773-task contract. The complete CN30–CN100 subset supports a controlled India 120-to-180 comparison for 216 matched architectures. A California–India contrast at 180 satellites is not made because a corresponding California population was not executed. The earlier 741-task execution and its correction remain documented in Section 4.8 and Appendix C as part of the provenance record, but none of its values enters the final results.

## 1.8 Contributions

The thesis makes three connected contributions.

1. **A traceable wildfire-task input.** The final table links Sentinel-3 product records to deduplicated fire-region tasks and preserves the information needed to reproduce priority and deadline assignments.
2. **A communication-aware architecture campaign.** The model separates physical opportunities, accepted transfers, task reception, and observation. It evaluates the same task rows across a Walker and Central Node grid with finite capacity and deterministic replay.
3. **A robustness-aware decision method.** Raw metrics are validated before Pareto filtering. Failure response and objective-weight sensitivity are reported independently, leaving the final portfolio conditional and auditable.

## 1.9 Report Structure

Chapter 2 brings together distributed coordination, routing over scheduled contacts, and active-fire sensing. Chapter 3 documents the conversion from Sentinel-3 detections to the frozen task table, and Chapter 4 describes the simulation. Chapter 5 fixes the experimental and post-processing rules before the results are examined. Chapter 6 retains the historical nominal comparison, while Chapter 7 presents the final verified campaign across 60, 120, and 180 satellites. Chapters 8 and 9 interpret the combined evidence, answer the research questions, and state the conditional architecture portfolio. Operational commands, file formats, and checkpoint details are placed in the appendices and repository.

# 2 Background and Related Work

## 2.1 Distributed Satellite Systems and Coordination

A distributed satellite system (DSS) contains space assets that share resources, information, or parts of a decision process. Those assets can belong to one constellation, cooperate across missions, or form a federation [7, 20]. These terms describe different aspects of a system and are used accordingly. “Constellation” refers mainly to orbital organization, “federation” to shared capabilities across assets or owners, and “decentralization” to the location of information and decision authority.

As fleets grow, spacecraft-by-spacecraft commanding becomes difficult to scale. Goal-based planning, automated monitoring, and coordination are proposed as alternatives [2]. For a time-critical task, however, better planning does not remove the communication delay. A geometrically suitable satellite can still miss the target if the request waits for the next ground-centred planning cycle.

A central planner can use a common system view and optimize one global objective, but only if information reaches it and its decisions return to the assets in time. Local planning removes some of that dependence, although each satellite then acts on a partial view. Decentralized allocation still needs messages for bids, task requests, or schedule updates [16, 29]. If those messages are assumed to arrive instantly, a schedule may look feasible even though the selected satellite would learn about the task after its useful access.

The Central Nodes used here sit between these two coordination extremes. They are preferred relays, comparable to information-sharing nodes in collaborative and federated concepts [11, 20]; they are not global schedulers. Keeping that boundary explicit allows the experiment to study communication centralization without also changing sensing or decision authority.

## 2.2 From Task Release to Observation

Earth-observation scheduling normally considers visibility, timing, instrument availability, storage, energy, and communication [6]. The present work isolates one part of that broader problem: whether a released follow-up request reaches a spacecraft before a useful access. Four states are retained:

1. **Task release:** a wildfire-derived request enters the simulated ground node.
2. **Task reception:** a satellite has received the complete request.
3. **Observation opportunity:** geometry places the satellite within the assumed access radius of the target.
4. **Observation fulfilment:** the opportunity occurs after reception and before the task deadline.

This sequence distinguishes two mechanisms. A task may be disseminated quickly but have no suitable access, or an access may be lost because the task arrived late. Reception, dissemination breadth, and observation fulfilment must therefore be reported separately. Image downlink is a later mission phase and is outside the experiment.

## 2.3 Time-Varying Connectivity and Space Networking

Satellite links are intermittent. A useful network representation is a temporal graph $G(t) = (V, E(t))$, where $V$ contains satellites and ground stations and $E(t)$ contains the contacts available at time $t$. A contact window records a physical opportunity. A transfer event records that the routing algorithm used some capacity in that window. Treating these as the same object would overstate network activity.

Store-carry-forward communication allows a task to cross a network without a simultaneous end-to-end path. An intermediate node keeps the data until a later contact. Bundle Protocol 7, Contact Graph Routing, and the CCSDS Schedule-Aware Bundle Routing standard formalize this idea for delay- and disruption-tolerant networks [1, 3, 4]. If a contact e links nodes i and j between times t_start and t_end at rate r, its ideal data bound is:

```
C_e = r * (t_end - t_start).
```

A transfer is accepted only if the task has been released, the remaining contact capacity can carry the packet, and the timing is still useful. A low hop count is not automatically fast: the next contact on a two-hop path may occur much later than the contacts on a longer route. Extra copies also cannot create a contact or capacity that does not exist.

Static graph quantities such as degree, density, component count, path length, and betweenness are retained only as explanatory diagnostics. They do not answer the mission question. For example, a failed network may still look connected in a snapshot even though every alternative path occurs too late. The robustness results are therefore obtained by replaying the task flow after the contact opportunities have been altered.

## 2.4 Central Nodes as Preferred Relays

Centralization may describe decision authority, information storage, or communication. Only the communication form is implemented in this thesis. A Central Node receives four properties together:

- eligibility for links used by the central-only topology;
- priority in the hybrid forwarding order;
- deterministic balanced placement across planes; and
- a 1.5 communication-distance factor for each central endpoint.

The fraction sweep therefore tests a package, not four independent treatments. If performance changes, this experiment alone cannot assign the change to link eligibility, forwarding order, placement, or range. Doing so would require four separate ablations. The conclusions are consequently limited to the implemented Central Node routing package and do not establish that a heterogeneous spacecraft design is generally superior.

Messina and Golkar show how collaborative methods can improve object-detection performance in federated satellite systems [11], while Scrocciolani and colleagues connect network structure to federated-system performance [20]. The present work extends this line in a narrower direction: it varies the fraction of preferred relays over a full constellation grid and checks whether the information arrives early enough for a later observation.

## 2.5 Active-Fire Products and Workload Construction

Active-fire products report thermal anomalies with acquisition time, location, fire radiative power (FRP), and quality information. FRP is the instantaneous radiant-energy release rate and is useful for distinguishing stronger thermal activity from a simple count of detections [25]. It does not by itself identify the cause of the anomaly or determine operational severity.

NASA FIRMS provides convenient access to active-fire detections from instruments including VIIRS. The 375 m VIIRS product improves the mapping of relatively small fires compared with coarser products [19]. Sentinel-3 SLSTR supplies a separate active-fire and FRP product with different sampling, overpass timing, channels, and confidence fields [26, 27]. Rows from the two systems are therefore not interchangeable.

The thesis uses the sources in complementary roles. FIRMS supports broad event screening and selected contextual checks. Sentinel-3 SLSTR is the only source used to construct the frozen simulation task table. This avoids double counting one event through two sensor conventions. It also keeps the task population reproducible: every architecture within an event window receives the same rows and release times.

## 2.6 Research Gap and Thesis Position

The literature covers the four required pieces separately: distributed satellite operation, decentralized allocation, scheduled space networking, and satellite active-fire sensing. The reviewed studies do not provide a constellation-level experiment that joins all four. This thesis addresses that gap by sending wildfire-derived requests through finite temporal contacts, checking reception before observation, and varying the preferred-relay fraction while recording both mission benefit and communication burden. Table 2.1 links each established research area to the specific step added in this thesis.

**Table 2.1** Positioning of the thesis relative to the four connected literature areas.

| Literature area | Established contribution | Step taken in this thesis |
|---|---|---|
| Federated and distributed satellite systems [2, 7, 11, 20] | Resource sharing, collaboration, scalable operations | Evaluate a declared central-node relay role over a controlled fraction sweep |
| Decentralized observation allocation [16, 29] | Distributed assignment and onboard decision concepts | Verify when the responsible satellites actually receive each task |
| Space DTN and scheduled routing [1, 3, 4] | Store-carry-forward over predicted contacts | Couple finite transfers to wildfire deadlines and later access |
| Active-fire remote sensing [19, 25–27] | Locations, confidence information, and FRP | Convert repeated detections into traceable requests with size and deadline |

Several boundaries follow from that design. CN0 under a central-only topology is not a peer-to-peer baseline. A task missed in the nominal case says nothing by itself about robustness to an injected failure. A graph correlation may suggest an explanation, but it does not establish a cause. These distinctions determine the wording of the results and conclusions.

# 3 Wildfire Task Dataset and Preprocessing

## 3.1 Purpose of the Task Dataset

The simulator starts from a task table, not from radiance measurements or images. Each row represents a fire region for which a follow-up observation may be requested. The main preprocessing decision is how to turn many thermal pixels into distinct tasks. Keeping every pixel would repeatedly request the same fire, whereas aggressive merging could hide separate events. Source-product identifiers are therefore preserved and duplicates are removed before any architecture is evaluated.

Two three-day windows were selected. The California window, 3–5 July 2025, includes the early Madre Fire; the India window, 14–16 April 2026, produces a much denser set of requests. After preprocessing, California has 32 tasks and India has 773. These are two engineering cases rather than a sample of global fire behaviour. Any California–India difference therefore includes the chosen events, geography, timing, density, and priority mix. The report describes this as a workload contrast, not a general regional effect.

Figure 3.1 shows where the retained tasks lie. The map uses the US Census Bureau boundary for California and Natural Earth for India [15, 24]. Selection itself is performed with event bounding boxes, not political polygons. A few accepted coordinates can therefore appear outside the drawn administrative outline. They remain visible in the figure because clipping them would conceal the effect of the stated filter.

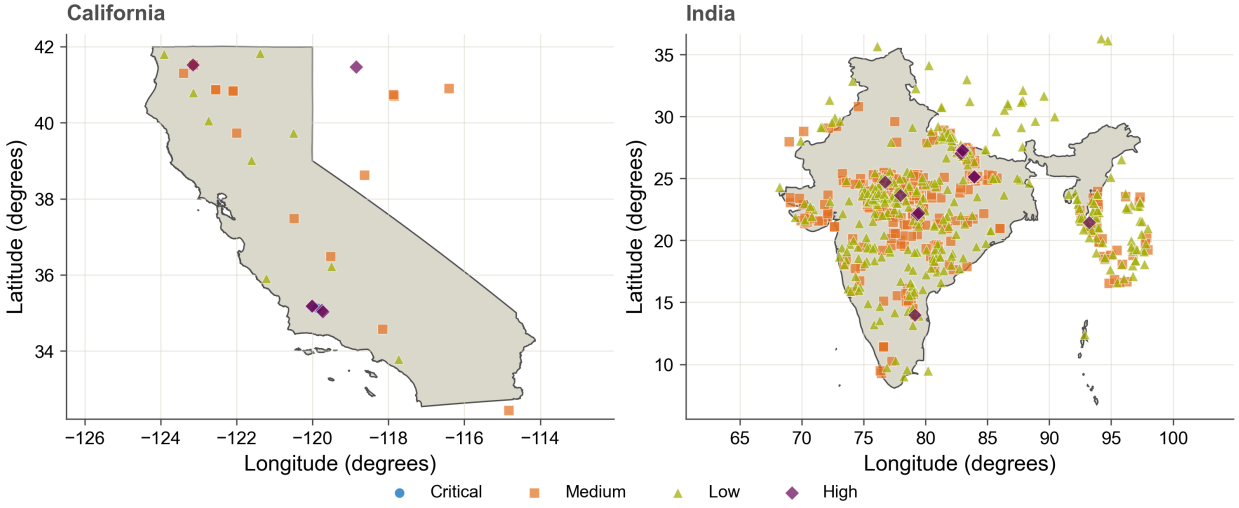



**Figure 3.1** Geographic distribution of all 805 final Sentinel-3 tasks on published administrative outlines. California contains 32 tasks and India 773; marker shape and colour identify the priority class.

## 3.2 Processing Sequence

Figure 3.2 records the actual processing order. Filtering decides which detections are eligible; clustering groups nearby detections into candidate fire regions; scoring assigns an engineering priority; and de-duplication decides which candidates remain separate tasks. Keeping these stages distinct is important because each one changes the eventual communication load in a different way.

Each downloaded `.SEN3` directory is one Sentinel-3 SLSTR Level-2 FRP granule. The code reads its sensing interval, extracts geolocated fire pixels from the primary FRP NetCDF file, applies the

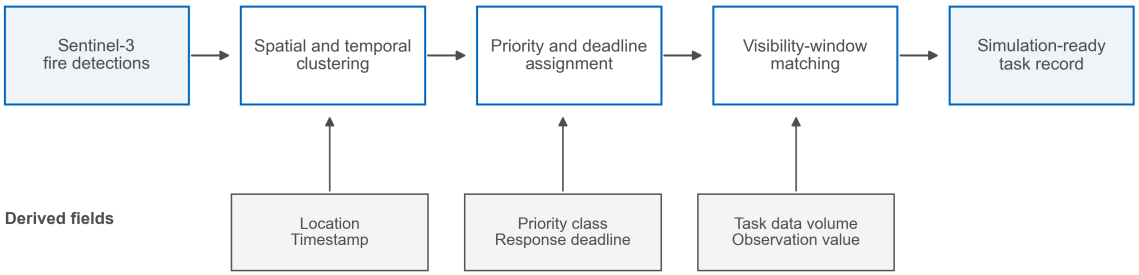



**Figure 3.2** Traceable task-generation workflow. Filtering occurs before clustering, and the exported task retains source-product provenance.

event-window and region filters, and rejects invalid coordinates or FRP values. The original product identifier is carried into the task output, so a simulated row can be traced back to the input granule.

De-duplication is applied at four levels:

1. Each Sentinel-3 product is read once.
2. Only the primary FRP NetCDF content is used for fire pixels.
3. Exact and near-duplicate hotspot rows are removed using position, time, and FRP.
4. Candidate tasks that represent the same fire region on the same day are merged, retaining a traceable representative and the strongest priority evidence.

The reduction in reported FRP during pipeline development reflects removal of repeated records; it is not evidence of a physical decrease in fire activity.

## 3.3 FRP Filtering and DBSCAN Clustering

The development pipeline applies an FRP inclusion threshold of 10 MW before spatial grouping. This threshold is an engineering rule for generating communication tasks, not a definition of wildfire occurrence.

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is used because the number of fire regions is not known in advance. DBSCAN connects detections that have enough neighbours within a chosen distance. Here, `eps` is the neighbourhood radius and `min_samples` is the minimum number of detections required for a dense cluster. The documented baseline uses a 10 km neighbourhood and `min_samples = 3`.

The filters are applied before DBSCAN. This prevents detections outside the selected date, region, quality, or FRP range from influencing a cluster. DBSCAN is then run on the remaining positions for the declared grouping unit. Dense detections form a cluster, while isolated detections follow the stated noise-point rule instead of being assigned to the nearest cluster without evidence. The exported row stores the centroid, approximate radius, contributing-detection count, and product provenance. The totals in Table 3.1 are therefore task totals after filtering and de-duplication, not raw pixel counts.

The clustering choice controls network load. A smaller radius creates more packets and can fragment one event. A larger radius creates fewer packets and can merge independent events. The baseline must therefore be interpreted together with the proposed 5 km and 20 km sensitivity checks.

## 3.4 Confidence and Priority

Sentinel-3 supplies numerical MWIR confidence information. FIRMS VIIRS uses categorical low, nominal, and high values. FIRMS confidence is used only during screening and is mapped to 0.33, 0.66, and 1.00 for visualization or contextual comparison. The final simulation score uses the Sentinel-3-derived value. It is a relative ranking input, not a calibrated probability of wildfire.

Invalid Sentinel-3 confidence values are flagged as missing. A neutral value of 0.50 is used in the score so that an unavailable field does not automatically suppress the task. The missing fraction and the original confidence summaries remain in the exported data.

The FRP component combines normalized total and maximum FRP:

$$S_FRP = 0.60 * N(\log(1 + FRP_total)) + 0.40 * N(\log(1 + FRP_max)).$$

The task score is:

$$S_task = 0.50 * S_FRP + 0.30 * S_conf + 0.20 * S_cluster.$$

The notation is used consistently: `S_FRP` is the intensity term, `S_conf` the confidence term, `S_cluster` the spatial-support term, and `S_task` the final score. The weights express an engineering preference for energetic strength while retaining confidence and spatial support. They were not fitted to emergency-service labels and must not be interpreted as an official severity model.

## 3.5 Final Task Population and Packet Assumptions

**Table 3.1** Final Sentinel-3 wildfire-task population after filtering and de-duplication.

| Region | Critical | High | Medium | Low | Total |
|---|---|---|---|---|---|
| California | 1 | 6 | 16 | 9 | 32 |
| India | 0 | 11 | 395 | 367 | 773 |
| Total | 1 | 17 | 411 | 376 | 805 |

India contributes 96 percent of all tasks, while urgent classes form a larger share of the California set. For this reason the results report both total communication volume and per-task quantities.

Priority determines the nominal response deadline and the size of the request message. Table 3.2 fixes these assumptions for all simulations:

**Table 3.2** Priority-dependent deadline and task-message size assumptions.

| Priority | Deadline | Packet size | Interpretation |
|---|---|---|---|
| Critical | 30 min | 2.0 Mbit | Immediate high-severity follow-up request |
| High | 60 min | 1.5 Mbit | Urgent follow-up request |
| Medium | 180 min | 1.0 Mbit | Time-sensitive routine request |
| Low | 360 min | 0.5 Mbit | Lower-urgency monitoring request |

These sizes represent structured requests, associated metadata, and protocol allowance. They are not image products. The values are author-defined engineering assumptions chosen to create a priority-dependent capacity load while remaining small enough for UHF store-carry-forward operation. Because deadline and packet size both change with priority, the controlled size experiment holds task identity and deadline fixed and multiplies every packet by 0.5, 1, 2, or 4. This separates data-volume sensitivity from urgency.

## 3.6 Dataset Validation and Limits

The Madre Fire provides one event-level plausibility check. Sentinel-3 and FIRMS both show strong detections near the reported event during the California window. Agreement at that location and time supports the corresponding California task, but it is not a validation of all 805 tasks, of the priority score, or of wildfire-classification accuracy. An equivalent named-incident link was not established for every India task.

The dataset has additional limits. Bounding-box filtering can retain detections outside a political boundary. Cloud, smoke, sampling, and overpass time affect what either sensor records, and FIRMS and Sentinel-3 totals cannot be compared directly without spatial and temporal matching. A different sensor, season, threshold, clustering rule, event mix, or priority mapping would create a different workload and could change the architecture ranking.

The task table is frozen before the architecture sweep. Every case within an event window therefore receives the same release times and task definitions. This makes architecture comparisons controlled within a region. It does not turn the two selected windows into population-level evidence for California or India.

# 4 System Model and Methodology

## 4.1 Method Objective and Overview

The method evaluates a simple operational question: when a wildfire task is released, can it reach useful satellites early enough for at least one of them to observe the target before the deadline? The simulation separates task data, orbital geometry, communication opportunities, routing, and evaluation so that a policy or failure can be changed without changing the fire sample or orbit. Figure 4.1 shows this separation and, importantly, the validation gate between raw simulation output and the evidence used for statistical comparison. Consequently, an architecture is not ranked merely because its run produced files; its inventory, schema, provenance, and completeness must first pass the stated checks.

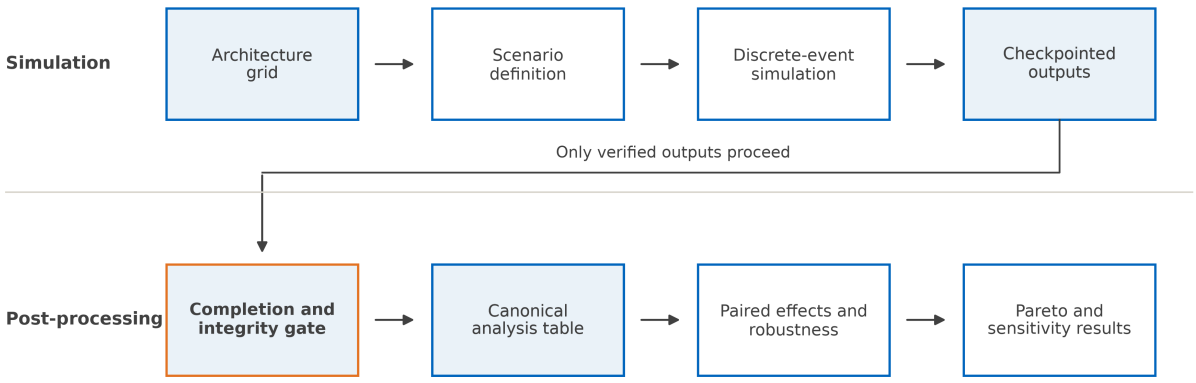



**Figure 4.1** Simulation and post-processing workflow. Architectures and scenarios produce checkpointed outputs; only outputs that pass the completion and integrity gate enter the canonical analysis table, paired comparisons, robustness analysis, and Pareto evaluation.

The experiment follows the sequence below so that later scenario changes do not alter the input tasks or geometry:

1. Freeze one regional task table, then generate the Walker architecture.
2. Select the requested Central Node fraction using the same balanced placement rule in every case.
3. Propagate spacecraft states and create observation and communication windows.
4. Apply the routing-policy or failure transformation.
5. Replay transfers chronologically with packet and capacity checks.
6. Credit observation only after full task reception and before the deadline.
7. Write case-, task-, and transfer-level evidence for validation and post-processing.

This sequence is the core method. Checkpoint file names, compression, worker memory, and restart commands are implementation details and are documented in the software appendix rather than used to explain the science.

## 4.2 Inputs and Architecture Grid

The simulator reads a fixed CSV rather than a live service. The task release epochs remain in UTC. Each regional run lasts at least 72 hours and extends to six hours after the latest task deadline when necessary. This produces approximately 73.1 hours for California and 73.7 hours for India.

The Walker grid varies the five factors and discrete levels listed in Table 4.1:

**Table 4.1** Full-factorial Walker-constellation and central-node design grid.

| Factor | Levels |
|---|---|
| Satellites, T | 60, 120, 180 |
| Planes, P | 3, 5, 10 |
| Altitude, H | 500, 800, 1000 km |
| Inclination, I | 60, 80, 98.6 deg |
| Central-node fraction, C | 0, 10, ..., 100 percent |

Walker phasing is fixed at F = 1. The first four factors create 81 base architectures. Eleven central-node fractions per base create 891 cases per region. A case identifier such as `T060_P10_H0800_I0986_CN040_F1` means 60 satellites, 10 planes, 800 km altitude, 98.6 deg inclination, 40 percent central nodes, and Walker phasing 1.

The absolute number of central nodes is `N_CN = T * C / 100`. The fraction and absolute count are both retained because CN40 represents 24 Central Nodes in a 60-satellite fleet, 48 in a 120-satellite fleet, and 72 in a 180-satellite fleet. Figure 4.2 illustrates the balanced assignment rule: the highlighted spacecraft are distributed across planes and along-track positions instead of being concentrated in one part of the constellation. The figure also emphasizes that the treatment changes the routing role, not the orbit or sensing capability of those spacecraft.

Central nodes are spread across planes as evenly as integer constraints permit and then spaced within each plane. Fixing placement removes a manual choice from the fraction sweep. It does not establish that the placement is optimal. All nodes retain the same sensing and orbit model; only the declared routing package changes.

## 4.3 Orbit and Observation Opportunities

The orbital layer uses circular two-body propagation on a spherical rotating Earth within the integrated TAT-C-based workflow [22]. The calculation uses an Earth radius of 6378.14 km, gravitational parameter 398600.44 km^3/s^2, Earth rotation rate 7.2921159e-5 rad/s, and a 60 s time step.

An observation opportunity exists when the satellite-to-task ground distance is at most 700 km. If task i is released at r_i, has deadline d_i, and satellite s receives it at a_is, a valid observation is the earliest candidate access t in O_is satisfying:

```
t >= max(r_i, a_is) and t <= d_i.
```

This rule prevents a pre-reception pass from being counted. The 700 km value is a mission-level access proxy, not a sensor field-of-view or pointing analysis. Cloud, smoke, illumination, slew, image quality, power, storage, and conflicts among simultaneous observations are outside the model. Results therefore remain conditional on the proxy until an observation-radius sensitivity is executed.

## 4.4 Communication Model

The ground segment contains one Munich station with a 10 deg minimum elevation. The active satellite-to-satellite model uses a UHF link with 2 W transmit power, 437 MHz carrier, 9.6 kHz band-

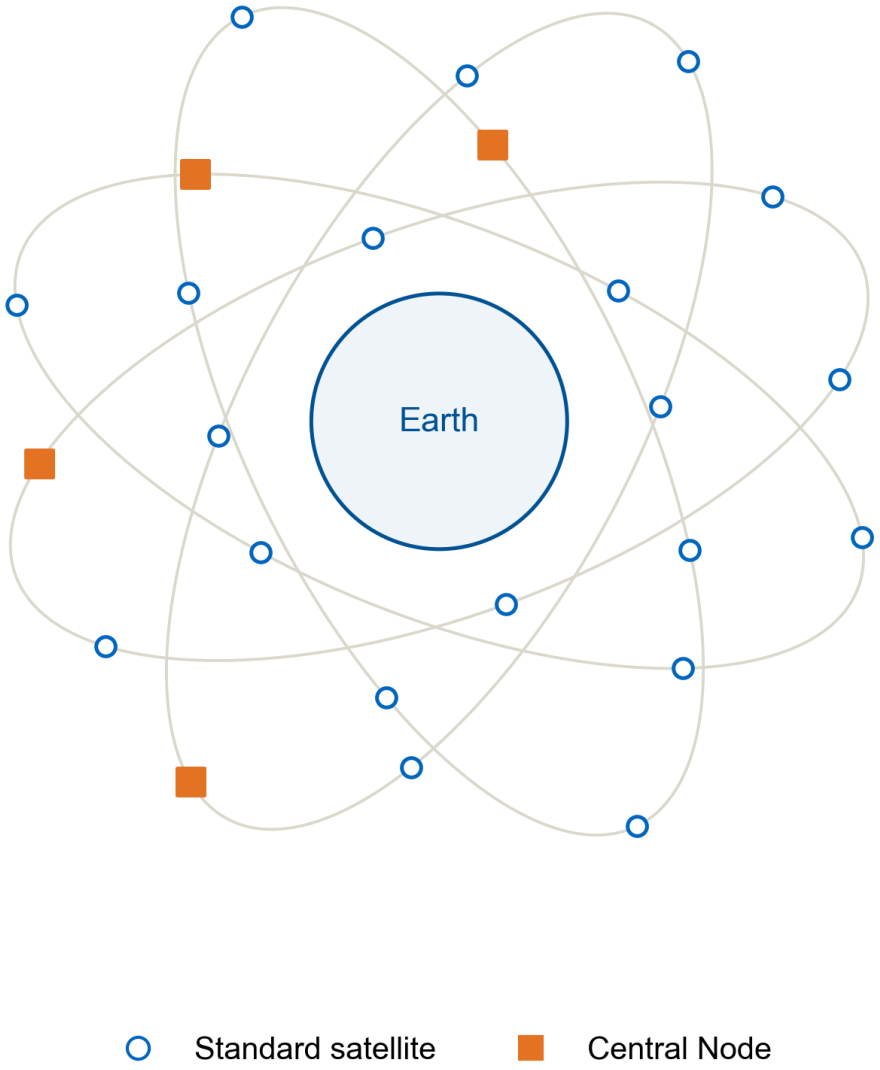



**Figure 4.2** Schematic Walker architecture with balanced central-node placement. Highlighted satellites have the routing role; all satellites keep the same orbit and sensing assumptions.

width, and 9.6 ksymbol/s QPSK. The effective rate is limited to approximately 19.2 kbit/s; satellite-to-ground windows use 100 kbit/s.

The central-node factor is applied as a distance multiplier for each central endpoint and as the equivalent link-budget boost in the rate calculation. Under the implemented UHF parameters, the sensitivity-based upper distance is approximately 2306 km for an ordinary-to-ordinary pair, 3460 km when one endpoint is central, and 5190 km when both endpoints are central. These are eligibility ceilings, not guaranteed contacts: Earth blockage, the analytical geometry, beam angles, and the instantaneous link-rate calculation still apply.

For a window with effective rate R, usable duration delta_t, and utilization factor eta, available capacity is:

```
B_window = R * delta_t * eta / 1e6 [Mbit].
```

The final verified campaign uses $\eta = 1$. A contact window records possible capacity. A transfer record proves that routing used part of that capacity. The analysis keeps the two quantities separate.

## 4.5 Temporal Routing and Topology Policies

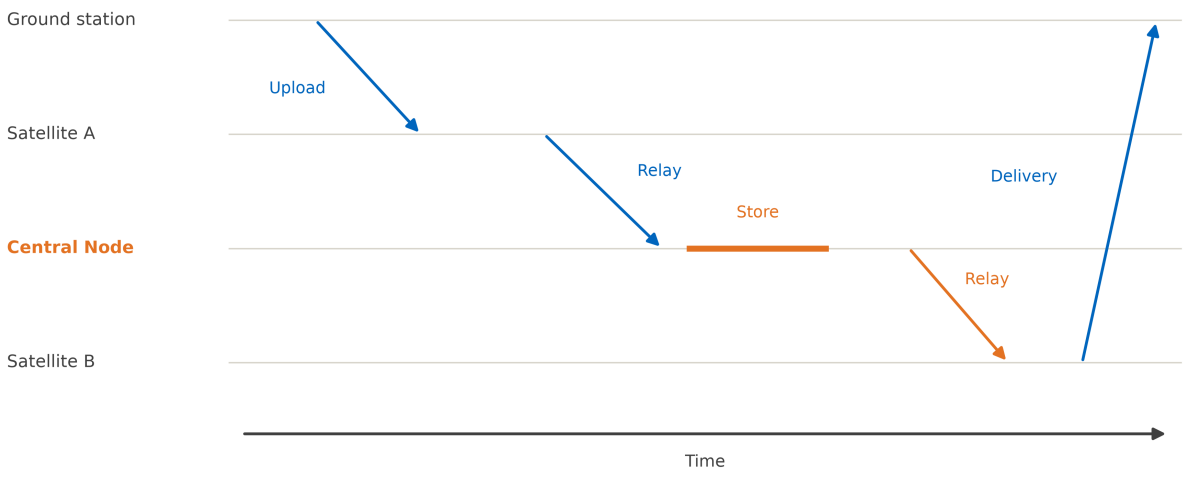



**Figure 4.3** Example temporal route. A contact is useful only if it occurs after task release and leaves the receiving satellite enough time for a later access before the deadline.

Three topology-policy variants are replayed:

- **Central-only:** an inter-satellite link is usable only when at least one endpoint is central. CN0 consequently has no inter-satellite links and depends on the ground uplink. It is not a peer-to-peer baseline.
- **Peer-only:** every feasible inter-satellite link is usable and the central label is ignored by the forwarding order.
- **Hybrid:** every feasible link is usable, but central-node forwarding is preferred when candidates otherwise satisfy the task constraints.

At release, the task is available at the ground node. The simulator visits contact windows in chronological order. A packet is indivisible: a transfer is accepted only when the remaining window capacity can

carry the complete packet. A receiver can forward only after full reception, and duplicate reception does not increase coverage.

The bounded nominal policy prioritizes deadline-relevant useful recipients and limits forwarding to three hops, three observer copies, and five central-node copies. The unlimited useful-deadline diagnostic removes those copy and hop caps while retaining the same useful-recipient and deadline definitions. It is used to show what the existing contact opportunities could support under a less restrictive forwarding rule; it is not presented as the nominal operational policy.

Routing is greedy and causal. It uses only information available at the current simulated time and does not solve a global temporal multicommodity-flow optimization. This distinction defines what conclusions can be drawn from a difference between bounded and unlimited forwarding.

## 4.6 Scenario and Failure Transformations

Every case is evaluated under 539 declared scenarios. The nominal case is complemented by policy, packet-size, failure, outage, degradation, and combined-stress replays. Each transformation changes one declared part of the baseline and then reruns routing and observation evaluation. Figure 4.4 makes the comparison logic explicit: every branch begins from the same architecture and baseline opportunities, applies a declared transformation, and is paired back to that baseline. Differences are therefore interpreted as within-case scenario effects rather than comparisons between unrelated architectures.

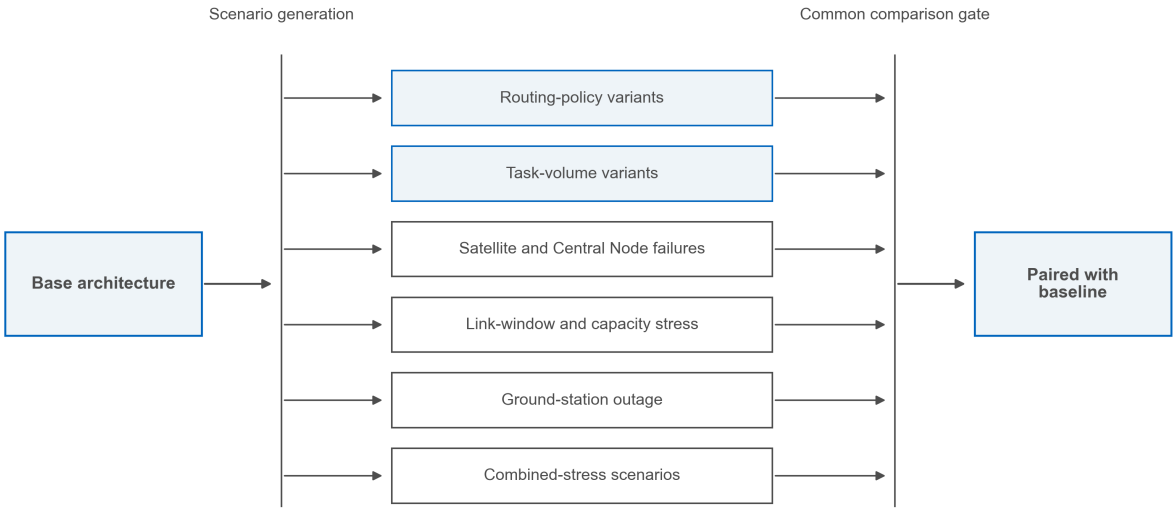



**Figure 4.4** Scenario logic. Each policy, packet, failure, or capacity change is applied to the same baseline case before routing and observation are replayed.

Table 4.2 separates routing and traffic interventions from component and contact disturbances, which prevents a policy sensitivity from being interpreted as a hardware-failure result. Random scenarios use deterministic seeds derived from the campaign seed, case identifier, scenario identifier, and replicate number. This makes a row reproducible and independent of worker count or completion order. Because the case identifier is part of the seed, two different architectures are not described as common-random-number pairs unless a dedicated paired-draw contract is added.

**Table 4.2** Scenario families and the interpretation supported by each replay.

| Scenario family | Transformation | Question answered |
|---|---|---|
| Routing policy | Central-only, peer-only, hybrid, unlimited diagnostic | Is performance limited by link eligibility or forwarding bounds? |
| Packet size | 0.5x, 1x, 2x, 4x | How sensitive is the network to request volume? |
| Random satellite failure | Remove sampled satellites and their links/accesses | How much performance remains after generic node loss? |
| Random/targeted central-node failure | Remove sampled or high-importance central nodes | Is the architecture dependent on specific relays? |
| Link-window outage | Remove sampled contact windows | Is the design sensitive to lost communication opportunities? |
| Ground-station outage | Disable one sampled interval | How dependent is release and injection on the ground segment? |
| Capacity degradation | Scale window capacity | How does lower effective throughput change outcomes? |
| Combined stress | Increase packet size and remove satellites | Do load and topology damage amplify each other? |

## 4.7 Outcomes and Interpretation

The metrics are grouped by the question they answer.

**Mission performance.** Observation fulfilment is the percentage of tasks with a valid post-reception access before deadline. Useful coverage is the average share of useful recipients reached before each task deadline. Strict completion is the share of tasks that reach every surviving useful recipient before deadline. Completion time is reported only when the required dissemination actually completes; censored tasks remain in the denominator of probability metrics.

**Communication burden.** Total transferred Mbit, transfer count, used-capacity percentage, central-node traffic share, peak node load, and the Gini coefficient of transmitted data describe how the result was achieved. A low delay obtained through much higher traffic is not treated as a free improvement.

**Robustness.** Stressed performance is compared with the same case's baseline using the raw stressed value, paired change, and retention ratio. Original and surviving denominators are stored separately so that deleting difficult recipients cannot create an artificial improvement.

**Decision support.** Exact Pareto filtering first removes cases that are no better on any declared objective and worse on at least one. Objective-weight sensitivity is then applied only to the non-redundant metric families. Rank-1 and top-5 acceptability over reproducible random weight draws describe stability under the declared preference distribution. They are not probabilities that an architecture is objectively correct.

Figure 4.3 provides one concrete interpretation. A task can be received by Satellite A and still fail if A has no access before the deadline. It can be forwarded to Satellite B and succeed if B receives the complete packet before its later access. A figure or metric that omits this event order would answer a different question.

## 4.8 Validation and Provenance

A case is not ranked merely because its output directory contains a success marker. Before a row enters the analysis, the post-processing checks the inventory, schema, task provenance, scenario count, duplicate keys, reconstructable metrics, and eligibility for a balanced comparison. The success marker answers only whether the programmed run finished. Separate checks of the task hash, regional count, and priority counts establish whether it finished with the intended scientific input.

The first India 180-satellite preparation failed this input audit. After India was filtered from the combined table, its rows still carried labels 32–804. The normalization table had new labels 0–772. When one labelled series was assigned to the other, pandas matched the labels instead of the row positions. The result was 32 invalid rows at the beginning and the loss of the final 32 Low-priority tasks from 16 April 2026. That execution is excluded from every final numerical claim. In the verified package, the regional index is reset before normalization. Execution is also tied to task and run-contract hashes, and all 282 selected cases are checked against the intended 773 tasks.

This correction determines which comparisons the final evidence can support:

- the 60- and 120-satellite California-India comparison uses the intended and matched regional inputs;
- the 180-satellite India extension uses the same 773-task population and supports a matched 120-to-180 comparison on the balanced CN30–CN100 block; and
- no California–India 180-satellite effect is claimed because no matched corrected California layer is available.

A completed list of scenario identifiers would not have exposed the mismatched task rows. For this reason, the corrected run contract also records the task-file hash, regional count, priority distribution, case allow-list, and positional normalization.

## 4.9 Reproducibility and Method Boundary

The stored evidence includes scenario and task metrics, selected transfer traces, Central Node placement, constellation nodes, manifests, and validation records in typed Parquet files. Checkpointing allows an interrupted campaign to continue. It changes execution time and recovery behaviour, not the experiment: changing the worker count leaves task rows, architecture identifiers, and deterministic seeds unchanged.

The model stops at the follow-up opportunity. It does not simulate the final image downlink, an emergency-service action, weather or cloud, detailed pointing, power and storage scheduling, lifecycle cost, or a flight-qualified relay design. The conclusions must therefore be read as results for the implemented Central Node routing role and for the two stated task populations.

# 5 Experimental Design and Reproducibility

This chapter sets the rules used to turn simulator output into thesis evidence. The datasets and their permitted claims are defined first (Section 5.1), followed by the architecture and scenario design (Sections 5.2–5.3), integrity and eligibility tests (Sections 5.5–5.6), and the statistical and decision analysis (Section 5.8). Chapters 6 and 7 follow this order. Commands and repository operation are left to Appendix E because they are needed for reproduction but are not part of the scientific argument.

## 5.1 Analysis Datasets and Permitted Inference

The report uses one final verified simulation campaign together with the earlier Reference Campaign. Within the final campaign, comparison blocks are selected according to the claim being tested rather than pooled without regard to task population or fleet coverage. Table 5.1 states the permitted use of each dataset before any result is interpreted.

**Table 5.1** Analysis datasets and the claims they can support.

| Dataset | Coverage | Permitted use |
|---|---|---|
| Reference Campaign | 891 California and 891 India architecture cases | Historical matched event-window baseline under the frozen nominal model |
| Final regional block | 594 California and 594 India cases at 60/120 satellites; 539 scenarios per case | Paired event-window effects, failure sensitivity, exact Pareto analysis, and cross-window candidates |
| Final India 180-satellite extension | 282 verified cases on 773 tasks; balanced 216-case CN30–CN100 block | Large-fleet response, matched India 120-to-180 effects, robustness, priority timing, and portfolio extension |

The Reference Campaign predates the final scenario contract. It is retained because it provides a complete like-for-like sweep under one frozen model, but its values are not averaged with the final campaign. The final verified campaign contains a matched regional block and an India 180-satellite extension. California and India share 594 architecture identifiers in the 60/120-satellite block, while the 180-satellite cases use the same 773-task India contract. The complete CN30–CN100 part can therefore be paired with the corresponding 120-satellite architectures for a fleet-size contrast. California–India conclusions remain limited to 60/120 satellites because no California 180-satellite population was executed.

## 5.2 Architecture Design

Four geometry factors produce 81 base architectures. Each base geometry is evaluated at eleven Central Node fractions, giving 891 identifiers per event window. Table 5.2 lists the factors and explains what their variation represents.

The factorial grid makes matched contrasts possible: a Central Node fraction can change while the Walker geometry remains fixed, and a California result can be paired with the same case identifier in India. The Central Node treatment is nevertheless a package. It changes eligibility, forwarding preference, deterministic placement, and link-budget assumptions together. Consequently, a response to Central Node fraction is a response to that implemented package, not an isolated causal coefficient for a single mechanism.

**Table 5.2** Architecture factors in the Full891 design.

| Factor | Levels | Experimental role |
|---|---|---|
| Total satellites | 60, 120, 180 | Fleet scale and resource proxy |
| Orbital planes | 3, 5, 10 | Spatial distribution and temporal contact geometry |
| Altitude | 500, 800, 1000 km | Access and communication opportunity |
| Inclination | 60, 80, 98.6 deg | Latitude-dependent access geometry |
| Central Node fraction | 0, 10, . . ., 100% | Exposure to the implemented Central Node routing package |
| Walker phasing | $F = 1$ | Fixed control |

## 5.3 Scenario Design

The final scenario contract declares 539 scenario identifiers for every architecture (Table 5.3). Deterministic settings are evaluated once. Random failure families use 30 seeded trials at each severity; combined-stress cells use 10. A replicate changes only the seeded failed nodes, removed windows, or outage interval while retaining architecture, task set, family, and severity.

**Table 5.3** Final verified scenario contract for each architecture case.

| Scenario family | Construction | Rows |
|---|---|---|
| Core, policy, and unlimited | Seven named scenarios | 7 |
| Task size | $0.5\times$, $1\times$, $2\times$, $4\times$ | 4 |
| Random satellite failure | Four levels $\times$ 30 trials | 120 |
| Random Central Node failure | Four levels $\times$ 30 trials | 120 |
| Link-window outage | Four levels $\times$ 30 trials | 120 |
| Ground-station outage | Four levels $\times$ 30 trials | 120 |
| Targeted Central Node failure | Four deterministic levels | 4 |
| Capacity degradation | Four deterministic levels | 4 |
| Combined stress | Two packet sizes $\times$ two failure levels $\times$ 10 trials | 40 |
| Total | | 539 |

Thirty trials per random-failure level are used for mean-response comparisons. The combined-stress cells have only ten trials, so they are used descriptively without a reported fifth-percentile or fine rank order. A convergence extension would be needed for that purpose. Central Node failure at CN0 is stored as `NOT_APPLICABLE`. Keeping this row prevents an experiment that has no meaning at CN0 from being confused with a failed simulation.

## 5.4 Scientific Analysis Workflow

Figure 5.1 gives the order applied to every result. Input identity and integrity are checked before a plot, score, or ranking is calculated. If a gate fails, the population is either restricted to a valid comparison block or the claim is withheld. Normalization and weighting are not used to conceal an incomplete input or output.

Only an eligible population proceeds to the statistical and decision stages. This prevents a plausible-looking plot from hiding duplicate rows, changed denominators, an incomplete fraction sweep, or mishandled censoring. Any one of these issues can change the Pareto set and the reported architecture order.

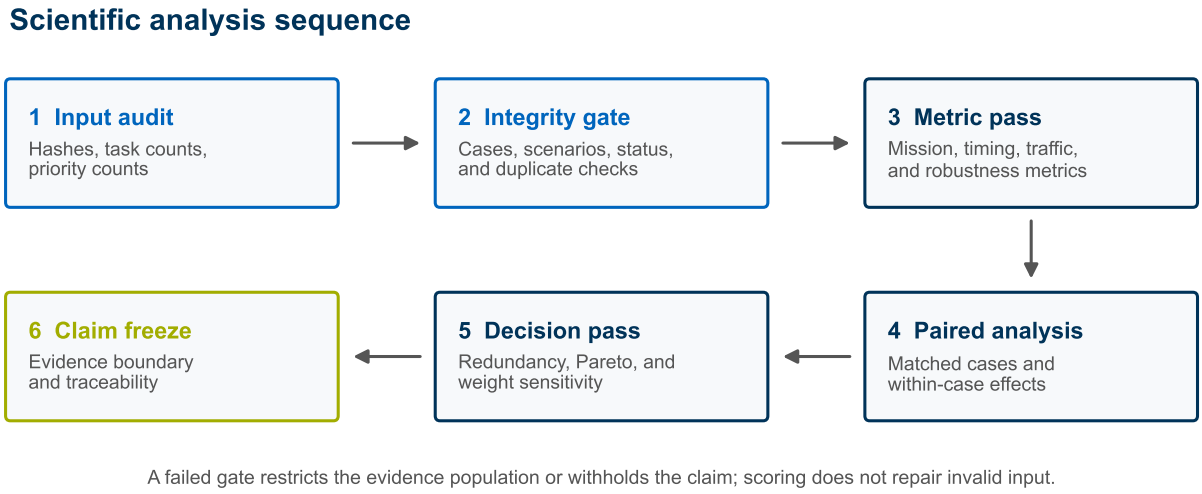



**Figure 5.1** Logical sequence from input audit to a frozen thesis claim. Implementation details are documented separately in Appendix E.

## 5.5 Integrity Gate and Task Provenance

The final campaign was inventoried before any mean or contrast was calculated. A complete regional inventory should contain $891 \times 539 = 480{,}249$ case–scenario status keys, and both regional archives meet that count after validation. Restarting the California campaign had created 4,269 duplicate intermediate keys. The de-duplication rule retains the final scientifically identical record, and no retained row has a failed status. This gate follows the model-credibility principle that an input, assumption, verification result, and intended claim must be traceable; a single global "validated" label is not enough [12, 14]. Table 5.4 lists the actual checks and their evidence.

**Table 5.4** Validation gates applied before result calculation.

| Gate | Test | Evidence |
|---|---|---|
| Inventory | Expected case directories, 539 scenario identifiers per case, markers, and explicit not-applicable states | Appendix C; completion-gate table |
| Uniqueness and status | Unique case–scenario keys after deterministic de-duplication; no failed retained row | Appendix C; integrity summary |
| Input contract | SHA-256 digest, regional task counts, priority counts, and metric denominators | Appendix C; task-population contract |
| Metric reconstruction | Independent recomputation from retained task/transfer evidence where available | Appendix C; metric audit |
| Pair eligibility | Shared case identifiers, complete factor sweep, and identical task population for each comparison | Tables in Chapters 6–7 |
| Decision eligibility | Missing values removed explicitly; redundancy checked; exact Pareto set computed before weighting | Decision tables and manifest |

The intended task file has the full SHA-256 digest

`e717a1dde84db0f47efdda09cfca2f437`
`909349b25a139a45a7c1928da80e533.`

The hashed file contains 32 California tasks and 773 India tasks. In the first India 180-satellite execution, regional filtering preserved source labels 32–804, while normalization created an index from

0–772. Label-based assignment then produced 32 invalid coordinate rows and omitted the final 32 source rows. The corrected package resets the positional index before normalization and checks both the regional count and task hashes in its run contract. Its verifier reports 282 complete cases, no missing or invalid cases, 539 scenarios per case, and all 773 India tasks. None of the earlier 741-task values is used in the final tables or figures.

## 5.6 Analysis Populations

Table 5.5 gives the final populations. The balanced blocks are used whenever marginal curves, paired effects, Pareto filtering, or rank comparisons require equal architecture coverage.

**Table 5.5** Final analysis populations after the integrity gate.

| Population | Bases | Cases | Scenarios/case | Rows |
|---|---|---|---|---|
| Matched California 60/120 | 54 | 594 | 539 | 320,166 |
| Matched India 60/120 | 54 | 594 | 539 | 320,166 |
| India 180, all completed | 27 | 282 | 539 | 151,998 |
| India 180, balanced CN30–CN100 | 27 | 216 | 539 | 116,424 |

The 180-satellite selection contains 20 cases at CN0, 22 at CN10, 24 at CN20, and all 27 base architectures at every level from CN30 through CN100. Low-CN rows remain available for inventory and individual-case inspection, but they are excluded from aggregate rankings that require equal base weights.

## 5.7 Metrics Used in the Main Analysis

The principal measures are defined here because their denominators determine the scientific interpretation. Table 5.6 gives their meaning and main boundary; Appendix A provides complete formulas and edge-case handling.

**Table 5.6** Principal result metrics and interpretation.

| Metric | Meaning | Important boundary |
|---|---|---|
| Observation fulfilment | Percentage of released tasks observed after the observing satellite received the task | Not a sensor-quality or end-to-end response metric |
| Strict useful completion ($S_{100}^{\text{useful}}$) | Percentage of tasks delivered before deadline to every simulator-defined useful recipient | Becomes harder as the useful-recipient set grows |
| Useful-recipient coverage | Mean fraction of useful recipients reached before deadline | Complements the strict all-recipient condition |
| P100 $T_{100}$ | Worst complete-dissemination time among tasks for which every useful recipient is reached | Undefined when any required delivery misses the deadline |
| Transferred data | Total modeled task-message traffic per case | Task-message burden, not image-data volume |
| Robustness retention | Stressed metric divided by the matched unlimited baseline | A relative sensitivity measure, not failure probability |

Timing analyses retain missed events as right-censored observations. Priority classes use their fixed deadline horizons, and any cross-window summary gives High, Medium, and Low equal weight rather than allowing the more numerous Low-priority tasks to dominate. The event-time summaries use the Kaplan–Meier estimator [8] and restricted mean event-free time within the declared deadline horizon [18].

## 5.8 Statistical and Decision Workflow

After the population and metric definitions are fixed, the following steps are applied in order:

1. **Describe the response.** Marginal means show the response over the enumerated grid. Base-clustered percentile-bootstrap intervals [5] show how stable that mean is across the tested base architectures; they are not geographic-population intervals.
2. **Form matched contrasts.** Results are subtracted only within the same architecture identifier when comparing event windows or scenarios. This removes between-architecture differences from the contrast.
3. **Check association and shape.** Spearman $\rho$ measures monotonic rank agreement without requiring a straight-line relation. A regression through the eleven Central Node marginal means is used as a compact description of shape. Cross-validated error is reported so that a high in-sample $R^2$ is not mistaken for a physical law.
4. **Measure robustness.** Each disturbed case is paired with the same architecture's unlimited baseline. Raw stressed performance, change, retention, and the response over severity are shown before ranking. This keeps degradation and robustness-envelope behaviour visible instead of collapsing them into one network score [23, 28].
5. **Screen redundancy.** Highly correlated metrics remain in the diagnostic tables, but a near-duplicate pair does not receive two full contributions to utility.
6. **Apply exact Pareto filtering.** A case is removed if another eligible case is at least as good in every declared objective and better in one. Raw objective directions are used before any preference weights, as in Pareto-robust space-system trade studies [17].
7. **Test preference sensitivity.** Dirichlet weight draws record how often a remaining case ranks first or in the top five. This is a stochastic multi-criteria acceptability analysis of the chosen preference range [10]; it is not a survey of stakeholder opinion. The broader principle is that a simulation conclusion should be checked against declared factor and preference changes [9].

For the robustness curves, the line at each severity is the mean of the matched base-architecture means. The shaded band is one standard deviation across those base means. It therefore shows architecture-to-architecture dispersion within the enumerated grid, not uncertainty for a wider geographic population and not variability from an unexecuted reliability model.

The measured robustness results are also placed in a likelihood–consequence screening matrix. Following the risk-matrix convention described in the NASA risk-management handbook [13], the two dimensions are kept explicit. Consequence is evidence-based: it is the worst mean strict-completion loss at 30% severity across the three eligible final-campaign blocks, scored as 1 for $< 2$ percentage points, 2 for 2–$< 5$, 3 for 5–$< 10$, 4 for 10–$< 20$, and 5 for $\geq 20$. Likelihood is not estimated by the simulation. It is a declared engineering screening assumption—Rare for targeted Central Node loss; Unlikely for ground-station outage, random Central Node loss, and random satellite loss; and Possible for capacity degradation and link-window outage. The product is classified Low at 1–4, Moderate at 5–9, and High at 10–25. The matrix supports prioritization and communication only; it is not a probabilistic safety assessment and should be recalculated when mission reliability data become available.

The workflow produces a portfolio rather than a universal winner. Weighted utility is used only inside the feasible Pareto set and never substitutes for raw mission, timing, traffic, and fleet values.

## 5.9 Scope Limits

The evidence uses a fixed 700.00 km access radius, one deterministic balanced Central Node placement, one ground station, one set of task-generation rules, and a bundled Central Node routing treatment. The two event windows are selected cases rather than random samples. Within the final campaign, the 180-satellite extension supports a controlled India 120-to-180 comparison only on the balanced CN30–CN100 cells and under the shared scenario definition. It does not create a California–India comparison at 180 satellites, and the incomplete CN0–CN20 selection prevents a fully balanced three-fleet sweep over all eleven fractions. These boundaries determine the wording of Chapters 7–9.

# 6 Nominal Results

This chapter uses the Reference Campaign to establish the behaviour of the frozen nominal model. The analysis begins with the two event-window baselines and then examines the Central Node and geometry responses. Correlation and regression are used only as compact summaries of those responses. The final sections check whether architecture order carries from one event window to the other and show the nominal Pareto trade space. The purpose is to expose the existing performance tensions, not to select a universal optimum from this earlier campaign.

## 6.1 Reference-Campaign Boundary

The earlier nominal campaign is retained as the historical like-for-like two-event-window architecture sweep. It contains 891 California cases and the same 891 identifiers for India, preserving the baseline comparison under that frozen model. These values are kept separate from the final verified campaign and are not averaged with the scenario results in Chapter 7. The contrast is a case-study comparison, not an estimate of a California, India, seasonal, or global population effect.

## 6.2 Validation Status

All case rows used in this chapter passed the validation procedure in Chapter 5. Specifically, all 1,782 expected case identifiers were present, every case had the expected task population and completed status, and 17,820 independently recomputed metric comparisons agreed with the stored values within $10^{-6}$. Appendix C records the checks and their evidence. This chapter preserves the Reference Campaign as the frozen nominal baseline. Chapter 7 then presents the final verified campaign, including the matched regional analysis and the verified 180-satellite evidence.

The analytical population contains 891 cases for California and 891 for India. Every case has the expected task count, and every required case status is completed. Core summary metrics were recomputed from lower-level outputs in 17,820 comparisons with zero failures at a tolerance of $10^{-6}$. The numerical evidence is therefore internally consistent even though external model-validity limitations remain.

## 6.3 Event-Window Campaign Summary

Table 6.1 gives event-window-normalized means over all 891 architectures. The India workload has the higher mean observation fulfilment, while the California workload reaches a larger share of its useful-recipient set. The mean number of transfer events per task is almost unchanged between the two windows. Data volume per task is higher for California, which contains a larger proportion of the bigger urgent packets.

**Table 6.1** Region-normalized summary across all 891 cases per event region.

| Region | Obs. (%) | $S_{100}^{\text{useful}}$ (%) | Receivers (%) | Mbit/task | Events/task |
|---|---|---|---|---|---|
| California | 47.47 | 0.62 | 13.87 | 5.88 | 6.19 |
| India | 55.07 | 0.46 | 12.06 | 4.62 | 6.20 |

The very low mean $S_{100}^{\text{useful}}$ values require careful interpretation. They do not mean that almost no task is observed: mean observation fulfilment is roughly 47–55 percent. They mean that completing

dissemination to every simulator-defined useful recipient before each deadline is rare under the targeted policy, three-hop limit, and copy limits. Observation requires only one suitably timed observer, while $S_{100}^{\text{useful}}$ requires all useful recipients.

## 6.4 Central-Node Fraction Effects

Figure 6.1 averages the 81 base geometries at each central-node level. The first step from CN0 to CN10 produces the largest observation gain. Further increases are smaller, and the observation mean falls sharply at CN100. First-reception latency behaves differently: it continues to decrease as the central-node fraction rises. Useful-recipient completion and receiver coverage instead reach their best aggregate values at modest fractions and then decline.

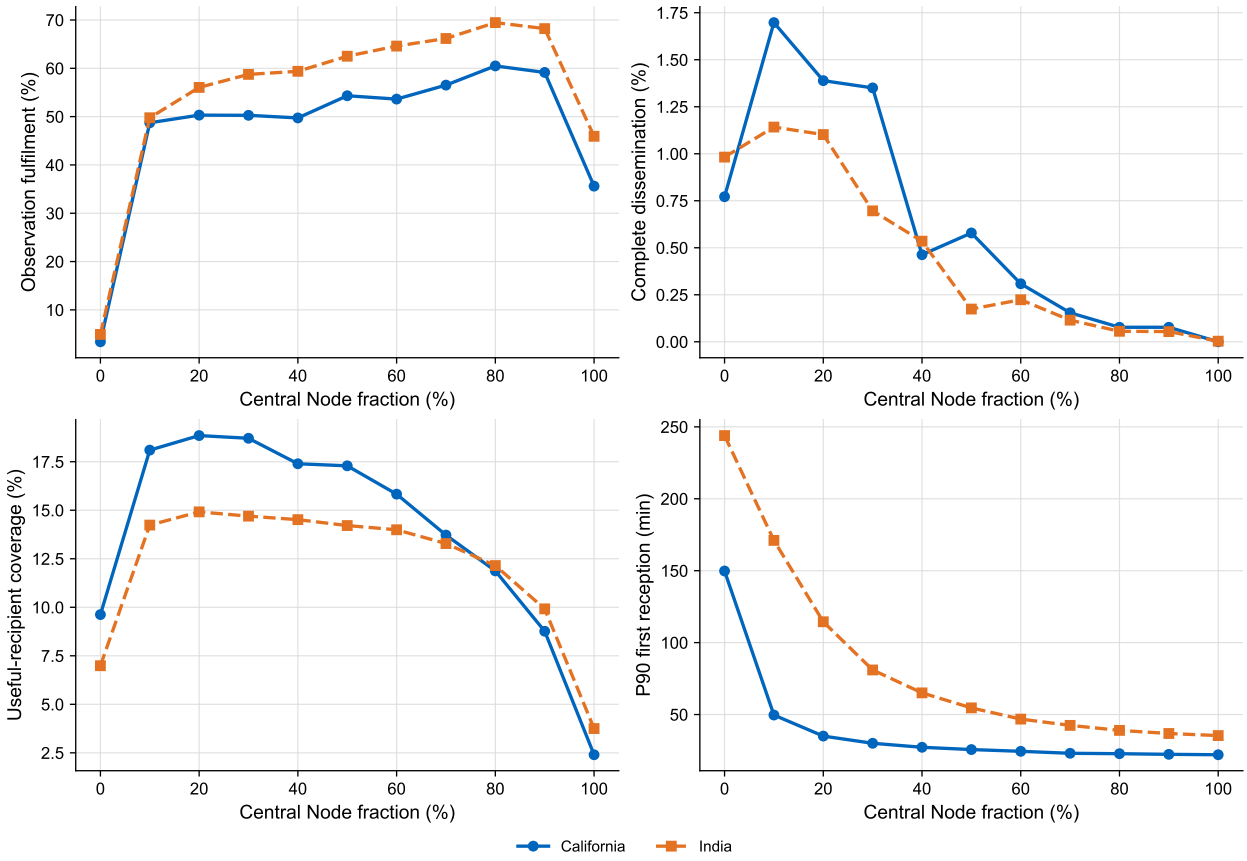



**Figure 6.1** Event-window-specific marginal response to central-node fraction across the full final grid. $S_{100}^{\text{useful}}$ denotes complete dissemination to all simulator-defined useful recipients.

Selected levels are reported in Table 6.2. The table illustrates the key engineering trade. For California, moving from CN0 to CN10 raises mean observation fulfilment from 3.40 to 48.73 percent and reduces P90 first-reception latency from 149.78 to 49.65 minutes. For India, the corresponding observation change is 4.92 to 49.77 percent and the latency change is 243.94 to 171.06 minutes. Further centralization continues to improve conditional first reception, but useful-recipient breadth does not continue to improve.

### 6.4.1 Observation fulfilment

The event-window-aggregated observation maximum occurs at 80 percent Central Nodes: 60.49 percent in California and 69.46 percent in India. At 90 percent, both means are slightly lower, and at 100

**Table 6.2** Selected event-window central-node marginal means across 81 base architectures per row.

| Region | CN (%) | Obs. (%) | $S_{100}^{\text{useful}}$ (%) | Receivers (%) | P90 first (min) |
|---|---|---|---|---|---|
| California | 0 | 3.40 | 0.77 | 9.62 | 149.78 |
| California | 10 | 48.73 | 1.70 | 18.10 | 49.65 |
| California | 20 | 50.31 | 1.39 | 18.85 | 35.01 |
| California | 50 | 54.32 | 0.58 | 17.29 | 25.65 |
| California | 80 | 60.49 | 0.08 | 11.87 | 22.77 |
| California | 100 | 35.61 | 0.00 | 2.40 | 22.04 |
| India | 0 | 4.92 | 0.98 | 6.99 | 243.94 |
| India | 10 | 49.77 | 1.14 | 14.24 | 171.06 |
| India | 20 | 56.04 | 1.10 | 14.92 | 114.49 |
| India | 50 | 62.50 | 0.17 | 14.21 | 54.63 |
| India | 80 | 69.46 | 0.06 | 12.15 | 38.98 |
| India | 100 | 45.93 | 0.00 | 3.75 | 35.37 |

percent they fall to 35.61 and 45.93 percent. This is the central result of the sweep: the first Central Nodes remove an information bottleneck, but full participation does not preserve that gain under the bounded forwarding package. The decrease is therefore meaningful rather than a cosmetic curve feature, although the bundled treatment does not identify whether forwarding concentration, copy limits, placement, or another package component is responsible.

The highest individual California observation result is 93.75 percent for `T060_P05_H0800_I0800_CN050_F1`. The highest India result is 99.09 percent for `T180_P10_H1000_I0600_CN090_F1`. Neither of these cases completes $S_{100}^{\text{useful}}$ for any task, emphasizing that a case can observe nearly all tasks without disseminating them to every useful recipient.

### 6.4.2 Useful-recipient completion and receiver coverage

Mean $S_{100}^{\text{useful}}$ is small at every aggregate level and reaches its marginal maximum at CN10 in both regions: 1.70 percent for California and 1.14 percent for India. Mean useful-receiver coverage instead peaks at CN20: 18.85 percent for California and 14.92 percent for India. Beyond this point, the number of useful receivers reached by the deadline decreases, particularly at CN100.

The strongest individual $S_{100}^{\text{useful}}$ cases are lower-resource 60-satellite configurations. California reaches 25.00 percent for `T060_P03_H0800_I0986_CN030_F1`; India reaches 21.48 percent for `T060_P05_H1000_I0600_CN020_F1`. These same cases also have the largest mean receiver fraction in their region, 47.58 and 45.56 percent, respectively.

The fact that 60-satellite cases dominate this metric is partly mathematical and partly operational. With fewer eligible recipients, completion requires fewer unique arrivals. Larger constellations create more possible observers but also enlarge the dissemination target. This is why $S_{100}^{\text{useful}}$ should be compared together with observation fulfilment and resource burden rather than used alone.

### 6.4.3 First-reception latency

Between CN0 and CN100, conditional P90 first-reception latency falls from 149.78 to 22.04 minutes for California and from 243.94 to 35.36 minutes for India. The shapes are not identical. California obtains most of its reduction by CN20, whereas India continues to improve strongly through CN50. Beyond those levels, each additional reduction is smaller.

The smallest per-case conditional P90 is 1.40 minutes in California and 8.52 minutes in India. These values do not by themselves identify the best architectures. A conditional latency can be small simply because difficult, unfinished tasks have been censored. Any latency comparison must therefore be

read together with its completion fraction, which is why the canonical tables retain the censoring columns.

## 6.5 Architecture Factor Effects

Table 6.3 collects the most decision-relevant factor-level means. Each entry pools the other factors, so it describes association within the grid rather than a controlled one-factor experiment. The table shows why geometry and centrality cannot be reduced to a single monotonic rule.

**Table 6.3** Selected factor-level marginal effects across the full nominal grid.

| Region | Factor | Level and interpretation | Mean obs. (%) | Mean P90 first (min) |
|---|---|---|---|---|
| California | Satellites | 60; best mean observation, widest receiver denominator advantage | 51.04 | 44.63 |
| California | Planes | 5; highest observation mean | 55.16 | 27.55 |
| California | Inclination | 60 deg; highest observation and lowest latency | 54.02 | 13.23 |
| California | CN fraction | 80%; highest mean observation | 60.49 | 22.77 |
| India | Satellites | 120; marginally highest observation mean | 55.57 | 78.79 |
| India | Planes | 10; highest observation mean | 62.27 | 68.98 |
| India | Inclination | 60 deg; strongest regional geometry | 72.42 | 54.22 |
| India | CN fraction | 80%; highest mean observation | 69.46 | 38.98 |

Adding satellites shortens the conditional P90 of first reception, but the fraction of useful receivers reached becomes smaller. Mean receiver coverage falls from 20.95 to 8.55 percent between 60 and 180 satellites in California and from 18.33 to 7.32 percent in India. This does not mean that the larger constellations send fewer messages. Their useful-recipient set grows faster than the bounded-copy policy can cover it.

Plane count affects the two event windows differently. California has its highest mean observation at five planes, while the India workload benefits strongly from ten. Inclination produces the largest event-window contrast: 60 degrees yields 72.42 percent mean observation in India, whereas 98.6 degrees yields 40.43 percent. This is consistent with task latitude and event epoch, but the simulator does not isolate which component is responsible.

Altitude has weaker mean association with observation fulfilment than inclination or plane count. Higher altitude reduces conditional P90 first reception in both regions, but the observation means differ by only a few percentage points across 500, 800, and 1000 km after pooling other factors.

## 6.6 Exploratory Correlation

Table 6.4 lists the strongest exploratory rank associations. Higher central-node fraction is associated with lower P90 first-reception latency in both regions. Satellite count is negatively associated with useful-receiver fraction, and inclination shows a strong negative association with India observation fulfilment over the sampled levels.

**Table 6.4** Selected exploratory Spearman associations across 891 cases per region.

| Region | Factor | Metric | $\rho$ |
|---|---|---|---|
| California | Central Node fraction | P90 first-reception latency | $-0.55$ |
| California | Satellite count | Mean useful-receiver fraction | $-0.56$ |
| California | Plane count | Observation fulfilment | $+0.39$ |
| California | Central Node fraction | Observation fulfilment | $+0.29$ |
| India | Central Node fraction | P90 first-reception latency | $-0.69$ |
| India | Satellite count | Mean useful-receiver fraction | $-0.61$ |
| India | Inclination | Observation fulfilment | $-0.59$ |
| India | Central Node fraction | Observation fulfilment | $+0.34$ |

Spearman $\rho$ asks whether the rank of one variable tends to rise or fall with the rank of another; the relation need not be linear. A value close to $+1$ or $-1$ indicates consistent monotonic order, while a value near zero indicates little such order. For India, $\rho = -0.69$ means that higher Central Node fractions are usually associated with lower first-reception latency in this grid. It is not a causal coefficient. Recipient opportunities and all components of the bundled routing treatment change with the fraction, and the calculation pools the remaining architecture factors.

## 6.7 Exploratory Regression of Central-Node Marginal Means

Figure 6.2 compares the best leave-one-level-out model for each event-window response. An exponential-with-offset model describes the steep latency fall better than the polynomial candidates, with leave-one-level-out RMSE of 23.57 minutes for California and 6.29 minutes for India. Quadratic models are preferred for receiver fraction in both event windows, capturing the intermediate peak and later decrease. In Table 6.5, $x$ is the Central Node fraction in percentage points. Observation and coverage equations return percentages, while the latency equations return minutes.

Table 6.5 gives the selected models. California observation has a large cross-validated error despite a high in-sample $R^2$, because a saturating exponential does not reproduce the CN100 decline. This is a warning against treating the fitted curve as a physical law. India observation and both receiver responses are better represented by a quadratic over the sampled range, but extrapolation beyond 0–100 percent is meaningless.

**Table 6.5** Best exploratory regression model by leave-one-central-node-level-out error.

| Region | Response | Fitted equation $y = f(x)$ | $R^2$ | LOOCV RMSE |
|---|---|---|---|---|
| California | Observation fulfilment | $-48.8869e^{-0.253451x} + 52.2948$ | 0.83 | 16.33 pp |
| California | Useful-recipient coverage | $-0.00414501x^2 + 0.318226x + 12.4641$ | 0.91 | 2.61 pp |
| California | P90 first reception | $124.746e^{-0.151333x} + 24.7261$ | 1.00 | 23.57 min |
| India | Observation fulfilment | $-0.0133479x^2 + 1.64310x + 19.6284$ | 0.78 | 14.21 pp |
| India | Useful-recipient coverage | $-0.00335199x^2 + 0.294200x + 9.08151$ | 0.85 | 2.31 pp |
| India | P90 first reception | $211.946e^{-0.0474877x} + 33.9073$ | 1.00 | 6.29 min |

The fit statistics are not uniformly “good”. The latency curves have excellent in-sample $R^2$, but California’s 23.57-minute leave-one-level-out error is still operationally large. Observation has lower $R^2$

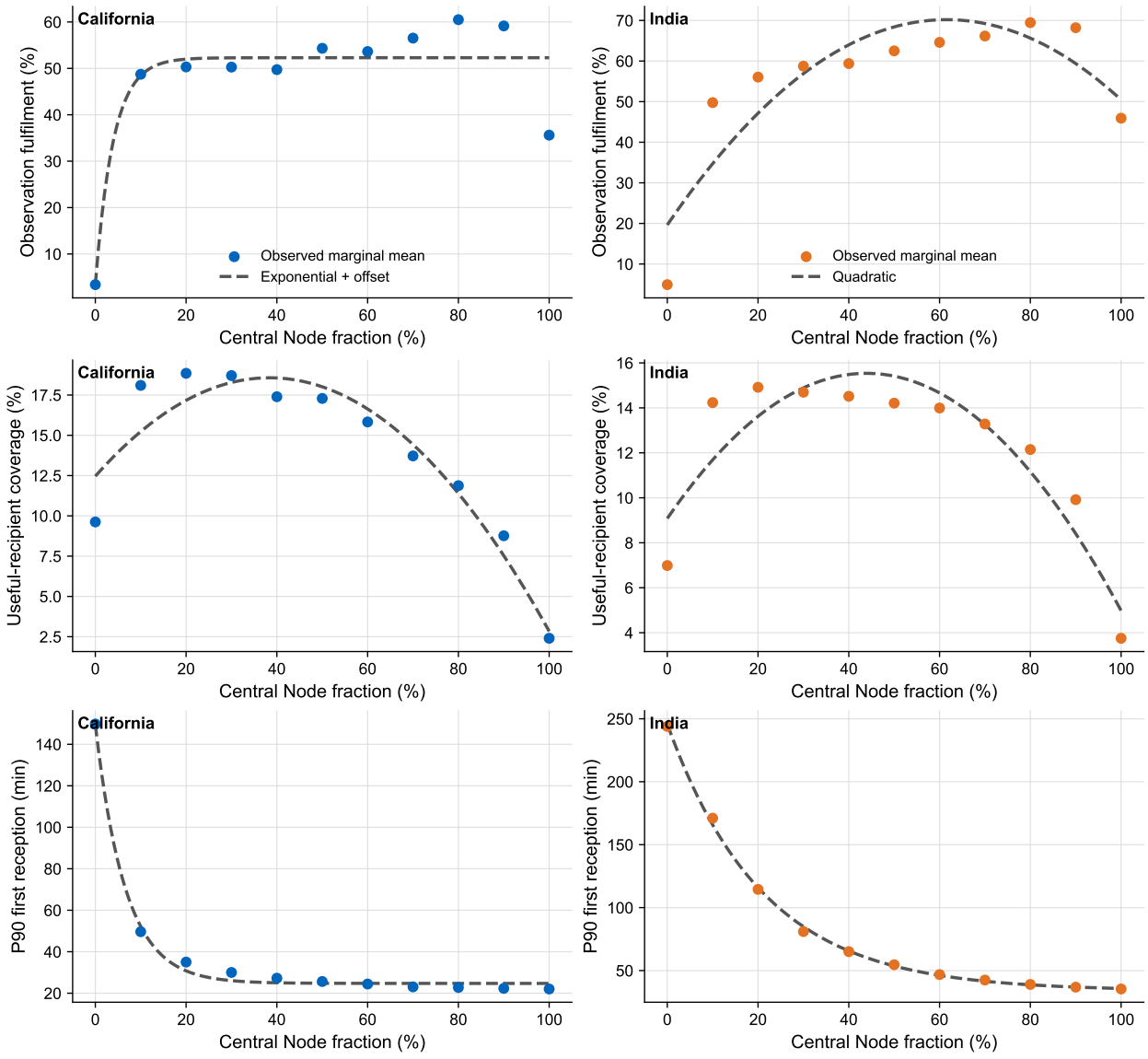



**Figure 6.2** Exploratory regression of the 11 event-window central-node marginal means. The displayed model minimizes leave-one-level-out RMSE among linear, quadratic, cubic, and exponential-with-offset candidates.

and 14.21–16.33-percentage-point prediction error because abrupt end-of-range changes are difficult to reconstruct when a Central Node level is omitted. Coverage is better behaved, with 2.31–2.61-percentage-point error. Thus the equations summarize saturation and hump-shaped response; they are not physical laws, they should not be extrapolated beyond the tested 0–100% range, and they cannot define an optimum without the competing completion and resource objectives.

## 6.8 Paired Event-Window Comparison

Figure 6.3 places the California and India observation result for each architecture on the same point. The scatter extends well to both sides of the diagonal. Some designs improve for India, others for California, and the ordering cannot be inferred reliably from one event window alone.

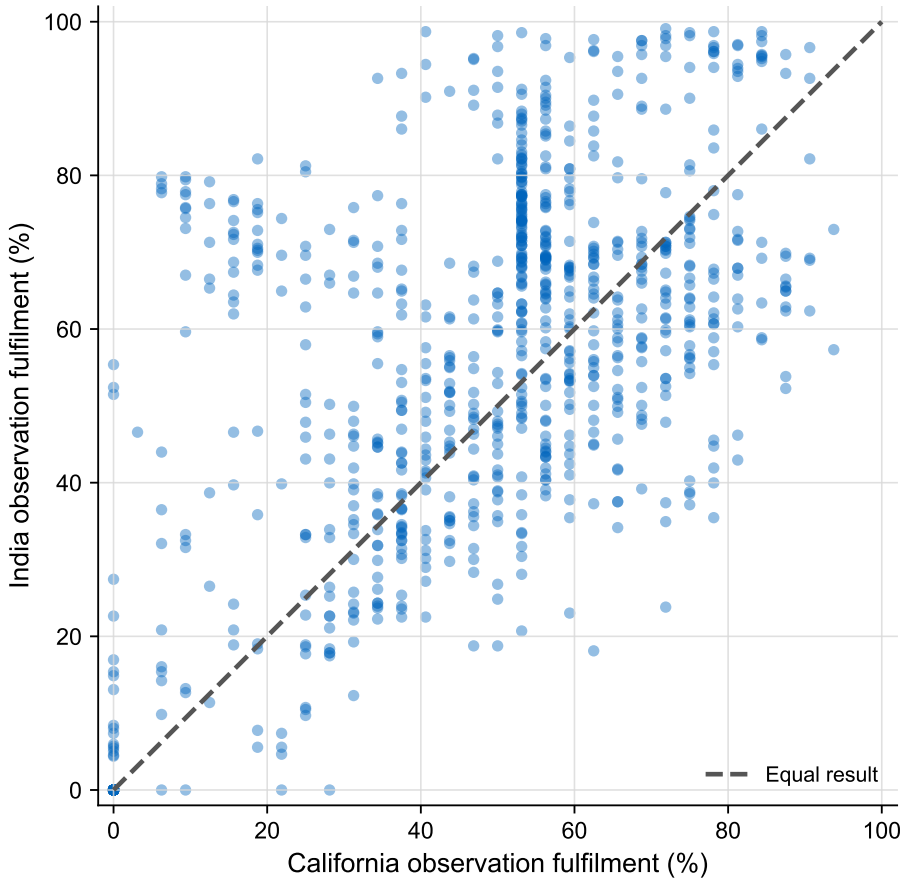



**Figure 6.3** Paired observation fulfilment for identical architecture identifiers under the California and India task sets.

Rank agreement is reported in Table 6.6. A value of 1 would mean that every architecture has the same relative order in both event windows; zero would indicate no monotonic agreement. Conditional P90 first reception and receiver coverage are comparatively stable at 0.78 and 0.74. Observation is only moderately stable at 0.50, and $S_{100}^{\text{useful}}$ agreement is 0.11. The last value means that an architecture selected for complete dissemination in one event window provides almost no ranking guarantee in the other. Different task counts, priority mixes, deadlines, and useful-recipient sets therefore alter the decision, not merely the absolute score.

**Table 6.6** Spearman rank agreement between California and India over 891 paired architecture identifiers.

| Metric | Rank agreement |
|---|---|
| Observation fulfilment | 0.50 |
| $S_{100}^{\text{useful}}$ | 0.11 |
| Mean useful-receiver fraction | 0.74 |
| Conditional P90 first-reception latency | 0.78 |

## 6.9 Exact Pareto Sets

Figure 6.4 plots observation fulfilment against the declared resource-burden proxy and outlines exact Pareto members computed in the full five-objective space. A point can be Pareto optimal even if it appears dominated in this two-dimensional projection, because it may provide better $S_{100}^{\text{useful}}$, receiver coverage, or conditional latency.

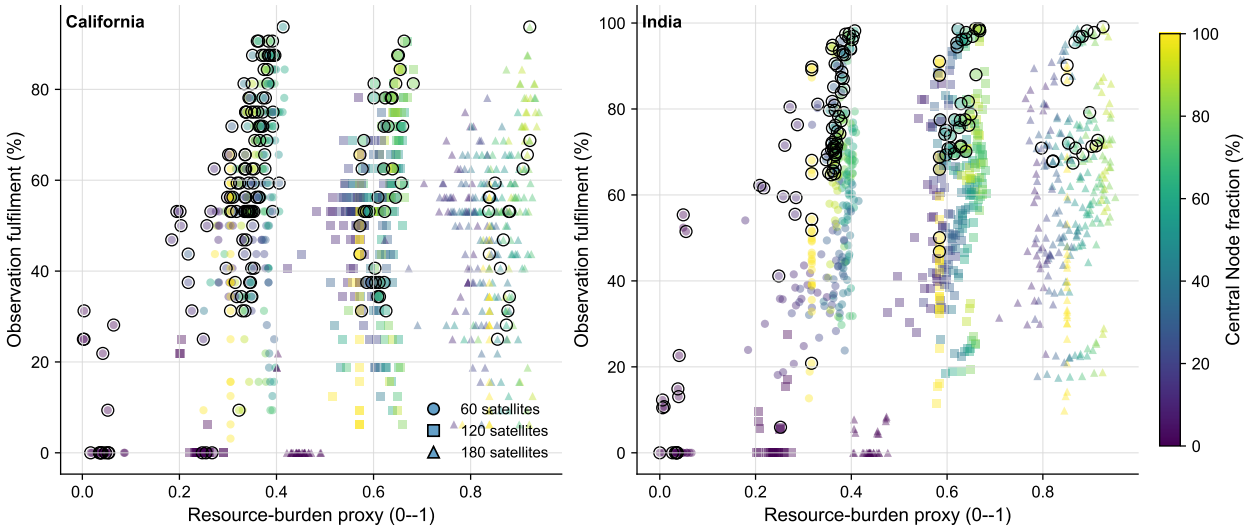



**Figure 6.4** Observation fulfilment versus resource burden, with fleet size shown by marker and Central Node fraction by colour. Black outlines identify cases on the exact five-objective Pareto set.

The exact fronts contain 183 California cases and 142 India cases, or 20.5 and 15.9 percent of the respective candidate sets. Table 6.7 explains which architectures form those fronts. Sixty-satellite cases dominate numerically (125 California and 85 India cases), with a median CN40 configuration. The 180-satellite front is much smaller (17 and 19 cases) and is concentrated at higher Central Node fractions. Smaller fleets therefore survive because they reduce resource burden, whereas larger fleets survive only when their mission benefit compensates for that burden. A single equally weighted score would hide this structure.

**Table 6.7** Composition of the exact nominal Pareto fronts.

| Region | Satellites | Pareto cases | Median CN (%) | Median observation (%) |
|---|---|---|---|---|
| California | 60 | 125 | 40 | 56.25 |
| California | 120 | 41 | 70 | 56.25 |
| California | 180 | 17 | 80 | 53.13 |
| India | 60 | 85 | 40 | 71.80 |
| India | 120 | 38 | 70 | 77.49 |
| India | 180 | 19 | 70 | 72.70 |

## 6.10 Nominal Result Synthesis

The nominal campaign shows a clear separation between early task delivery and broad dissemination. Adding a small central-node population greatly improves observation and first reception relative to CN0 under the central-only topology. Beyond that first step, latency continues to improve, but observation, useful-recipient coverage, and $S_{100}^{\mathrm{useful}}$ do not follow a monotonic trend. The bounded copy count and three-hop limit are therefore part of the observed architecture response rather than incidental implementation details.

The paired event-window results add a second constraint: the same architecture does not retain the same rank for California and India. This chapter answers the nominal routing-package and case-study workload questions under the earlier frozen model. It does not establish a population-level regional effect or tolerance to explicit failures; those experiments belong to the final verified campaign in Chapter 7.

# 7 Sensitivity and Robustness Results

This chapter presents the final verified simulation campaign as one connected analysis: integrity and population definition, matched 60/120-satellite results, the India 180-satellite extension, and an integrated decision synthesis. Mission, timing, traffic, and robustness are kept separate so that a gain in one quantity cannot hide deterioration in another.

## 7.1 Final Campaign Evidence Scope

The cross-window block contains 54 base geometries and 594 cases per event window, using the intended 32 California or 773 India tasks. The same campaign includes 282 verified India cases at 180 satellites. Aggregate 180-satellite analysis is restricted to the balanced block of 216 cases covering all 27 base architectures at CN30–CN100. This supports a matched India 120-to-180 comparison on those cells, while California–India conclusions remain restricted to 60/120 satellites.

## 7.2 Final Evidence Structure and Integrity

The final post-processing treats the 60/120-satellite regional cases and the verified India 180-satellite cases as parts of one final campaign. Table 7.1 records the comparison blocks used for each claim. The matched regional block supports California–India conclusions, while the balanced CN30–CN100 portion of the 180-satellite extension supports the paired India 120-to-180 comparison under the common 773-task contract. Both blocks contribute to the final architecture portfolio.

**Table 7.1** Final analysis populations and scientific use.

| Population | Bases | Cases | Scenarios/case | Rows | Scientific use |
|---|---|---|---|---|---|
| California 60/120 block | 54 | 594 | 539 | 320,166 | Matched event-window effects and portfolio analysis |
| India 60/120 block | 54 | 594 | 539 | 320,166 | Matched event-window effects and portfolio analysis |
| India 180, all verified | 27 | 282 | 539 | 151,998 | Complete selected-case inventory on 773 tasks |
| India 180, balanced | 27 | 216 | 539 | 116,424 | CN30–CN100 architecture, matched fleet, robustness, timing, and portfolio extension |

Every verified 180-satellite case has a success marker, 539 unique scenario identifiers, no failed retained row, and a consistent metric task count of 773. The supplied verifier reports 282 complete cases, zero missing or invalid cases, and matching task, case-allow-list, and run-contract hashes. The population therefore passes both the computational and scientific input gates.

### 7.2.1 Task-population consistency and balanced block

The intended India input has 773 tasks. The earlier preparation kept row labels 32–804 after filtering India from the combined table, but normalization created a table labelled 0–772. Pandas matched the assigned series by label rather than by row order. This left 32 invalid rows at the start and removed the final 32 Low-priority tasks. In the verified package, the index is reset before normalization and output is accepted only if the task-file, normalized-task, allow-list, and run-contract hashes agree.

The earlier 741-task values are excluded; every final 180-satellite number comes from the verified 773-task execution.

The 180-satellite selection does not contain all 27 base architectures at CN0–CN20. It contains 20 cases at CN0, 22 at CN10, and 24 at CN20. All 27 bases are present at each fraction from CN30 through CN100, creating the balanced 216-case block used for aggregate inference. Table 7.2 records the exact availability by Central Node fraction. The lower-CN results remain in the canonical table, but they are not used for rank or marginal-mean comparisons that require equal base coverage.

**Table 7.2** India 180-satellite case availability by Central Node fraction in the final verified campaign.

| CN (%) | 0 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Cases | 20 | 22 | 24 | 27 | 27 | 27 | 27 | 27 | 27 | 27 | 27 |
| Missing from 27-base block | 7 | 5 | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

## 7.3 Matched 60/120-Satellite Results

### 7.3.1 Matched event-window contrast

The confirmatory population contains 594 matched architectures per event window. Table 7.3 reports India minus California for identical case identifiers. The 95% intervals resample the 54 base architectures, preserving the linked central-node fractions within each base. They are stability intervals over the enumerated design grid, not population confidence intervals for either geographic region.

**Table 7.3** Matched India-minus-California contrast on the confirmatory 60/120-satellite population.

| Metric | Cases | Mean change | 95% stability interval |
|---|---|---|---|
| Observation fulfilment | 594 | $+4.66$ pp | $[+1.58, +7.78]$ pp |
| Strict useful completion | 594 | $-2.98$ pp | $[-5.39, -0.33]$ pp |
| Mean useful coverage | 594 | $-5.95$ pp | $[-7.64, -4.41]$ pp |
| P100 $T_{100}$ | 541 | $+175.07$ min | $[+166.94, +182.97]$ min |
| Transferred data | 594 | $+27,957$ Mbit | $[+25,193, +30,779]$ Mbit |
| Robustness retention AUC | 541 | $-0.04$ | $[-0.05, -0.04]$ |

India therefore achieves slightly higher follow-up observation fulfilment but weaker and later complete dissemination. The result is not contradictory: observation requires that at least one informed spacecraft later reaches the target, whereas $S_{100}^{\text{useful}}$ requires all declared useful recipients to receive the task before the deadline. Architecture rank agreement is 0.73 for observation, 0.77 for strict completion, 0.72 for coverage, 0.78 for $T_{100}$, and 0.61 for robustness. A design chosen on one event-window workload does not retain a guaranteed rank on the other.

### 7.3.2 Central Node fraction and forwarding policy

Figure 7.1 shows that the central-node response depends on the forwarding policy. Under the bounded nominal policy, mean observation peaks near an intermediate fraction and falls sharply at high fractions. Under the unlimited useful-deadline diagnostic, dissemination continues to improve toward CN100 while communication demand rises.

For the bounded policy, the CN90-to-CN100 observation change is $-21.93$ percentage points in California and $-21.43$ in India. The decline establishes a package–policy interaction but not a unique mechanism: eligibility, forwarding preference, deterministic placement, and the link-budget multiplier

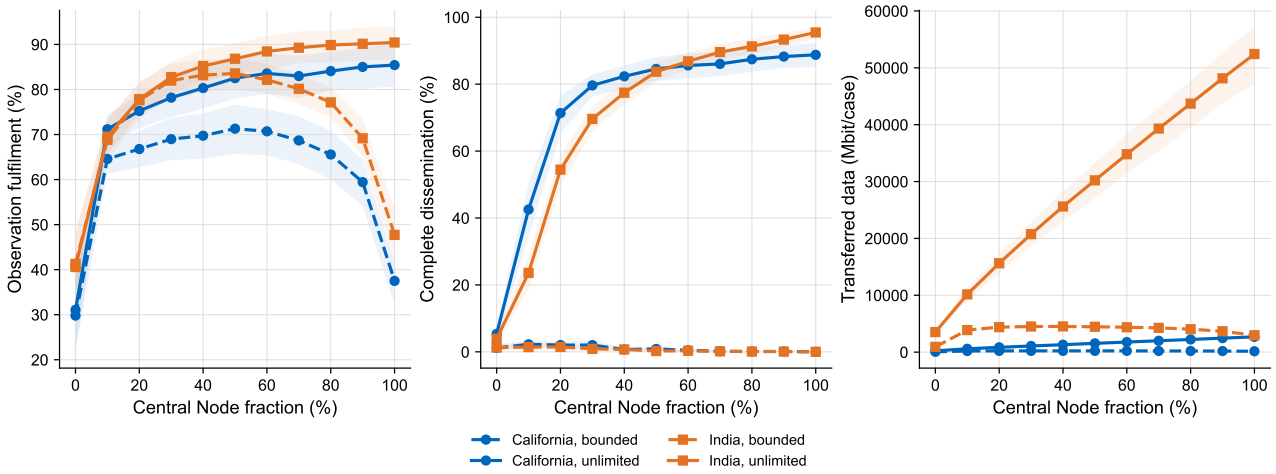



**Figure 7.1** Central Node response in the final matched regional block. Solid lines show unlimited routing; dashed lines show bounded routing. Bands are base-clustered stability intervals.

change together. Under peer routing relative to central-only routing, observation changes by $+5.53$ points in California and $+4.70$ in India, while strict completion changes by $+11.68$ and $+13.58$ points. The corresponding traffic increases are approximately 1,061 and 19,347 Mbit per case.

### 7.3.3 Packet-size sensitivity, timing, and robustness

Packet growth has a much larger effect in the dense India workload. Doubling packet size reduces strict completion by 3.18 points in California and 15.36 in India. Quadrupling packet size reduces it by 14.94 and 37.02 points. These are paired within-case contrasts and therefore isolate scenario changes within the declared architecture and task population.

The censor-aware analysis keeps missed events as right-censored observations and reports results by the fixed priority horizons of 30, 60, 180, and 360 minutes. At CN40 under the unlimited diagnostic, equal weighting of the shared High, Medium, and Low classes gives observation attainment/speed of 0.81/0.50 for California and 0.77/0.43 for India. For strict completion, the corresponding values are 0.82/0.51 and 0.71/0.32. Stratification prevents the much larger Low-priority population from silently defining the event-window comparison.

At 30% severity, outage of the sole Munich ground station causes the largest individual mean strict-completion loss: $-20.35$ points in California and $-17.50$ in India. This is operationally important because removing a ground-contact interval blocks injection of new tasks for all spacecraft, whereas a random satellite or link loss removes only part of the available temporal graph. Figure 7.2 shows this separation: the ground-station curve declines most strongly, while the node, link, and capacity disturbances retain a larger share of their matched baselines over the evaluated range. The result identifies a single-point vulnerability in the tested network, but it does not quantify the benefit, location, or cost of an additional station. Fourfold packets combined with 20% satellite failure reduce strict completion by 17.86 points in California and 42.01 in India. The India loss is more than a minor perturbation: the denser workload consumes the remaining transfer opportunities after nodes are removed, so isolated failure tests understate the risk of simultaneous load growth and component loss.

### 7.3.4 Metric redundancy, Pareto efficiency, and main shortlist

Transferred data and transfer-event count carry almost the same ordering; both correlation coefficients round to 1.00 at the precision used in the report. Both remain in the diagnostic output, but they do not contribute two full metric-family weights. Their unrounded Pearson and Spearman values are available in the machine-readable audit table. After exact Pareto filtering, 137 of 548 eligible California

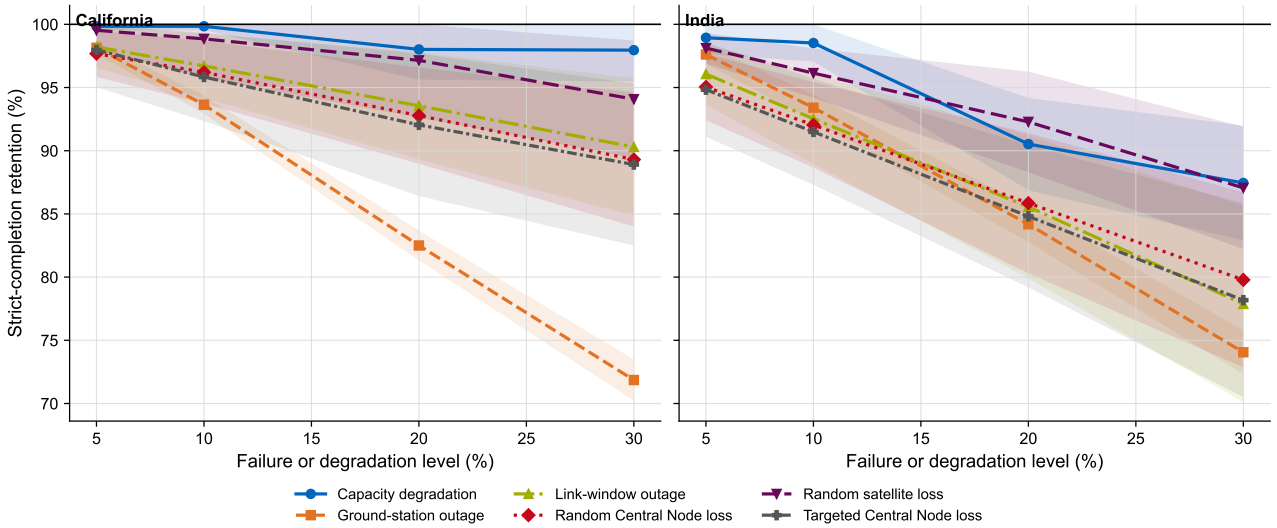



**Figure 7.2** Strict-completion retention in the final matched regional block. Lines are means of matched base-architecture means; shaded bands show $\pm 1$ standard deviation across those base means. Colour, marker, and line style identify the failure family; the black line is the matched no-failure baseline.

cases and 140 of 561 eligible India cases remain. The 10,000 reproducible Dirichlet draws show how rankings react to different weights. They do not represent 10,000 stakeholder opinions.

**Table 7.4** Confirmatory five-case cross-window shortlist from the 60/120-satellite population.

| Architecture | CA Obs. | CA $S_{100}^{\text{useful}}$ | India Obs. | India $S_{100}^{\text{useful}}$ | India $T_{100}$ | India Mbit |
|---|---|---|---|---|---|---|
| T060_P05_H0800_I0800_CN050_F1 | 100.00 | 93.75 | 82.15 | 82.15 | 253.48 | 19,326 |
| T060_P10_H0800_I0986_CN040_F1 | 96.88 | 100.00 | 99.61 | 83.70 | 291.88 | 18,238 |
| T060_P10_H1000_I0986_CN090_F1 | 96.88 | 100.00 | 100.00 | 100.00 | 167.65 | 32,947 |
| T060_P10_H1000_I0986_CN100_F1 | 96.88 | 100.00 | 100.00 | 100.00 | 146.65 | 35,700 |
| T120_P10_H1000_I0986_CN100_F1 | 100.00 | 100.00 | 100.00 | 100.00 | 105.45 | 71,400 |

Table 7.4 keeps five rows as a shortlist because they make the choice visible. The CN40 design combines strong California rank stability with comparatively low India traffic. CN90 and CN100 favour India completion instead. Moving from the comparable 60-satellite CN100 case to the 120-satellite CN100 case reduces completion time, but approximately doubles the India task traffic.

## 7.4 India 180-Satellite Results and Fleet-Scale Extension

### 7.4.1 Central-node response in the balanced block

Figure 7.3 covers the balanced CN30–CN100 block. With bounded routing, mean observation decreases from 88.15% at CN30 to 43.52% at CN100. Strict completion moves from 0.25% to zero, and useful coverage from 11.20% to 2.04%. This is more than a flat, saturated response: the implemented package performs worse when nearly every spacecraft receives the Central Node role. The curve cannot identify one congestion mechanism because placement, preference, link eligibility, link budget, hop limit, and copy limit are changed together. It does establish that the role cannot be assigned to the whole fleet without changing the routing behaviour.

The unlimited diagnostic moves in the opposite direction. From CN30 to CN100, observation rises from 90.07% to 92.36%, strict completion from 88.98% to 97.16%, and useful coverage from 95.74% to 99.17%. Mean P100 $T_{100}$ falls from 257.08 to 142.03 minutes. The contacts created at high participation are therefore useful, but the bounded forwarding rules do not exploit them effectively. This

gain also requires traffic: mean transferred data increases from 45,476 to 105,760 Mbit per case. A 133% traffic increase buys 2.29 points of observation and 8.18 points of strict completion. Table 7.5 reports the selected marginal means underlying this comparison. CN100 is attractive only if that faster, broader completion is worth the added communication burden.

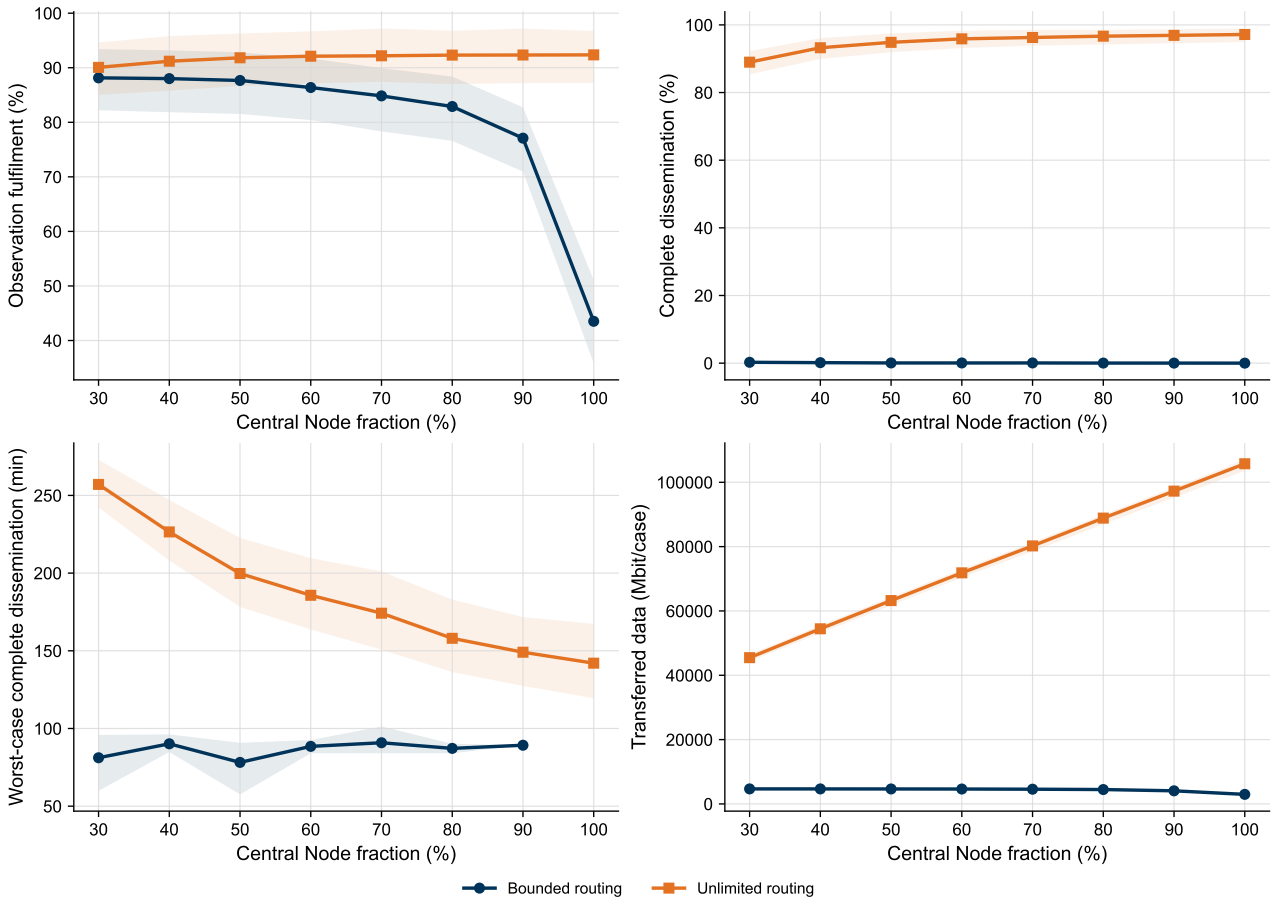



**Figure 7.3** India 180-satellite Central Node response for 773 tasks and the balanced 216-case block of the final campaign. Bounded routing is the implemented copy/hop-limited package; unlimited routing removes those forwarding limits as a diagnostic. Shading shows base-clustered stability intervals.

**Table 7.5** Selected India 180-satellite marginal means from the final campaign.

| Policy | CN (%) | Obs. (%) | $S_{100}^{\text{useful}}$ (%) | Coverage (%) | $T_{100}$ (min) | Data (Mbit) |
|---|---|---|---|---|---|---|
| Bounded | 30 | 88.15 | 0.25 | 11.20 | 81.16 | 4,671 |
| Bounded | 50 | 87.67 | 0.06 | 10.81 | 78.18 | 4,655 |
| Bounded | 100 | 43.52 | 0.00 | 2.04 | – | 2,970 |
| Unlimited | 30 | 90.07 | 88.98 | 95.74 | 257.08 | 45,476 |
| Unlimited | 50 | 91.82 | 94.84 | 98.45 | 199.74 | 63,225 |
| Unlimited | 100 | 92.36 | 97.16 | 99.17 | 142.03 | 105,760 |

### 7.4.2 Matched India 120-to-180 fleet-size effect

Because both fleet sizes use the same 773 tasks, each 180-satellite case in the balanced block can be paired with the corresponding 120-satellite case by plane count, altitude, inclination, and Central Node fraction. Table 7.6 reports 180 minus 120 satellites under the unlimited useful-deadline scenario. The 95% stability intervals resample the 27 base-geometry clusters while retaining their eight Central Node fractions.

**Table 7.6** Matched India 180-minus-120-satellite effects for 216 CN30–CN100 architectures under unlimited routing.

| Metric | T120 mean | T180 mean | Mean change with 95% stability interval |
|---|---|---|---|
| Observation fulfilment (%) | 90.57 | 91.80 | $+1.23$ pp $[+0.75, +1.82]$ |
| Strict useful completion (%) | 91.01 | 94.99 | $+3.99$ pp $[+2.74, +5.31]$ |
| Useful-recipient coverage (%) | 96.80 | 98.34 | $+1.55$ pp $[+1.07, +2.03]$ |
| P100 $T_{100}$ (min) | 231.23 | 186.55 | $-44.68$ min $[-52.09, -37.89]$ |
| Transferred data (Mbit/case) | 50,060 | 75,887 | $+25,827$ $[+25,318, +26,193]$ |

Figure 7.4 shows that the larger fleet improves the three mission measures and shortens complete dissemination across the balanced Central Node range, but the improvement requires substantially more traffic. The controlled comparison therefore supports a qualified scale benefit, not a claim that the larger fleet dominates after resource burden is included.

### 7.4.3 Policy and packet-size effects

Peer and hybrid policies are again nearly identical. Relative to central-only routing, each improves observation by 0.28 points and strict completion by approximately 0.94 points, but adds about 29,300 Mbit per case. The smaller marginal mission change at 180 satellites does not mean the policies are free: the traffic difference is substantial.

Halving packet size improves observation by 1.05 points and strict completion by 2.59 points. Doubling packet size reduces them by 6.13 and 10.49 points. Quadrupling packet size reduces observation by 22.32 points and strict completion by 33.67 points while adding about 133,628 Mbit per case. Removing the bounded forwarding and copy limits changes observation by $+11.98$ points and strict completion by $+94.92$ points, with about 71,546 Mbit additional traffic. Figure 7.5 summarizes these paired effects.

### 7.4.4 Failure robustness and combined stress

At each family's maximum evaluated severity of 30%, mean strict-completion retention is 76.26% for ground-station outage, 93.99% for targeted Central Node failure, 95.28% for random Central Node failure, 95.43% for capacity degradation, 95.47% for link-window outage, and 95.67% for random satellite failure. Figure 7.6 shows that the ground-station series separates progressively from the other failure families as severity increases; the result is therefore a pattern across the tested levels rather than a conclusion based only on the 30% endpoint. Ground access is the dominant tested single vulnerability in the sensitivity experiment as well. Targeted Central Node loss is worse than random Central Node loss, as expected when the removed nodes are selected for structural importance, but the difference is small compared with the ground-station effect. The evidence supports prioritizing a ground-network trade study before fine optimization of failure-aware Central Node placement. It does not yet support a specific number or location of additional stations.

Under combined stress, twofold packets with 10% and 20% satellite failure retain 86.44% and 83.81% of baseline strict completion. Fourfold packets reduce retention to 60.97% and 57.49%. Figure 7.7 makes the interaction visible. The retained ten combined-stress replicates support descriptive clustered means; they do not justify fine-grained low-tail ordering.

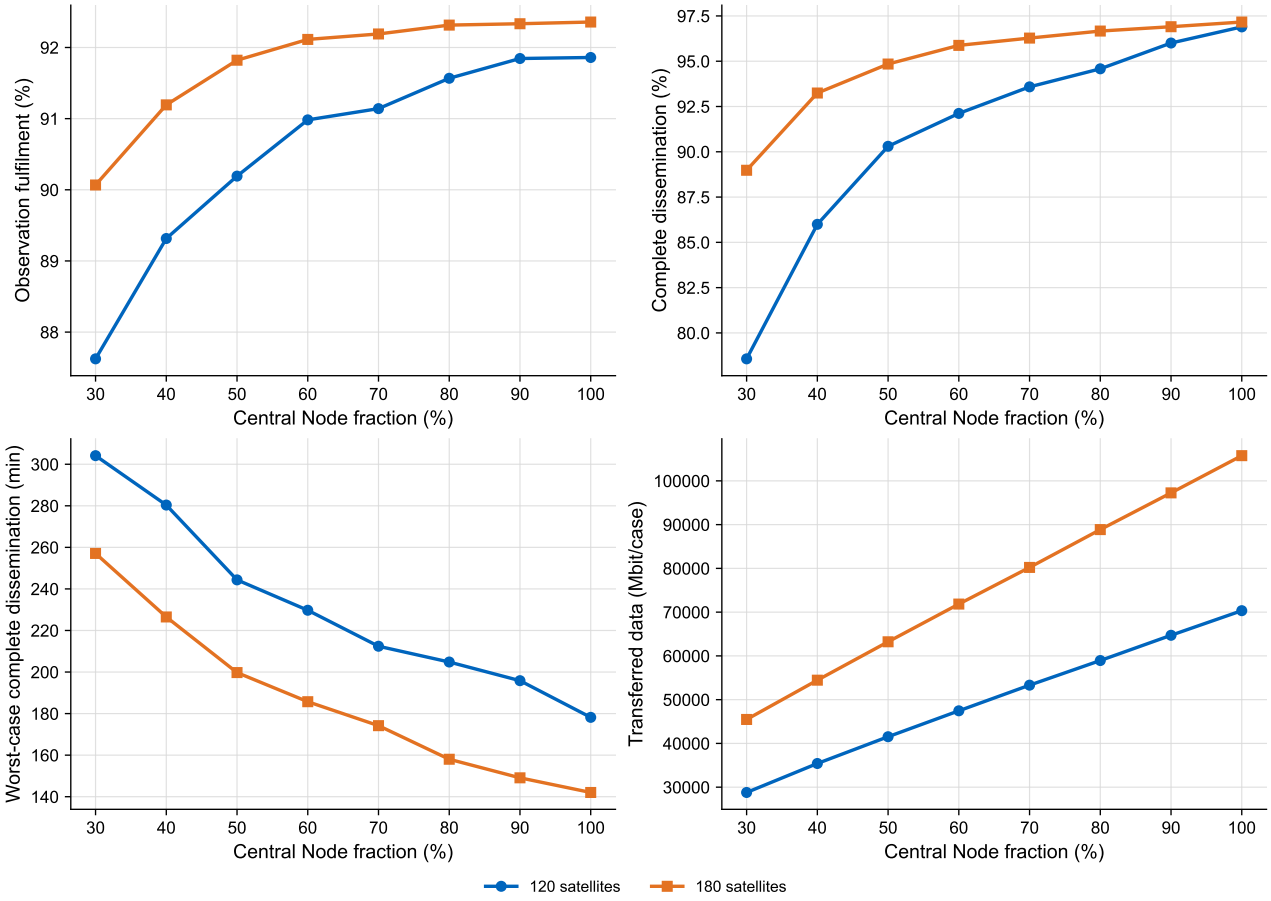



**Figure 7.4** Matched India 120- and 180-satellite response under unlimited routing for the 216 common CN30–CN100 architectures.

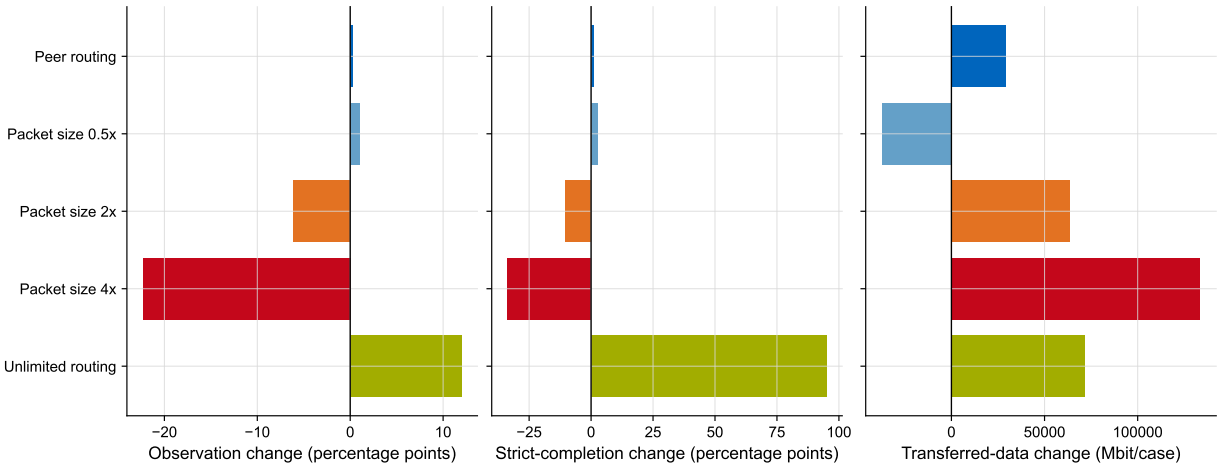



**Figure 7.5** Paired India 180-satellite effects relative to the corresponding reference scenario. Bars to the right denote an increase; bars to the left denote a decrease. Mission metrics are in percentage points and traffic is in Mbit per case.

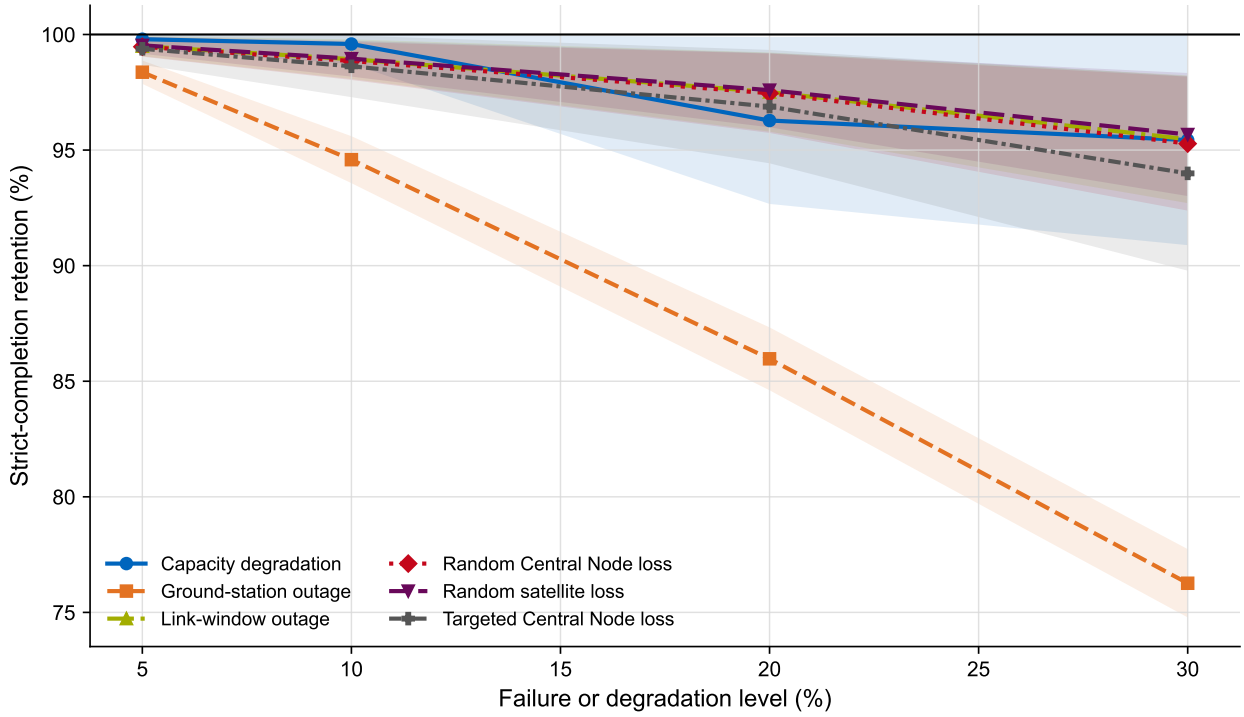



**Figure 7.6** India 180-satellite strict-completion retention across failure families. Lines are means of matched base-architecture means; shaded bands show $\pm 1$ standard deviation across those base means. Colour, marker, and line style distinguish the disturbance; the black line is the matched no-failure baseline.

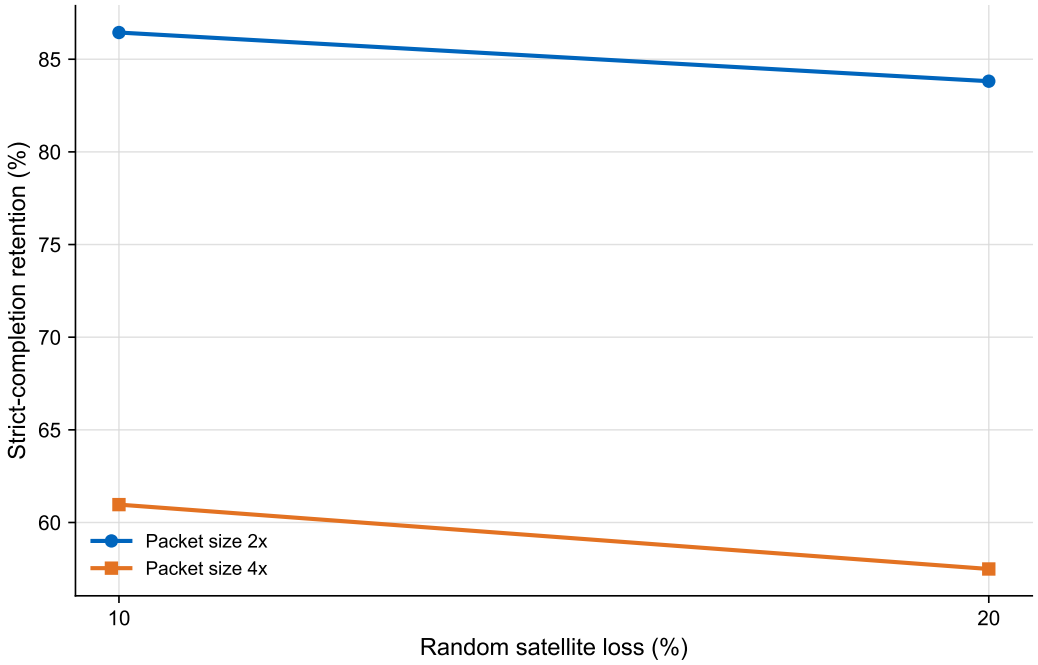



**Figure 7.7** India 180-satellite strict-completion retention under combined packet and satellite-failure stress.

### 7.4.5 Likelihood–consequence screening

Figure 7.8 combines the measured 30%-severity consequence with the declared likelihood categories defined in Section 5.8. Under those assumptions, link-window outage and ground-station outage enter the High screening class. Capacity degradation, random satellite loss, and random Central Node loss are Moderate; targeted Central Node loss is Low because the declared occurrence category is Rare even though its measured consequence is substantial. This ranking is conditional on the likelihood assumptions. The simulation supplies the consequence values but does not estimate event probabilities, so the matrix must not be interpreted as a reliability prediction.

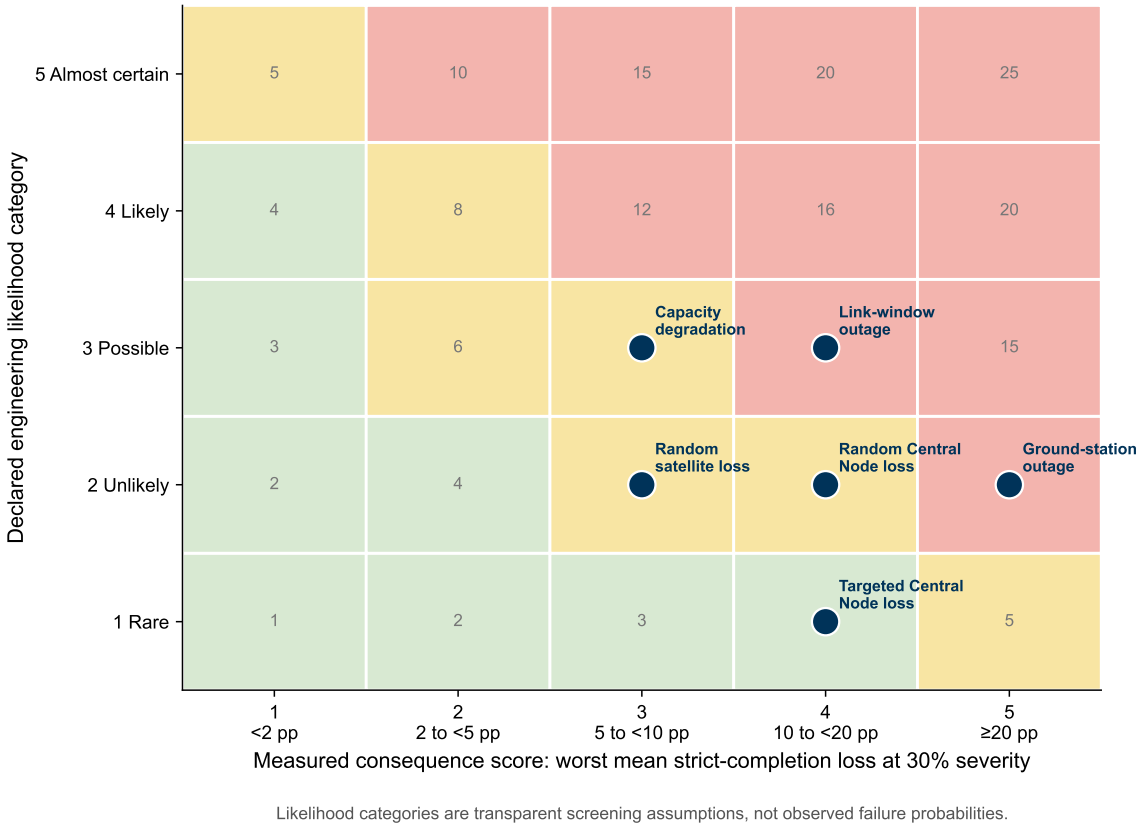



**Figure 7.8** Likelihood–consequence screening of the six individual disturbance families. The horizontal category is based on the worst mean strict-completion loss at 30% severity across the eligible final-campaign blocks. Vertical categories are declared engineering assumptions, not observed failure probabilities.

### 7.4.6 Priority-stratified timing

The priority analysis uses task deadlines as censoring limits and reports High, Medium, and Low classes separately; the India population contains no Critical task. From CN30 to CN100 under the unlimited diagnostic, High-priority strict-completion probability rises from 50.95% to 65.66% within 60 minutes. Medium-priority probability rises from 80.87% to 95.90% within 180 minutes, and Low-priority probability from 99.23% to 100% within 360 minutes. The corresponding standardized speed increases from 0.16 to 0.31 for High, 0.38 to 0.65 for Medium, and 0.57 to 0.77 for Low. Figure 7.9 displays both quantities because higher deadline attainment and faster conditional completion answer different questions: the former counts how often the deadline is met, whereas the latter distinguishes earlier from later completion among the available time horizons.

These results show improvement with higher CN participation under the unlimited diagnostic. They now cover the complete intended population of 11 High-, 395 Medium-, and 367 Low-priority India tasks.

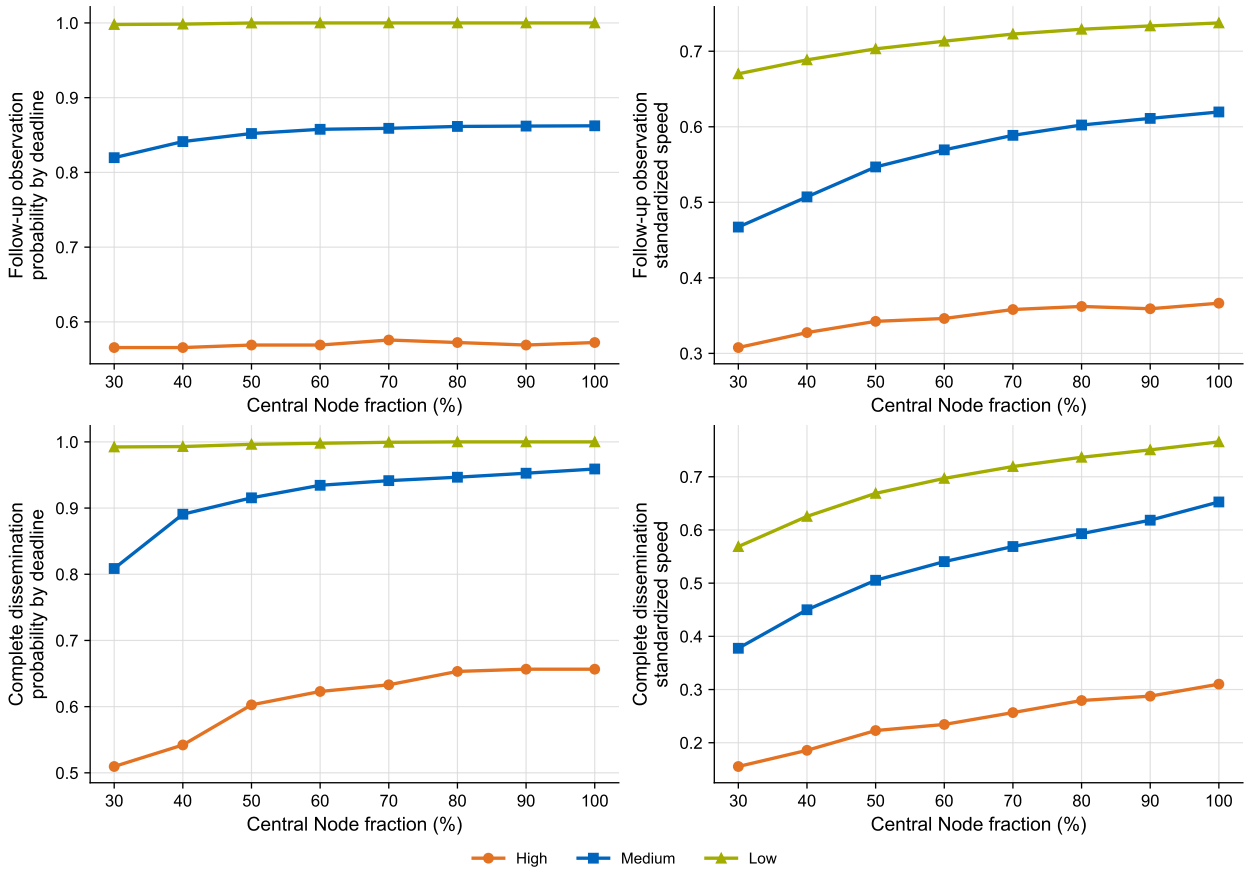



**Figure 7.9** India 180-satellite priority-stratified deadline attainment and standardized speed under unlimited routing. Each class uses its fixed deadline as the time horizon.

### 7.4.7 Pareto set and rank acceptability

The redundancy audit again identifies transferred data and transfer-event count as severe duplicates. Exact Pareto filtering leaves 61 of the 216 balanced cases. Figure 7.10 shows why a single two-variable ranking is insufficient: architectures with similar mission performance can require markedly different traffic, and some points outlined as Pareto efficient in the full objective set do not appear optimal in either two-dimensional projection alone. The large frontier and the distribution of winner frequencies show that no 180-satellite architecture is preference-free.

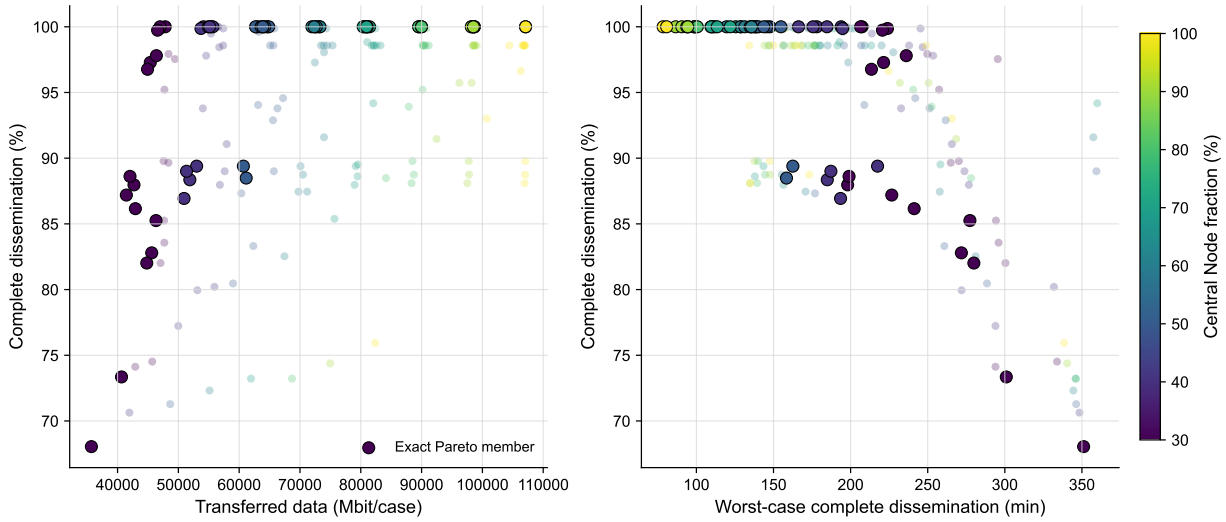



**Figure 7.10** India 180-satellite trade space shown in two two-variable projections. Colour denotes Central Node fraction and black outlines denote exact Pareto members computed from the full objective vector.

Figure 7.11 answers a different question from Figure 7.10: each orange set is recomputed using only the two displayed objectives. Of the 216 eligible architectures, 10 are non-dominated for completion versus traffic, 6 for observation versus traffic, 1 for completion versus time, and 6 for observation versus Central Node count. The one-member completion–time front occurs because one architecture is at least as good in both displayed objectives as every other candidate. These pairwise fronts help explain individual trades, but the 61-member full Pareto set remains the correct feasible set for the six-objective decision problem.

**Table 7.7** Highest India 180-satellite rank-1 acceptability cases across 10,000 random family-weight vectors.

| Architecture | CN (%) | Obs. (%) | $S_{100}^{\text{useful}}$ (%) | $T_{100}$ (min) | Data (Mbit) | Rank-1 / top-5 (%) |
|---|---|---|---|---|---|---|
| T180_P10_H0500_I0986_CN050_F1 | 50 | 100 | 100 | 138.39 | 64,862 | 26.55 / 46.38 |
| T180_P10_H0800_I0986_CN040_F1 | 40 | 100 | 100 | 166.22 | 55,659 | 19.70 / 29.28 |
| T180_P10_H0500_I0800_CN090_F1 | 90 | 100 | 100 | 86.56 | 98,692 | 14.31 / 37.73 |
| T180_P05_H0800_I0800_CN100_F1 | 100 | 100 | 100 | 78.43 | 107,100 | 9.62 / 23.57 |
| T180_P10_H0800_I0800_CN100_F1 | 100 | 100 | 100 | 80.69 | 107,100 | 8.85 / 33.34 |

Table 7.7 shows that the top two cases use CN50 and CN40 and trade slower dissemination for lower traffic. The faster CN90/CN100 alternatives require approximately 98,700–107,100 Mbit per case. These candidates extend the final architecture portfolio for an India workload; their scope is marked explicitly because a matched California 180-satellite population was not executed.

## 7.5 Integrated Result Synthesis

The regional and 180-satellite blocks of the final campaign support a consistent system-level interpretation. Central Nodes create useful temporal paths, but their value depends on forwarding constraints

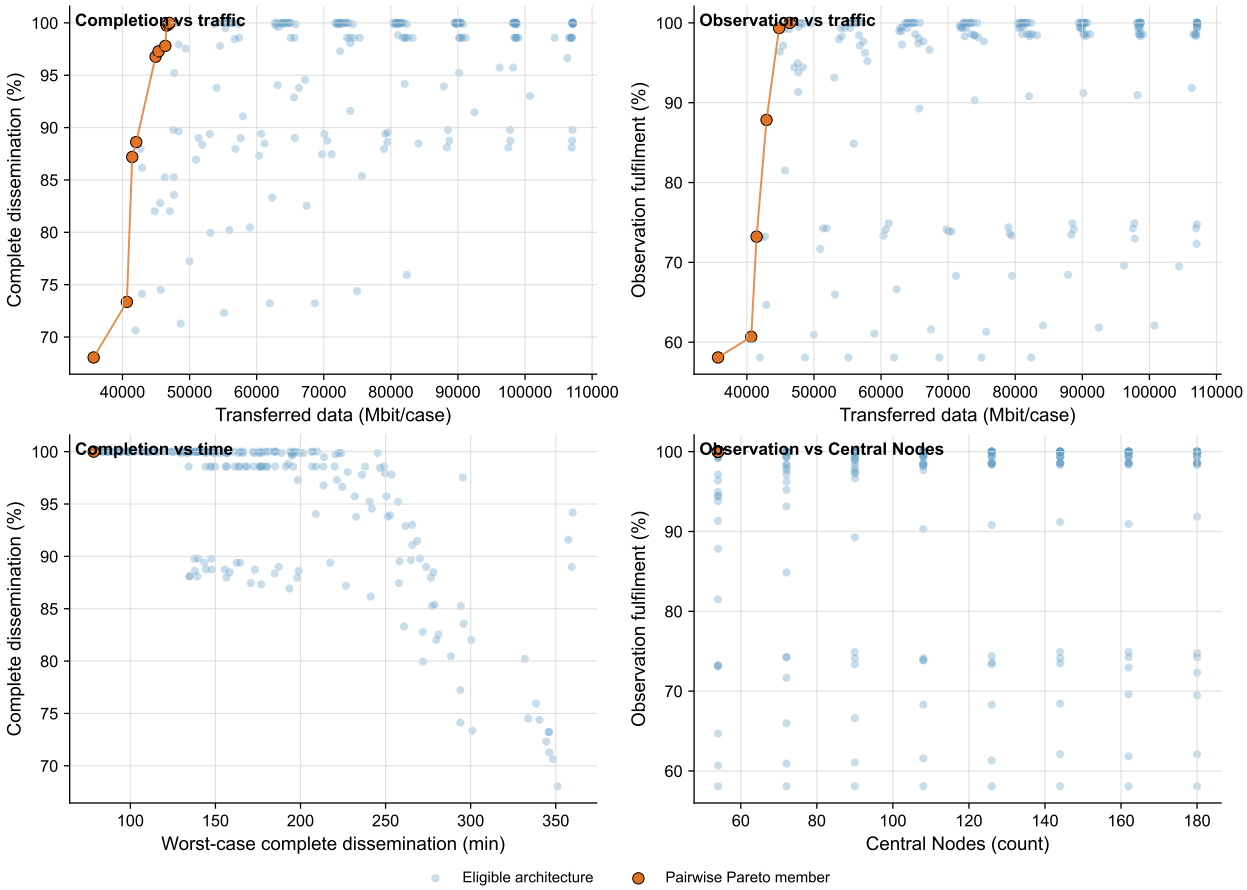



**Figure 7.11** Recomputed two-objective Pareto fronts for the balanced India 180-satellite block. Pale points are all eligible architectures; orange points and connecting lines are the non-dominated set for the two objectives shown in that panel.

and workload. Mission gains can saturate while traffic continues to grow. Packet size is a strong design driver, especially for the India load case. Ground access is the largest tested single vulnerability across the campaign, while combined load and failure produce losses that isolated disturbances do not reveal. Exact Pareto and rank-acceptability results support portfolios rather than universal winners.

The final campaign identifies a controlled India 120-to-180 effect for the 216 matched CN30–CN100 cells: the larger fleet improves mission completion and timing, but requires about 25.8 Gbit more task-message traffic per case. Low-CN 180-satellite coverage is incomplete, and no California 180-satellite grid exists. The result is therefore a qualified India scale comparison, not a universal three-fleet or cross-region law.

# 8 Discussion

## 8.1 Interpretation Strategy

The final interpretation treats the verified simulation evidence as one campaign while selecting valid comparison blocks for each claim. The matched 60/120-satellite block supports California–India event-window statements. The India 180-satellite extension uses the same 773-task contract, so its balanced CN30–CN100 cases can be paired with the corresponding 120-satellite block. This creates a controlled India scale comparison without implying that an unexecuted California 180-satellite population exists.

The interpretation is therefore conditional but complete. The available evidence is sufficient to answer the research questions within the declared scope. The report identifies which statements are supported by matched regional evidence, which apply to the India large-fleet extension, and which require a new experiment.

## 8.2 Method Contribution

The communication framework was successfully adapted to a multi-task wildfire use case. Remote-sensing detections were converted into requests carrying location, release time, priority, deadline, and packet size. The simulator enforced the sequence release, reception, and post-reception access before awarding a follow-up observation. This prevents geometric opportunity from being counted when the relevant spacecraft learns about a task too late.

The contribution is a communication-constrained follow-up and dissemination workflow, not a complete wildfire-response chain or a satellite scheduling optimizer. The model begins after creation of a thermal-event task and ends at a modeled follow-up opportunity. Instrument pointing, cloud and smoke screening, image production, downlink of imagery, alert distribution, and human response remain outside the evidence. This boundary aligns the study identity with the implemented mechanism.

## 8.3 Workload Context

The matched regional block shows that the event-window workloads change both values and architecture order. India has higher observation fulfilment but weaker complete dissemination, lower useful coverage, substantially later P100 $T_{100}$, and lower robustness retention. Observation can remain high when one informed satellite later reaches the target even though full dissemination to all useful recipients is incomplete. The larger India task population also increases absolute traffic and contention.

Priority-stratified results show why pooled latency alone would be misleading. Deadlines and packet sizes are functions of priority, so the censoring process is not independent of task class. Direct deadline attainment, class-specific Kaplan–Meier summaries, and RMST with fixed class horizons therefore form the defensible timing result. Equal-weight class summaries are used when a cross-window aggregate is required.

Here, “California” and “India” are shorthand for two selected event windows. They are not random samples from either region. Matching keeps the spacecraft architecture fixed, but it cannot control season, event density, spatial pattern, task composition, or priority mix. The difference is therefore interpreted as a contrast between two workload cases, not as a population-level California–India effect.

## 8.4 RQ1: Central Node Fraction and Forwarding Policy

RQ1 does not have one Central Node curve; it has different curves for different forwarding rules. In the matched regional block, bounded observation improves at low and intermediate fractions and then falls. The 180-satellite India block shows the same interaction more clearly: bounded observation drops from 88.15% at CN30 to 43.52% at CN100, whereas the unlimited diagnostic improves in observation, strict completion, coverage, and timing toward CN100. High participation is useful when forwarding is permissive, but can be harmful when the copy and hop limits act on the full package.

The experiment changes four things together when the Central Node label is applied: link eligibility, forwarding preference, deterministic placement, and link budget. The high-CN decline can therefore be attributed only to the implemented package. Congestion or routing concentration may explain the pattern, but the present outputs cannot distinguish that explanation from the other package components. Component ablations and detailed forwarding traces would be required before assigning a cause to “centralization” itself.

At 180 satellites, peer and hybrid routing each improve strict completion by about 0.94 percentage points relative to Central Node-only routing. They also add roughly 29,300 Mbit per case. The improvement is measurable, but small compared with the added communication volume in this experiment.

## 8.5 RQ2: Sensitivity to Workload, Packet Size, Geometry, and Failures

The sensitivity results change both the metric values and the order of architectures. On the matched 60/120-satellite grid, the India workload has higher observation fulfilment than California, but complete dissemination is weaker and later. Packet size produces one of the largest changes. Fourfold packets lower strict completion by 37.02 points in the matched India block and by 33.67 points in the 180-satellite extension. In that extension, they also add about 133,628 Mbit per case.

No constellation factor has one monotonic effect that can be separated from the rest of the grid. Satellite count, plane count, altitude, and inclination all change the temporal contact opportunities, and their effects differ with Central Node fraction and event window. The marginal plots average over those other factors. They show how the tested grid behaves, not an isolated causal coefficient. Geometry clearly changes the feasible routes and the ranking, but the preferred level still depends on the remaining design choices.

Failures also alter the ranking. Across the final verified campaign, ground-station outage is the most damaging single family tested. This identifies the one-station Munich baseline as an important vulnerability, but it does not determine how many replacement stations are needed or where they should be located.

Combined load and failure are also important: the worst 180-satellite cell retains only 57.49% of its baseline strict completion. RQ2 is therefore answered for the workloads, packet multipliers, geometries, and disturbances that were executed. Observation radius, alternative placement, task construction, and package ablations remain outside the evidence.

## 8.6 RQ3: Architecture Trade-Offs and Decision Stability

No design dominates all objectives. The matched regional block yields large event-window Pareto sets and a five-case cross-window core. The 180-satellite extension contains 61 exact Pareto cases among 216 balanced candidates, and its highest rank-1 acceptability is only 26.55%. These results show that the preferred design depends on how mission completion, observation, timing, fleet burden, Central Node count, traffic, and robustness are valued.

The pairwise fronts make that dependence concrete. In the balanced 180-satellite block, completion versus traffic retains 10 non-dominated cases and observation versus traffic retains 6, while comple-

tion versus complete-dissemination time retains only 1. These smaller fronts are useful for explaining a single trade, but none replaces the 61-case frontier computed from the full objective vector. A point can be dominated in one two-objective view and still remain efficient when the other mission and resource objectives are restored.

The final architecture product is a tiered portfolio. Its five-case core is supported in both event windows, while the 180-satellite candidates extend the portfolio for the India workload. Within that large-fleet tier, CN40/CN50 can offer lower traffic, whereas CN90/CN100 provides faster complete dissemination at substantially higher communication burden. The tiers are reported together, with the evidence scope stated for each candidate.

The preferred Central Node fraction moves when the forwarding policy, task size, failure family, event window, or objective weights change. Random family-weight draws produce different winners across the campaign. This is interpreted as sensitivity of the decision, not as stakeholder agreement. The reference coefficients are engineering choices made for the analysis. The 700.00 km access radius, priority rules, deadlines, packet sizes, and clustering settings are likewise fixed assumptions.

## 8.7 Importance, Non-Significance, and Engineering Meaning

Three findings are large enough to change an architecture choice.

First, routing bounds reverse the high-CN trend: the bounded package becomes worse while the unlimited diagnostic becomes better. Second, larger packets cause double-digit losses in strict completion and large traffic increases. Message size is therefore a design variable, not a software detail. Third, ground-station outage causes the largest individual robustness loss tested across the final verified campaign. For the implemented baseline, the single station is a more immediate weakness than randomly removing a comparable share of spacecraft.

The likelihood–consequence matrix adds prioritization without disguising its assumptions. Ground-station and link-window outage are High screening items under the declared likelihood categories; the remaining families are Moderate or Low. The consequence axis is measured, whereas the likelihood axis is not. A different reliability assessment can therefore move a disturbance vertically and change its class. The matrix supports the order of follow-up engineering work, but the measured loss and its across-architecture standard deviation remain the primary simulation evidence.

Several smaller differences need restraint. Peer and hybrid routing are numerically almost identical at 180 satellites, so the two labels are not treated as evidence of separate mechanisms. The roughly one-point peer-routing gain in strict completion comes with another 29,300 Mbit per case.

Altitude has a weaker pooled association than inclination, plane count, or Central Node fraction. This does not make altitude irrelevant, because interactions and the tested 500–1000 km range can hide a conditional effect. Similarly, the near-perfect correlation between traffic and event count is an accounting redundancy, not a new physical result.

The reported intervals have a deliberately limited interpretation. They describe the stability of a mean across the enumerated base architectures or seeded disturbance trials. They are not confidence intervals for all fires, all constellations, or either geographic region. Changes are therefore judged against mission thresholds and resource burden rather than using the interval as a test of a geographic-population null hypothesis.

## 8.8 Interpretation of the 180-Satellite Campaign Extension

The India 180-satellite extension has 282 verified cases. Each uses all 773 intended tasks and contains the expected 539 scenario identifiers. The task, normalized-task, case-allow-list, and run-contract hashes agree with the verifier. Aggregate curves and rankings use only CN30–CN100, where all 27 base architectures are present. This avoids giving unequal weight to the partially completed low-CN levels.

The balanced block gives a direct India 120-to-180 comparison for 216 architectures under unlimited routing. The 180-satellite fleet adds 1.23 points of observation, 3.99 points of strict completion, and 1.55 points of useful coverage. P100 $T_{100}$ is 44.68 minutes shorter, but traffic is 25,827 Mbit per case higher. This is a performance–traffic trade, not unconditional dominance by the larger fleet. It also says nothing about a California–India comparison at 180 satellites, and CN0–CN20 is excluded from balanced aggregate claims.

The task inconsistency exposed a gap in the original restart checks. Scenario identifiers alone were not enough: the run also had to be tied to input hashes, regional and priority counts, the case allow-list, and a reset positional index. A success marker proves that a programmed job ended, but not that old and new partitions used identical scientific inputs. The verified package performs these checks and is the only source used for the final 180-satellite results.

## 8.9 Expected-Relationship and Objective Assessment

Table 8.1 evaluates the three literature-motivated relationships against the verified results and makes the conditional nature of the Central Node finding explicit.

**Table 8.1** Assessment of the three literature-motivated relationships stated in the introduction.

| Relationship | Assessment | Evidence |
|---|---|---|
| E1 | Supported conditionally | Central Nodes create useful temporal paths, but gains saturate or reverse under the bounded policy; the unlimited diagnostic continues to improve at increasing traffic cost. |
| E2 | Supported | Faster and broader dissemination generally requires more transferred data and, at high Central Node fractions, more differentiated nodes. |
| E3 | Supported | Matched event-window effects, packet sensitivity, robustness, Pareto membership, and architecture ranks differ with workload and preference assumptions. |

Objective 1 is met by the traceable task pipeline. Objective 2 is met by the temporal simulation and reception-before-observation logic. Objective 3 is met by the separate mission, burden, and robustness comparisons. Objective 4 is met by exact Pareto filtering and preference-sensitive shortlists. No objective is claimed for radius, placement, task-model, component-ablation, multi-station, or replicate-extension questions that were not simulated.

## 8.10 Threats to Validity

### 8.10.1 Construct validity

Observation is a geometric 700.00 km access proxy rather than a sensor-quality or imaging-feasibility model. Packet sizes describe task messages, not imagery. The Central Node role is a bundled communication treatment rather than a separately costed spacecraft class. P100 $T_{100}$ measures complete useful-recipient dissemination, not first awareness or end-to-end alert latency.

### 8.10.2 Internal validity

The factorial grid controls the declared Walker factors but not alternative task rules, node placements, or package components. High-CN aggregate patterns are associations under one implementation. The verified India population includes all intended 773 tasks, but the 180-satellite case selection remains incomplete at CN0–CN20.

### 8.10.3 External validity

External validity is limited to what was run. The evidence covers the listed constellations, routing rules, deadlines, link assumptions, one ground station, and disturbance families in two event windows. The 180-satellite findings apply to the 773-task India window. A larger fleet, a California 180-satellite layer, or another event window would be a new experiment rather than an extension of these numbers.

### 8.10.4 Statistical conclusion validity

The base-clustered intervals refer to the enumerated architecture blocks, not to a geographic population. Thirty failure replicates and ten combined-stress replicates are sufficient for the mean-response analysis used here, but not for a fine low-tail ranking without convergence checks. Correlated metrics are screened before weighting, and exact Pareto membership is calculated before applying a compensatory score.

## 8.11 Decision Implication

The engineering output is one tiered portfolio. CN40/CN50 provides a lower-resource balance. CN90/CN100 gives stronger India completion but uses more traffic. The 120-satellite CN100 case is defensible only if its faster complete dissemination justifies the larger fleet and communication burden. The 180-satellite comparison adds the next scale step: better completion and timing than the matched 120-satellite cases at a cost of about 25.8 Gbit more task traffic per case. Its candidates are included in the same portfolio with an India-workload scope label.

Before an operational commitment, the mission owner should provide preference weights and feasibility thresholds, confirm the task-message product, and evaluate at least observation radius, placement, package components, and alternative ground networks on a small representative set. Those steps refine the decision; they are not evidence already contained in this thesis.

## 8.12 Integrated Evidence-to-Decision Synthesis

The contribution of this thesis is clearest when the complete evidence chain is considered rather than any single performance curve. Existing work supplies the component ideas: distributed and federated satellite systems share resources and information [7, 11, 20]; decentralized allocation assigns observation work across spacecraft [16, 29]; scheduled space networking carries data through intermittent contacts [1, 3, 4]; and active-fire products provide time-tagged thermal detections [19, 25–27]. The step taken here is to connect those elements in one traceable experiment and ask a mission-level question: did a released task reach a suitable spacecraft before a later useful access?

That connection produces three findings that are not visible from geometric coverage alone. First, observation fulfilment and complete dissemination can move in opposite directions. The selected India event window has higher observation fulfilment than the selected California window, but lower strict dissemination completion and receiver coverage. Second, adding spacecraft improves all three percentage-based mission measures in the matched India 120-to-180 comparison, yet it also raises task traffic by about 25.8 Gbit per case. Third, increasing the Central Node fraction does not have a universal effect: bounded forwarding reverses at high fractions while the unlimited diagnostic continues to improve. The architecture question is therefore a coupled timing, routing, and resource problem rather than a coverage-maximization problem.

Figure 8.1 places the two principal matched comparisons on one common percentage-point scale. The left panel shows why an event window cannot be described as simply “better” or “worse”. The right panel shows a consistent mission benefit from the larger India fleet, but the communication-cost result must still be considered separately. The whiskers quantify stability across the enumerated base-architecture clusters; they are not probability statements about either geographic region.

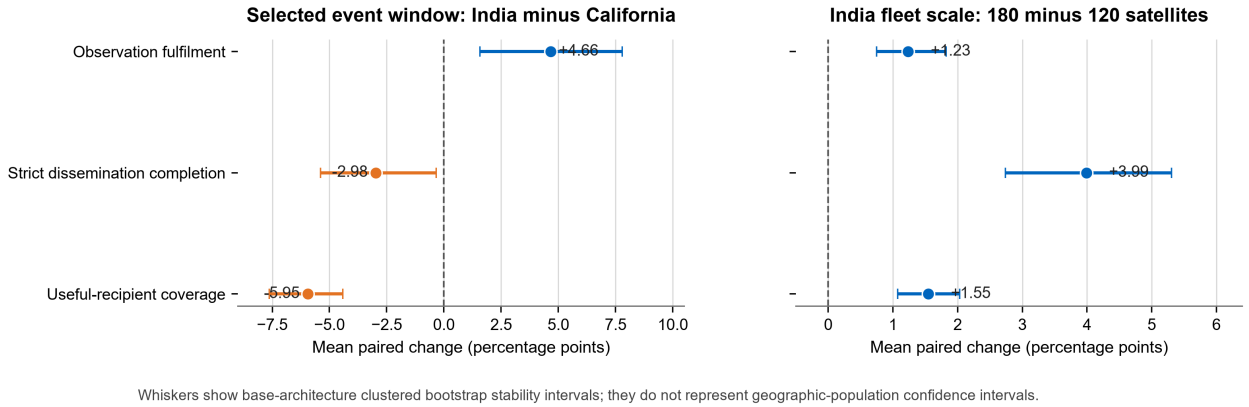



**Figure 8.1** Integrated matched effects for the three percentage-based mission metrics. The event-window panel uses 594 matched 60/120-satellite cases. The fleet-scale panel uses 216 matched India cases across 27 base architectures. Whiskers are base-architecture clustered bootstrap stability intervals.

The resulting contribution has three layers. Methodologically, the workflow preserves release, reception, dissemination, and observation as separate events. Empirically, it identifies conditional responses to workload, routing bounds, packet size, failures, and fleet scale. For decision support, it retains non-dominated portfolios and exposes when a recommendation changes with mission preferences. This is an incremental architecture-study contribution, not a claim that Central Nodes are universally superior or that one constellation is operationally optimal.

### 8.12.1 Validity and Uncertainty Budget

The numerical results are deterministic outputs from declared simulation inputs, but the engineering interpretation still contains several forms of uncertainty. Treating all of them as sampling error would be misleading. Table 8.2 separates represented variability from assumptions that were held fixed. This distinction also explains why the report uses paired differences, base-clustered bootstrap intervals, robustness replicates, and preference sensitivity for different purposes.

**Table 8.2** Uncertainty budget and its consequence for interpretation.

| Source | Representation in this thesis | Consequence for the claim |
|---|---|---|
| Architecture variation | Enumerated Walker factors and Central Node fractions; pairing within common base identifiers | Supports comparison inside the tested grid, not inference to every possible constellation. |
| Event-window variation | Two frozen workloads with matched 60/120-satellite architectures | Establishes a workload-case contrast; it does not estimate a California or India population effect. |
| Failure realization | Thirty seeded trials for single failures and ten for combined stress | Supports mean robustness and across-trial stability; fine tail probabilities and rare-event reliability are not estimated. |
| Base-architecture heterogeneity | Clustered bootstrap over base architectures [5] | Shows whether a mean effect is stable across the enumerated designs; it is not a geographic confidence interval. |
| Task construction | Fixed clustering, priority, deadline, packet-size, and 700.00 km access rules | Results are conditional on the implemented task product and observation proxy. |
| Central Node implementation | Eligibility, preference, placement, and link-budget changes applied as one package | Package effects are identifiable, but component-specific causal effects are not. |
| Decision preference | Declared weight profiles and 10,000 reproducible family-weight draws | Measures preference sensitivity; it is not a survey of stakeholder beliefs. |

The strongest internal evidence comes from comparisons that preserve the same task contract and match the same architecture identifiers. That is why the 594 regional pairs and the 216 India fleet-scale pairs carry more inferential weight than pooled marginal means. The bootstrap intervals answer a narrow question: whether the sign and approximate size of a mean paired effect remain stable when base architectures are reweighted. They do not license a null-hypothesis statement about all fires, and no such statement is required for the engineering decision made here.

Robustness trials answer a different question. Repeated failure seeds quantify sensitivity to the declared disturbance model. They do not convert an assumed failure severity into an empirical component-failure probability. Similarly, the risk matrix combines measured consequence with declared likelihood categories and must remain a screening tool, consistent with the model-validity and evidence-traceability principles used in the campaign [12–14].

The remaining uncertainty is therefore visible rather than hidden. Internal consistency is supported by task hashes, case allow-lists, recomputed metrics, matched identifiers, and frozen analysis code. External validity remains bounded by the observation proxy, two event windows, one ground station, one Central Node package, and the tested constellation levels. This division supports the numerical conclusions while preventing them from being presented as universal laws.

### 8.12.2 Mechanism Interpretation and Competing Explanations

The most striking response is the high-fraction decline under bounded forwarding. The observation is secure: CN90-to-CN100 reduces observation fulfilment by about 22 percentage points in both matched event windows, and the 180-satellite bounded curve declines from 88.15% at CN30 to 43.52% at CN100. The mechanism is not uniquely identified because the Central Node label changes several model components together. Table 8.3 therefore separates explanations that are consistent with the outputs from explanations that have actually been demonstrated.

**Table 8.3** Competing explanations for the high-Central-Node response.

| Candidate explanation | Evidence consistent with it | Evidence required to distinguish it |
|---|---|---|
| Copy or hop limits become binding | Bounded performance reverses while the unlimited diagnostic improves toward CN100 | Per-task copy exhaustion, hop-depth, and drop-reason traces across the fraction sweep. |
| Forwarding concentration creates contention | Higher fractions change preferred paths and task traffic; large packets cause major completion losses | Queue occupancy, waiting-time, attempted-transfer, and contact-capacity utilization traces. |
| Deterministic placement changes temporal reachability | Central Nodes are distributed across planes and the selected identities change with the fraction | Randomized or optimized placements repeated under the same architecture and routing policy. |
| Link eligibility and range, rather than routing priority, drive the effect | The role changes both admissible links and a 1.5 endpoint range factor | Component ablations that vary eligibility, range, preference, and placement one at a time. |
| Workload composition changes which paths matter | Regional rank agreement is low for strict completion and packet sensitivity is large | Additional event windows with the same task-generation contract and controlled priority mixtures. |

The bounded-versus-unlimited comparison is still diagnostically valuable. It falsifies a simpler explanation that high Central Node participation is intrinsically harmful under every forwarding rule. When the limits are relaxed, high fractions improve completion and timing. The decline therefore belongs to an interaction between the package and the bounded policy. It cannot be attributed to orbital geometry alone, because the same base architectures display a different response when the forwarding rule changes.

The packet-size result provides a second mechanism clue. Fourfold packets reduce strict completion by 37.02 points in the matched India block and 33.67 points in the 180-satellite extension. Larger packets consume more of each finite contact and make the same temporal route harder to complete before a deadline. This establishes capacity sensitivity at the package level. It does not prove whether queueing, contact fragmentation, or copy allocation is the dominant internal cause.

Accordingly, the thesis makes a mechanism-bounded claim: routing limits, message size, and the implemented Central Node package interact strongly enough to alter architecture ranking. Detailed trace instrumentation and component ablation are the next tests. This wording preserves the engineering meaning of the observed reversal without turning a plausible explanation into an unmeasured causal result.

### 8.12.3 Operational Architecture Decision Path

The simulation does not select one flight design automatically. It narrows the decision to a portfolio and identifies which additional information would change the choice. Table 8.4 translates the evidence into architecture gates for a mission design review.

**Table 8.4** Evidence-to-decision path for applying the thesis results.

| Decision gate | Evidence from this thesis | Defensible provisional choice | Information required before commitment |
|---|---|---|---|
| Task-message product | Packet size is among the strongest sensitivity drivers | Keep the request compact and transmit only information needed for scheduling | Confirm operational fields, encoding, authentication, and update frequency. |
| Forwarding policy | Peer/hybrid improves completion but adds substantial traffic; bounded high-CN response reverses | Use bounded routing only with explicit copy/hop verification; retain a peer-capable fallback | Trace queues, drops, contact utilization, and recovery behaviour. |
| Central Node tier | CN40/CN50 offers a lower-resource balance; CN90/CN100 can improve unlimited completion and timing | Carry both a moderate and high-participation candidate into the next trade | Separate range, placement, eligibility, and priority by component ablation. |
| Fleet scale | India 180 improves mission metrics and P100 $T_{100}$, but adds about 25.8 Gbit per case | Select 180 only if the completion and timing gains justify spacecraft and network burden | Add cost, energy, storage, launch, operations, and downlink constraints. |
| Ground segment | Ground-station outage is the strongest single failure family tested | Treat the one-station baseline as a design risk | Compare additional station locations, availability assumptions, and service cost. |
| Decision weights | Exact Pareto and rank acceptability show multiple defensible winners | Preserve the non-dominated portfolio rather than fixing an analyst-selected winner | Elicit thresholds and weights from mission operators and programme stakeholders. |

The first decision is therefore not “how many Central Nodes are best?” but “what operational task must be delivered, by when, and at what communication cost?” Once that contract is fixed, infeasible architectures can be removed with explicit thresholds. Exact Pareto filtering should then precede preference weighting so that a compensatory score cannot rescue a design that is worse on every retained objective. Rank acceptability is used last to show how often each remaining candidate wins under alternative, declared preferences [10].

Responsibilities are also separable. The mission scientist defines the useful observation and deadline. The communications engineer validates contact capacity and forwarding behaviour. The spacecraft architect assesses the mass, power, antenna, and processing implications of the Central Node package. The ground-segment designer addresses the single-station vulnerability. The programme decision authority supplies thresholds and preferences. The thesis provides a common evidence set for those roles, but it does not substitute analyst weights for their decisions.

For the current evidence, the defensible output remains tiered. A moderate-Central-Node candidate is appropriate when traffic and differentiated-node burden are constrained. A high-participation candidate is appropriate only under a forwarding policy that avoids the bounded reversal and when faster complete dissemination justifies the added traffic. The 180-satellite tier is an India-workload scale option, not a universal replacement for the 120-satellite cases.

### 8.12.4 Adversarial Stress Test and Calibrated Claim Set

The strongest counter-argument is that the apparent architectural lessons could be artifacts of one simulator, one bundled Central Node implementation, one observation-radius proxy, one ground station, and two selected event windows. Under that interpretation, the high-fraction reversal might disappear with different placement or copy rules; the regional contrast might reflect the chosen days rather than geography; and the portfolio weights might encode analyst preference rather than mission value. This counter-argument is valid against any universal claim of superiority.

It does not invalidate the narrower thesis claim. The workflow demonstrates that reception timing must be combined with later access before a follow-up can be credited. The factorial and matched comparisons then show that, within the verified experiment, routing bounds, task size, event workload, fleet scale, and failure family materially alter the result and often alter the ranking. Those are conditional findings, and their usefulness is precisely that they reveal which assumptions must be tested before a design is frozen.

Table 8.5 converts this adversarial review into a claim ledger that separates supported, suggestive, and unsupported statements and identifies what evidence could overturn each conclusion.

**Table 8.5** Calibrated claim set after adversarial review.

| Claim status | Claim supported by the present evidence | Boundary or falsifying extension |
|---|---|---|
| Supported | Geometric opportunity alone overstates useful follow-up when task reception occurs too late. | A verified replay in which reception timing never changes fulfilment would contradict this result. |
| Supported | The implemented Central Node package interacts with forwarding bounds and can reverse at high fractions. | The package claim would fail if the reported bounded/unlimited outputs or matched identifiers could not be reproduced. |
| Supported | Packet size and ground-station outage are decision-relevant sensitivities in the tested campaign. | Their importance may change under a different message product or multi-station network. |
| Supported | No tested architecture dominates every mission, timing, traffic, burden, and robustness objective. | A new candidate that dominates the retained portfolio would replace the present decision set. |
| Suggestive, not causal | Copy exhaustion, routing concentration, or queue contention may explain the bounded high-CN decline. | Requires trace-level diagnostics and one-factor component ablations. |
| Not supported | Central Nodes are universally superior, one region is inherently easier, or 180 satellites are always preferable. | These statements exceed the tested workloads, package, and objective set. |

The five-perspective cross-check leads to a bounded positive judgement. The research question, method, result, and conclusion form a traceable chain; the task-population correction and integrity gates address the most serious reproducibility risk; the main numerical comparisons are paired and uncertainty-qualified; and the architecture recommendation is conditional rather than absolute. No unresolved contradiction was found between the reported tables, figures, and conclusions. The remaining limitations concern transfer to new event windows, component-level mechanism attribution, and operational implementation. They require follow-on experiments, but they do not negate the thesis contribution within its declared scope.

The central result can therefore be stated without overclaiming: a preferred-relay architecture can improve time-critical task dissemination, but its value depends on forwarding limits, workload, fleet scale, communication burden, and failure tolerance. A mission-ready choice requires these quantities to be evaluated together and traced back to the task contract that produced them.

# 9 Conclusions and Outlook

## 9.1 Conclusions

This thesis developed and evaluated a communication-constrained workflow for disseminating time-critical wildfire follow-up tasks through a distributed satellite system. The completed work connects remote-sensing-derived task generation, Walker constellation design, temporal communication, Central Nodes, reception-before-observation logic, explicit failure replay, and multi-objective post-processing.

The final verified simulation campaign is treated as one body of evidence. It contains 594 matched California and 594 matched India cases at 60 and 120 satellites, together with 282 verified India cases at 180 satellites. Every retained case follows the same 539-scenario contract. The complete CN30–CN100 part of the 180-satellite extension contains 216 cases and supports a direct comparison with the corresponding India 120-satellite architectures. This structure allows all final results to be discussed together while keeping each comparison within the data actually executed.

On matched California and India architectures, the India workload has 4.66 percentage points higher observation fulfilment but 2.98 points lower strict useful completion, 5.95 points lower useful-recipient coverage, approximately 175 minutes longer worst-case complete dissemination, and 0.04 lower robustness-retention AUC. These simultaneous changes establish that observation, dissemination completeness, timing, and communication burden cannot be represented by one metric.

Central Nodes are useful, but more are not always better. With bounded routing, high participation can sharply reduce observation. When copy and hop limits are removed, completion improves toward higher fractions while traffic rises. Peer-capable routing improves strict completion relative to Central Node-only routing. Larger packets are particularly harmful for the India workload. The outage of the sole ground station is the largest individual vulnerability tested, and combined packet growth with satellite failure produces losses that are not visible in nominal performance alone.

The 180-satellite results strengthen the same campaign-level interpretation. From CN30 to CN100, unlimited strict completion rises from 88.98% to 97.16% and P100 $T_{100}$ falls from 257.08 to 142.03 minutes, while traffic rises from 45,476 to 105,760 Mbit per case. Fourfold packets reduce strict completion by 33.67 points. Ground-station outage retains $76.26 \pm 1.48$% of baseline strict completion across base architectures at 30% severity, whereas fourfold packets combined with 20% satellite failure retain 57.49%. The 61-case exact Pareto set and the maximum rank-1 acceptability of 26.55% confirm that the larger-fleet choice also depends on mission preferences. Recomputed two-objective fronts contain between 1 and 10 cases, showing that a simple trade plot can be more selective than the full decision problem but cannot replace it.

Across 216 common India architectures under unlimited routing, moving from 120 to 180 satellites increases observation fulfilment by 1.23 points, strict completion by 3.99 points, and useful-recipient coverage by 1.55 points. Worst-case complete dissemination is 44.68 minutes faster, but task traffic increases by 25,827 Mbit per case. The larger fleet therefore improves the tested mission outcomes without dominating the resource objective.

Taken together, the results reveal why no single architecture dominates the tested objectives. The final engineering output is one tiered architecture portfolio: cross-window candidates supported in both event windows, supplemented by large-fleet candidates whose evidence applies to the India workload. The distinction records comparison scope; it does not represent the 180-satellite results as a different or provisional simulation.

## 9.2 Answers to the Research Questions

**RQ1: How do Central Node fraction and forwarding policy change timely task dissemination and follow-up observation in the implemented distributed satellite system?** The two factors act together. Intermediate fractions work best with the bounded package, while the unlimited diagnostic continues to benefit from higher fractions and also produces more traffic. Since eligibility, forwarding preference, placement, and the link-budget boost are changed as one package, the evidence supports the complete implemented treatment rather than an isolated component effect.

**RQ2: How do workload, packet size, constellation geometry, and injected failures change performance and the relative ranking of architectures?** Each factor changes the result values and, in several comparisons, the architecture order. On matched 60/120-satellite cases, India has slightly higher observation fulfilment but weaker, later, and less robust complete dissemination. Fourfold packets, ground-station outage, and combined packet/failure stress produce the largest changes. Geometry changes the temporal opportunities, but no factor level is best in every tested context.

**RQ3: Which architectures provide defensible trade-offs among observation fulfilment, deadline-complete dissemination, communication burden, resource proxies, and robustness?** Exact Pareto filtering and weight sensitivity identify a five-case cross-window core and 61 Pareto-efficient 180-satellite extensions. Together they form a tiered portfolio rather than competing reports. A final choice still requires declared mission priorities, the workload to be served, and an agreed set of preference weights.

## 9.3 Scientific and Engineering Contributions

The completed thesis work comprises:

1. conversion of Sentinel-3 fire detections into traceable satellite tasks with priorities and deadlines;
2. implementation of a temporal simulation in which physical access, communication, task reception, and follow-up observation remain separate events;
3. definition and verification of a 539-scenario contract, including not-applicable states, deterministic de-duplication, and task-population auditing;
4. priority-dependent time analysis with censoring instead of discarding unfinished tasks;
5. paired evidence for workload, routing, packet size, failures, and combined stress;
6. redundancy checks, exact Pareto filtering, and rank acceptability for the architecture decision; and
7. demonstration that a valid restart must check input hashes, regional and priority counts, and index normalization in addition to completed scenario identifiers.

These contributions provide a documented path from an input task to a reported claim, retain unsuccessful outcomes, and distinguish a technically completed job from a scientifically comparable simulation.

## 9.4 Recommended Architecture Decision

The portfolio begins with the 60-satellite candidates that are supported in both event windows. `T060_P10_H0800_I0986_CN040_F1` combines strong California rank stability with relatively low India traffic. The 60-satellite CN90/CN100 alternatives at 1000 km and 98.6-degree inclination place greater

emphasis on India completion. The 120-satellite CN100 design is faster for India, but is justified only when that improvement warrants roughly twice the traffic of the comparable 60-satellite CN100 case.

For an India workload where a 180-satellite fleet is feasible, `T180_P10_H0500_I0986_CN050_F1` is the first large-fleet portfolio member to inspect. It has 26.55% rank-1 acceptability under the random family-weight analysis, completes the reported mission metrics, and uses less traffic than the faster CN90/CN100 alternatives. Its evidence applies to the balanced 773-task India block and is therefore reported alongside, rather than substituted for, the cross-window candidates.

Raw mission, timing, traffic, fleet, Central Node, and robustness values should be compared before approved preferences are applied. The weights used here are analyst-defined scenarios, not elicited stakeholder consensus. Likewise, the risk-screen likelihood categories are declared engineering assumptions rather than simulated probabilities. The sole-ground-station result motivates a ground-network trade study but does not quantify the benefit or location of a specific additional station.

## 9.5 Limits and Future Work

The completed campaign establishes an India 120-to-180 effect on the 216 matched CN30–CN100 architectures, but it does not establish an unrestricted 60-to-120-to-180 law. The 180-satellite coverage is unbalanced below CN30, and a matched California 180-satellite population would be required for a large-fleet event-window contrast.

The next research steps are:

1. complete the 15 missing India 180-satellite low-CN cells and execute a matched California 180-satellite grid using the verified task-hash and positional-index contract;
2. repeat representative cases for 500, 700, and 900 km observation radii to quantify the sensitivity of opportunity and recipient definitions;
3. separate Central Node eligibility, forwarding preference, placement, and link-budget enhancement in a component-ablation design;
4. compare balanced, random, and graph-centrality-based Central Node placement using several seeds;
5. evaluate two or more ground-station networks and explicit outage correlation, because the one-station baseline is the dominant tested vulnerability;
6. vary packet construction, task clustering, priority thresholds, and deadline rules to test whether the packet-size effect survives plausible wildfire-product definitions;
7. retain trace-level queue, forwarding, and dropped-copy decisions for diagnosing the bounded high-CN decline;
8. extend random-failure and combined-stress trials until mean, fifth-percentile, retention-AUC, and rank-acceptability estimates meet predefined convergence thresholds;
9. add cloud, smoke, instrument pointing, imaging duration, image downlink, and alert-delivery models to extend the endpoint beyond a geometric follow-up opportunity;
10. sample additional seasons and wildfire event windows before making geographic or climatological generalizations;
11. elicit stakeholder weights and feasibility thresholds from mission operators rather than treating analyst-defined weights as preferences; and
12. replace the present satellite/Central Node/traffic proxies with lifecycle cost, power, mass, and operations estimates for an implementable mission trade study.

Within the analysed evidence, routing participation, task workload, forwarding limits, packet size, constellation geometry, and failure conditions jointly determine whether information delivery becomes a deadline-complete follow-up opportunity. A transparent portfolio with explicit evidence scope is therefore more defensible than a universal Central Node fraction or fleet-size recommendation.

# A Detailed Metric Definitions

## A.1 Notation and Denominators

Table A.1 defines the symbols, units, and denominators used by the reported mission, timing, traffic, and robustness measures.

**Table A.1** Metric notation used in the final verified analysis.

| Symbol | Unit/type | Definition |
|---|---|---|
| $r_i$ | UTC time | Release time of task $i$ |
| $d_i$ | UTC time | Priority-dependent deadline |
| $b_i$ | Mbit | Task-message size |
| $a_{is}$ | UTC time | First complete receipt of task $i$ by satellite $s$ |
| $O_{is}$ | time set | Candidate observation opportunities |
| $t_i^{\mathrm{obs}}$ | UTC time | First valid post-reception observation before deadline |
| $E_i^0$ | node set | Original useful-recipient set before failure |
| $E_i^g$ | node set | Surviving useful-recipient set in scenario $g$ |
| $R_i^g(d_i)$ | node set | Unique surviving useful recipients reached by deadline |
| $Q_i^g$ | percent | Surviving useful-recipient coverage |
| $S_{100,i}^g$ | Boolean | Every non-empty surviving useful set reached by deadline |
| $F_{\mathrm{obs}}^g$ | percent | Tasks observed after reception and before deadline |
| $T_{100,i}^g$ | min | Release-to-last-useful-reception time when complete |
| $B_w$ | Mbit | Usable capacity of communication window $w$ |
| $B$ | Mbit/case | Total message volume across accepted transfers |
| $E$ | count/case | Accepted transfer-event count |
| $P$ | 0–1 | Normalized performance pillar |
| $R$ | 0–1 | Normalized resource-efficiency pillar |

Original and surviving denominators are stored separately. This prevents node removal from improving a percentage merely because the denominator became smaller. A central-node failure scenario at CN0 is `NOT_APPLICABLE`, not a successful zero-effect trial.

## A.2 Task-Level Outcomes

Observation success is

$$I_i^{\mathrm{obs}} = \mathbb{I}[\exists s, t \in O_{is} : r_i \le a_{is} \le t \le d_i]\,. \tag{A.1}$$

The case-level observation percentage is $100N^{-1}\sum_i I_i^{\mathrm{obs}}$. Observation latency is $t_i^{\mathrm{obs}} - r_i$ only for successful tasks and must be paired with the success fraction.

For a surviving useful set $E_i^g$,

$$Q_i^g = 100\frac{|R_i^g(d_i)|}{|E_i^g|}, \tag{A.2}$$

$$S_{100,i}^g = \mathbb{I}[E_i^g \neq \emptyset \ \wedge \ R_i^g(d_i) = E_i^g]. \tag{A.3}$$

$S_{100}^{\text{useful}}$ is the mean of this strict indicator. It must not be called "all-satellite reach" unless the useful set has separately been proven to equal the full fleet.

When all useful recipients are reached,

$$T_{100,i} = \max_{s \in E_i}(a_{is} - r_i). \tag{A.4}$$

P100 $T_{100}$ is the largest completed $T_{100,i}$ within a case. Missing completion is censored; it is not set to zero. The final campaign does not expose a separate first-reception field in the canonical case table. Earlier nominal first-reception results therefore remain a separate evidence layer.

## A.3 Communication and Network Metrics

Window capacity is $B_w = R_w \Delta t_w \eta_w / 10^6$. Used capacity is accepted transferred Mbit divided by eligible available capacity. Undirected physical capacity is counted once per pair when represented by a symmetric adjacency matrix.

The principal burden fields are total transferred Mbit, transfer-event count, used-capacity percentage, peak node-sent Mbit, central-node traffic share, and the Gini coefficient of node-sent data. A link window is a possible contact; a transfer is realized policy use. Counts of these two objects answer different questions.

## A.4 Robustness Effects

For a benefit $F$, paired retention and percentage-point change are

$$\rho_F(g) = \frac{F_g}{F_0}, \qquad \Delta F(g) = F_g - F_0, \tag{A.5}$$

where $F_0$ is the same architecture's unlimited baseline. Percentage-point change is preferred for the chapter summaries because a small baseline can make ratios unstable. For latency, the report uses $\Delta T = T_g - T_0$ and retains the completion/censoring count.

Random failures are summarized over their seeded replicates. Replicate variation describes the sampled failure selection under the model; it does not include uncertainty in orbital propagation, task generation, or the link budget.

## A.5 Pareto Dominance and Normalization

For benefit objectives $f_1, \ldots, f_k$, case $a$ dominates $b$ if

$$f_j(a) \geq f_j(b) \quad \forall j, \qquad f_j(a) > f_j(b) \quad \text{for at least one } j. \tag{A.6}$$

Burden and latency variables are inverted only after their raw values and censoring are checked. Min–max normalization over the 890 eligible California cases is

$$\widetilde{x}_c = \frac{x_c - x_{\min}}{x_{\max} - x_{\min}}. \tag{A.7}$$

For a lower-is-better metric, benefit is $1 - \widetilde{x}_c$. If a timeliness value is censored, its benefit is defined as zero rather than excluding the case. The eight-objective frontier uses observation, strict completion, coverage, inverse P100 $T_{100}$, inverse satellite count, inverse central-node count, inverse data, and inverse transfer events.

## A.6 Objective Utility

The performance and resource-efficiency pillars are

$$P = 0.25\widetilde{F}_{\text{obs}} + 0.25\widetilde{S}_{100} + 0.25\widetilde{Q} + 0.25\widetilde{T}, \tag{A.8}$$

$$R = 0.40(1 - \widetilde{N}_{\text{sat}}) + 0.20(1 - \widetilde{N}_{\text{CN}}) + 0.20(1 - \widetilde{B}) + 0.20(1 - \widetilde{E}). \tag{A.9}$$

The high-level score is $U(\lambda) = \lambda P + (1 - \lambda)R$. The coefficients are declared preferences. They do not convert the resource terms into money and they do not replace feasibility requirements.

Randomized sensitivity uses

$$\lambda \sim \text{Uniform}(0, 1), \tag{A.10}$$

$$\vec{w}_P \sim \text{Dirichlet}(1, 1, 1, 1), \tag{A.11}$$

$$\vec{w}_R \sim \text{Dirichlet}(4, 2, 2, 2), \tag{A.12}$$

for 10,000 draws with seed 20260825. A winner frequency is conditional on these distributions; it is not a posterior probability or confidence interval.

## A.7 Final Censor-Aware Event Measures

For event type $e \in \{\text{follow-up observation}, \text{S100 dissemination}\}$, let $X_i^e$ be the event time and $C_i = d_i - r_i$ the deadline-based censoring time. The retained duration and event indicator are

$$Y_i^e = \min(X_i^e, C_i), \qquad \delta_i^e = \mathbb{I}[X_i^e \le C_i]. \tag{A.13}$$

At ordered event time $t_j$, with $d_j$ events among $n_j$ at-risk tasks, the Kaplan–Meier estimate is

$$\widehat{S}_e(t) = \prod_{t_j \le t} \left(1 - \frac{d_j}{n_j}\right). \tag{A.14}$$

Direct deadline attainment, $N^{-1}\sum_i \delta_i^e$, is the primary completion measure. Kaplan–Meier and restricted mean event-free time are calculated within priority/deadline class, using $\tau_k \in \{30, 60, 180, 360\}$ minutes for Critical, High, Medium, and Low tasks. Within a class, the reported event probability is $1 - \widehat{S}_{e,k}(\tau_k)$ and the restricted mean is $\int_0^{\tau_k} \widehat{S}_{e,k}(t)\,\mathrm{d}t$. A lower RMST is favourable only when event probability is considered alongside it.

For the paired California–India event-window summary, High, Medium, and Low class estimates are standardized with fixed weights $(1/3, 1/3, 1/3)$. The California-only Critical class is reported separately because India contains no Critical task. An unstratified pooled Kaplan–Meier curve is not primary evidence unless non-informative censoring across the coupled deadline/packet-size classes is independently justified.

## A.8 Base-Clustered Resampling-Stability Intervals

Let $b = 1, \ldots, 81$ identify Walker base architectures. A bootstrap draw samples base identifiers with replacement and retains every eligible CN level, scenario, and replicate belonging to each sampled base. For a reported quantity $\widehat{\theta}$, 1,000 draws produce the 2.5th and 97.5th percentile limits. This preserves within-base dependence. The interval measures stability over the evaluated architecture grid and must not be described as population sampling uncertainty.

Paired event-window contrasts use identical case identifiers. For metric $m$ and case $c$,

$$\Delta_{c,m}^{\text{region}} = m_{c,\text{India}} - m_{c,\text{California}}. \tag{A.15}$$

Positive values therefore favour the India event-window workload for benefit metrics and indicate a larger India-window burden for cost metrics. This sign convention is printed in every final paired table. It controls architecture but does not identify a population-level regional effect.

## A.9 Robustness Retention and Severity Area

For benefit metric $F$, stressed severity $g$, and the matched unlimited baseline $0$,

$$\Delta F_c(g) = F_c(g) - F_c(0), \qquad \rho_{F,c}(g) = \frac{F_c(g)}{F_c(0)}. \tag{A.16}$$

The normalized retention area is calculated by trapezoidal integration after adding $(0, 1)$:

$$A_{F,c} = \frac{1}{g_{\max}} \int_0^{g_{\max}} \rho_{F,c}(g)\, \mathrm{d}g. \tag{A.17}$$

These quantities are not interchangeable. Absolute stressed performance shows the remaining capability, paired change gives the loss or gain, retention scales that change by the baseline, the fifth percentile describes a low trial outcome, and $A_{F,c}$ summarizes the severity curve. When the baseline is zero, retention is undefined; the case remains in the absolute and difference tables instead of being deleted.

## A.10 Redundancy Gate and Final Decision Families

For the unlimited useful-deadline scenario, the redundancy check calculates both Pearson $r$ and Spearman $\rho_s$ between candidate performance metrics. A pair is flagged at $\max(|r|, |\rho_s|) \geq 0.95$ and classified as severe at 0.98. This is a practical screen against double weighting. It does not prove that the two metrics represent the same physical quantity.

After exact Pareto filtering, non-redundant normalized components are organized into the five final families

$$U_{\text{ref}} = 0.25M + 0.20O + 0.20T + 0.20R + 0.15A, \tag{A.18}$$

where $M$ is mission dissemination, $O$ observation success, $T$ timeliness, $R$ resource efficiency, and $A$ robustness retention. The coefficients are the reference proposal for supervisor review. Resource efficiency is the equal mean of inverse-normalized satellite count, central-node count, and transferred Mbit unless a revised approved value model is recorded.

For the stochastic weight analysis, $\mathbf{w}^{(k)} \sim \text{Dirichlet}(1, \ldots, 1)$ over the retained family scores for $k = 1, \ldots, 10{,}000$. Rank-1 acceptability is the percentage of draws in which a case has the largest utility. Top-5 acceptability is the percentage in which it appears among the five largest. These are preference-space stability measures, not probabilities of scientific truth.

# B Campaign Configuration and Case Identifiers

## B.1 Case Identifier Grammar

Table B.1 decodes representative identifiers so that fleet size, geometry, Central Node fraction, and phasing can be traced from a result row back to its architecture.

```
T{satellites:03d}_P{planes:02d}_H{altitude_km:04d}_
I{inclination_x10:04d}_CN{fraction:03d}_F1
```

**Table B.1** Example case identifiers.

| Identifier | Interpretation |
|---|---|
| `T060_P03_H0500_I0600_CN000_F1` | 60 satellites, 3 planes, 500 km, 60 deg, no central nodes |
| `T120_P10_H0800_I0986_CN030_F1` | 120 satellites, 10 planes, 800 km, 98.6 deg, 30% CN |
| `T180_P10_H1000_I0600_CN100_F1` | 180 satellites, 10 planes, 1000 km, 60 deg, all nodes central |

## B.2 Final Campaign Frozen Controls

Table B.2 records the fixed physical and operational settings that are not part of the architecture sweep.

**Table B.2** Core physical and operational controls of the final verified campaign.

| Control | Value | Note |
|---|---|---|
| Campaign seed | 20260811 | Scenario seeds derived by SHA-256 |
| Orbital time step | 60 s | Discrete opportunity evaluation |
| Observation radius | 700 km | Fixed; no radius sweep in the final campaign |
| Ground station | Munich | 10-deg minimum elevation |
| Sat–sat rate | V19 UHF calculation | Effective cap near 19.2 kbit/s |
| Sat–ground rate | 100 kbit/s | Fixed window rate |
| Central link-budget boost | 1.5 | Physical-rate multiplier in link calculation |
| Nominal topology | `central_only` | CN0 has ground uplink but no ISL |
| Nominal routing | targeted useful/deadline | Greedy temporal replay |
| Nominal hop/copy limits | 3 / 3 observer / 5 CN | Removed in unlimited diagnostic |
| Window utilization | 1.0 | Capacity multiplier before failure |

The legacy field `sat_sat_data_rate_bps=250000` is present in configuration provenance but is not used by the active V19 link-budget calculation. It must not be reported as the satellite-to-satellite rate of the final verified campaign.

## B.3 Scenario Counts

Table B.3 reconciles the deterministic and seeded scenario families with the 539 rows declared for every architecture.

**Table B.3** Scenario rows per architecture.

| Family | Levels | Replicates/level | Rows |
|---|---|---|---|
| Core/policy/unlimited | 7 | 1 | 7 |
| Task size | 4 | 1 | 4 |
| Random satellite failure | 4 | 30 | 120 |
| Random central-node failure | 4 | 30 | 120 |
| Link-window outage | 4 | 30 | 120 |
| Ground-station outage | 4 | 30 | 120 |
| Targeted CN failure | 4 | 1 | 4 |
| Capacity degradation | 4 | 1 | 4 |
| Combined stress | 4 | 10 | 40 |
| Total | | | 539 |

## B.4 Seed Derivation

Table B.4 states which factors share a random draw and which receive a new draw, preserving paired comparisons across architectures and regions.

For every stochastic row, the seed material is the UTF-8 concatenation

```
campaign_seed|case_id|scenario_id|replicate
```

The SHA-256 digest is calculated and its first eight bytes define the numeric random seed. This makes each replicate stable even when worker count, job order, or resume timing changes.

**Table B.4** Random-seed sharing scope in the completed campaign.

| Comparison | Shared random draw? | Reason |
|---|---|---|
| Same case/scenario/replicate after resume | Yes | Complete seed key is identical |
| Different replicate of one case/scenario | No | Replicate number changes |
| Different CN fraction or architecture | No | Case identifier changes |
| Different scenario identifier | No | Scenario identifier changes |
| California versus India | Only if every seed-key field is identical | No cross-window sharing is assumed without an explicit paired contract |

The current campaign therefore guarantees reproducibility, not common random numbers across architectures. Common random numbers can reduce variance only when the corresponding draws are deliberately shared and recorded across alternatives [21]. A future paired variance-reduction run must therefore introduce a separate shared-draw identifier.

## B.5 Checkpoint and Completion Rules

The runner writes scenario metrics in Parquet batches of 20 rows. It writes to a temporary path first and renames that path only after the batch is complete. On restart, the runner reads the scenario keys already stored and calculates the missing keys. An interruption close to a batch boundary can still leave duplicate keys, so the post-processing applies one deterministic key-level de-duplication rule before validation.

A case is classified as complete only if `_SUCCESS.json` exists and every expected scenario identifier is either completed or explicitly marked not applicable. Readable rows without that marker are not promoted into the complete analysis population.

## B.6 Representative Parquet Reading Commands

Python with pandas and PyArrow can inspect a partition without conversion:

The literal directory name `full891_fresh_complete` in the following command is a legacy repository identifier for the clean, validated campaign restart. It does not denote a separate simulation method, scenario family, or evidence population.

```
from pathlib import Path
import pandas as pd

case_dir = Path("outputs/full891_fresh_complete/runs/California") / CASE_ID
metrics = pd.concat(
    [pd.read_parquet(p) for p in sorted(case_dir.glob("case_metrics_part_*.parquet
    "))],
    ignore_index=True,
)
print(metrics.shape)
print(metrics[["scenario_id", "status",
               "observation_success_percent"]].head())
```

DuckDB provides an interactive alternative:

```
SELECT scenario_id, status, observation_success_percent
FROM read_parquet('case_metrics_part_*.parquet')
ORDER BY scenario_id
LIMIT 20;
```

The compact CSV tables in this report package are intended for quick inspection and plotting. They are derived evidence and do not replace the raw Parquet archive.

# C Validation and Evidence Traceability

## C.1 Evidence Hierarchy

Table C.1 defines the precedence rule used when a summary file, a configuration file, and the validated case evidence disagree.

**Table C.1** Evidence hierarchy used to resolve conflicting artefacts.

| Rank | Source | Permitted use |
|---|---|---|
| 1 | Raw campaign archive, manifests, markers, and checksums | Authoritative row-level status and provenance |
| 2 | Validated canonical/post-processed tables | Quantitative thesis tables and figures |
| 3 | Frozen simulator code and configuration | Algorithm and configuration description |
| 4 | Frozen wildfire-task CSV and task fingerprints | Counts, task fields, positions, priorities, epochs, and population identity |
| 5 | Meeting notes and presentation material | Requirements, rationale, and development chronology |
| 6 | Older or smoke runs | Diagnostics and historical context only |

## C.2 Final Structural Validation Ledger

Table C.2 reports the expected and observed regional inventory, including scenario-key completeness and task-population checks.

**Table C.2** Full-region structural inventory of the final verified campaign.

| Check | California | India | Interpretation |
|---|---|---|---|
| Base architectures | 81 | 81 | Complete |
| Architecture identifiers | 891 | 891 | Complete grid |
| Declared scenarios per case | 539 | 539 | Common contract |
| Unique scenario-status rows | 480,249 | 480,249 | Exact expected count |
| Completed scientific rows | 470,205 | 470,205 | Retained where eligible |
| Not applicable rows | 10,044 | 10,044 | Principally CN failure at CN0 |
| Failed retained rows | 0 | 0 | None |
| Architecture checks passed | 19,179 | 19,179 | All recorded checks |
| Resume duplicates removed | 4,269 | 0 | California keys were scientifically identical |

The combined inventory contains 960,498 unique status rows, 940,410 completed scientific rows, 20,088 explicit `NOT_APPLICABLE` rows, and 38,358 passed architecture checks. Structural completion is necessary but does not by itself establish comparability; task population and the balance of the analysis block are separate gates.

The final evidence release therefore uses two explicit campaign populations:

- a confirmatory California–India population with 594 matched 60/120-satellite cases per event window; and

- an India 180-satellite extension with 282 verified cases and 151,998 scenario-status rows, all based on the intended 773 tasks. Its balanced CN30–CN100 block contains 216 cases from 27 base architectures.

The 180-satellite availability is 20 cases at CN0, 22 at CN10, 24 at CN20, and all 27 base architectures at each CN level from 30 to 100. The 216-case block is used whenever a complete central-node sweep or exact decision comparison is required.

## C.3 Task-Population Audit and Root Cause

The frozen intended input has the SHA-256 value

`e717a1dde84db0f47efdda09cfca2f437`
`909349b25a139a45a7c1928da80e533`.

It contains 32 California and 773 India tasks. The 741-task population in the first 180-satellite execution was traced to an index-alignment defect in task normalization. Filtering India from the combined CSV retained row labels 32–804. The normalization function created a new table indexed 0–772 and assigned filtered Series without first resetting their labels. Pandas aligned by label: the first 32 normalized rows became invalid and the final 32 source rows were omitted. All omitted records were Low-priority tasks from 16 April 2026; the High- and Medium-priority task sets were unchanged.

The corrected execution resets the regional index and freezes the task population through the source, normalized-task, allow-list, and run-contract hashes. Its verification record reports `PASS`, 282 of 282 selected cases complete, zero missing or invalid cases, 539 scenario identifiers per case, and 773 India tasks. The earlier output is retained only as provenance evidence and is not used in final numerical tables. The common task contract permits a matched India 120-to-180 comparison on the complete CN30–CN100 block.

## C.4 Final Claim Ledger

Table C.3 links each final claim to the population that supports it and records the boundary beyond which it is not asserted.

**Table C.3** Supported and bounded claims in the final verified analysis.

| Claim | Status | Boundary |
|---|---|---|
| Full two-region structural scenario inventory | Supported | 891 identifiers and 480,249 unique status rows per region |
| Uniform 60/120 matched regional effects | Supported | 594 cases per region; intended task populations |
| India T180 corrected findings | Supported | 282 verified cases on the intended 773-task population |
| Balanced T180 CN/Pareto analysis | Supported | 216 cases, CN30–CN100, 27 bases |
| Matched India 120-to-180 fleet-size effect | Supported conditionally | 216 common CN30–CN100 architectures; unlimited-routing baseline |
| Resume duplicates changed sample size | Rejected | 4,269 California duplicate keys removed deterministically |
| Routing-policy and packet-size effects | Supported | Paired within each eligible analysis population |
| Random and targeted failure robustness | Supported | Time-aware replay with declared seeds |
| Combined-stress fifth-percentile generalization | Not supported | Ten replicates support descriptive means only |
| Exact Pareto membership and rank acceptability | Supported conditionally | Separate cross-window and India 180 feasible sets |
| Final-campaign first-reception result | Not supported | Separate field absent from the canonical case table |
| Observation-radius stability | Not supported | Radius fixed at 700.00 km |
| Routing-package component effects | Not supported | Principal sweep changes bundled components together |
| Placement robustness | Not supported | One deterministic balanced placement tested |
| Population-level California–India effect | Not supported | Only two selected event windows |
| Universal optimum or monetary cost | Not supported | Case study, analyst weights, and resource proxies only |
| End-to-end wildfire alert latency | Not supported | Chain ends at follow-up observation opportunity |

## C.5 Output-to-Claim Traceability

Table C.4 maps the primary machine-readable outputs to their use in the thesis, providing a direct audit route from claim to evidence file.

For every quantitative claim, the release retains a compact CSV or Parquet table, the population filter, and the frozen analysis settings. Regional differences always use India minus California. The decision record also states the objective directions, normalization population, reference weights, random-weight seed, draw count, and eligibility rules. Distinct output prefixes identify the regional and 180-satellite blocks of the same final campaign so that an unsupported comparison cannot be formed accidentally.

**Table C.4** Primary final outputs and their thesis use.

| Output | Use |
|---|---|
| `data/final_results/completion_gate.csv` | Full inventory and confirmatory eligibility |
| `uniform_60_120_matched_region_delta_summary.csv` | Confirmatory regional effects and stability intervals |
| `priority_stratified_event_time_summary.csv` | Confirmatory priority/deadline evidence |
| `uniform_60_120_robustness_summary.csv` | Confirmatory failure response and retention |
| `uniform_60_120_exact_pareto.csv` and rank tables | Confirmatory non-dominated set and shortlist |
| `corrected_180_scope_gate.csv` | Corrected 282-case integrity and task-population gate |
| `corrected_180_cn_fraction_intervals.csv` | Balanced 180-satellite Central Node response |
| `corrected_180_matched_120_180_summary.csv` | Matched India fleet-size effects and stability intervals |
| `corrected_180_paired_scenario_effects_summary.csv` | Corrected policy and packet-size effects |
| `corrected_180_robustness_summary.csv` | Corrected failure and retention evidence |
| `corrected_180_priority_event_summary.csv` | Corrected priority/deadline evidence |
| `corrected_180_exact_pareto.csv` and rank tables | Corrected Pareto and acceptability results |
| `corrected_180_postprocessing_manifest.json` | Inputs, settings, verification, and inference boundary |

## C.6 Independent Review Cross-Check

The final cross-check compared the manuscript with the consolidated review ledger. It confirmed the following boundaries throughout the text: the completed study concerns dissemination and a follow-up opportunity, not scheduling; the Central Node treatment is a package; priority-specific censoring is retained; robustness comes from replayed failures; and the Munich result concerns the tested single-station baseline. The task model, placement, observation radius, and analyst-defined weights remain explicit assumptions. The mission chain stops before image delivery or emergency response, and results from the ten-trial combined-stress cells remain descriptive.

## C.7 Automated Checks and Reproducibility Boundary

The release checker tests figure and citation references, placeholders, unexpectedly large files, secret-like assignments, and manifest hashes. The portable report package contains the validated compact tables, figures, analysis manifests, and LaTeX source, but not the multi-gigabyte Parquet campaign. A complete external scientific archive would also need the raw data, immutable SHA-256 manifests, an environment lock, the simulator commit, and a small public test fixture. Data-service credentials are not included in the repository.

# D Supplementary Result Tables

## D.1 Confirmatory Numerical Reference

Tables D.1, D.2, and D.3 preserve the confirmatory regional effect estimates, cross-window rank agreement, and the resulting conditional shortlist in a compact numerical form.

**Table D.1** Matched India-minus-California results for the validated 60/120-satellite population.

| Metric | Matched cases | Mean change | 95% stability interval |
|---|---|---|---|
| Observation fulfilment | 594 | $+4.66$ pp | $[+1.58, +7.78]$ pp |
| Strict useful completion | 594 | $-2.98$ pp | $[-5.39, -0.33]$ pp |
| Mean useful coverage | 594 | $-5.95$ pp | $[-7.64, -4.41]$ pp |
| P100 $T_{100}$ | 541 | $+175.07$ min | $[+166.94, +182.97]$ min |
| Transferred data | 594 | $+27,957$ Mbit | $[+25,193, +30,779]$ Mbit |
| Robustness retention AUC | 541 | $-0.04$ | $[-0.05, -0.04]$ |

**Table D.2** Cross-window Spearman rank agreement on the confirmatory population.

| Metric | Spearman $\rho$ |
|---|---|
| Observation fulfilment | 0.73 |
| Strict useful completion | 0.77 |
| Mean useful coverage | 0.72 |
| P100 $T_{100}$ | 0.78 |
| Transferred data | 0.99 |
| Robustness retention AUC | 0.61 |

**Table D.3** Five-case conditional shortlist calculated on the validated 60/120-satellite population.

| Architecture | CA Obs. | CA $S_{100}^{\text{useful}}$ | India Obs. | India $S_{100}^{\text{useful}}$ | India $T_{100}$ | India Mbit |
|---|---|---|---|---|---|---|
| T060_P05_H0800_I0800_CN050_F1 | 100.00 | 93.75 | 82.15 | 82.15 | 253.48 | 19,326 |
| T060_P10_H0800_I0986_CN040_F1 | 96.88 | 100.00 | 99.61 | 83.70 | 291.88 | 18,238 |
| T060_P10_H1000_I0986_CN090_F1 | 96.88 | 100.00 | 100.00 | 100.00 | 167.65 | 32,947 |
| T060_P10_H1000_I0986_CN100_F1 | 96.88 | 100.00 | 100.00 | 100.00 | 146.65 | 35,700 |
| T120_P10_H1000_I0986_CN100_F1 | 100.00 | 100.00 | 100.00 | 100.00 | 105.45 | 71,400 |

## D.2 India 180-Satellite Campaign Extension

Tables D.4 and D.5 report the balanced Central Node response and the matched fleet-size effect. Tables D.6, D.7, D.11, and D.12 summarize policy, packet-size, disturbance, combined-stress, and deadline-sensitive results. The engineering risk screen and the recomputed two-objective fronts are documented in Tables D.9 and D.10, respectively.

**Table D.4** India 180-satellite Central Node response on the balanced 216-case CN30–CN100 block.

| Policy | CN | Obs. (%) | $S_{100}^{\text{useful}}$ (%) | Coverage (%) | P100 $T_{100}$ (min) | Data (Mbit) |
|---|---|---|---|---|---|---|
| Bounded | 30 | 88.15 | 0.25 | 11.20 | 81.16 | 4,671 |
| Bounded | 50 | 87.67 | 0.06 | 10.81 | 78.18 | 4,655 |
| Bounded | 100 | 43.52 | 0.00 | 2.04 | – | 2,970 |
| Unlimited | 30 | 90.07 | 88.98 | 95.74 | 257.08 | 45,476 |
| Unlimited | 50 | 91.82 | 94.84 | 98.45 | 199.74 | 63,225 |
| Unlimited | 100 | 92.36 | 97.16 | 99.17 | 142.03 | 105,760 |

**Table D.5** Matched India 180-minus-120-satellite effects under unlimited routing.

| Metric | T120 mean | T180 mean | Mean change |
|---|---|---|---|
| Observation fulfilment (%) | 90.57 | 91.80 | $+1.23$ pp |
| Strict useful completion (%) | 91.01 | 94.99 | $+3.99$ pp |
| Useful-recipient coverage (%) | 96.80 | 98.34 | $+1.55$ pp |
| P100 $T_{100}$ (min) | 231.23 | 186.55 | $-44.68$ min |
| Transferred data (Mbit/case) | 50,060 | 75,887 | $+25,827$ |

**Table D.6** India 180-satellite paired policy and packet-size effects.

| Contrast against nominal $1\times$ | $\Delta$ Obs. (pp) | $\Delta$ $S_{100}^{\text{useful}}$ (pp) | $\Delta$ data (Mbit) |
|---|---|---|---|
| Peer-only routing | $+0.28$ | $+0.94$ | $+29,304$ |
| $0.5\times$ packet size | $+1.05$ | $+2.59$ | $-37,467$ |
| $2\times$ packet size | $-6.13$ | $-10.49$ | $+63,857$ |
| $4\times$ packet size | $-22.32$ | $-33.67$ | $+133,628$ |
| Unlimited useful-deadline forwarding | $+11.98$ | $+94.92$ | $+71,546$ |

**Table D.7** India 180-satellite strict-completion retention at 30% severity.

| Failure family | $S_{100}^{\text{useful}}$ retention (%) |
|---|---|
| Ground-station outage | 76.26 |
| Targeted Central Node failure | 93.99 |
| Random Central Node failure | 95.28 |
| Capacity degradation | 95.43 |
| Link-window outage | 95.47 |
| Random satellite failure | 95.67 |

**Table D.8** Strict-completion retention at 30% severity, reported as mean $\pm$ standard deviation across matched base-architecture means.

| Block | Region | Disturbance | Retention (%) | Loss (pp) |
|---|---|---|---|---|
| Regional 60/120 | California | Capacity degradation | $97.96 \pm 2.49$ | 1.22 |
| Regional 60/120 | California | Ground-station outage | $71.86 \pm 1.63$ | 20.35 |

| Block | Region | Disturbance | Retention (%) | Loss (pp) |
|---|---|---|---|---|
| Regional 60/120 | California | Link-window outage | $90.30 \pm 5.44$ | 4.77 |
| Regional 60/120 | California | Random Central Node loss | $89.30 \pm 5.30$ | 6.87 |
| Regional 60/120 | California | Random satellite loss | $94.08 \pm 4.65$ | 4.43 |
| Regional 60/120 | California | Targeted Central Node loss | $88.91 \pm 6.41$ | 7.37 |
| Regional 60/120 | India | Capacity degradation | $87.45 \pm 4.54$ | 7.23 |
| Regional 60/120 | India | Ground-station outage | $74.05 \pm 1.73$ | 17.50 |
| Regional 60/120 | India | Link-window outage | $77.88 \pm 7.78$ | 10.33 |
| Regional 60/120 | India | Random Central Node loss | $79.78 \pm 6.92$ | 11.79 |
| Regional 60/120 | India | Random satellite loss | $87.06 \pm 4.84$ | 8.63 |
| Regional 60/120 | India | Targeted Central Node loss | $78.18 \pm 7.66$ | 13.21 |
| India 180 | India | Capacity degradation | $95.43 \pm 4.55$ | 4.16 |
| India 180 | India | Ground-station outage | $76.26 \pm 1.48$ | 22.46 |
| India 180 | India | Link-window outage | $95.47 \pm 2.76$ | 4.14 |
| India 180 | India | Random Central Node loss | $95.28 \pm 2.90$ | 4.33 |
| India 180 | India | Random satellite loss | $95.67 \pm 2.66$ | 3.97 |
| India 180 | India | Targeted Central Node loss | $93.99 \pm 4.21$ | 5.59 |

The standard deviations in Table D.8 measure dispersion across the tested base architectures after the within-base scenario aggregation. They do not represent uncertainty across unobserved wildfire events or hardware populations.

**Table D.9** Likelihood–consequence screening used in Figure 7.8.

| Disturbance | Worst loss (pp) | Consequence | Likelihood | Score | Class |
|---|---|---|---|---|---|
| Capacity degradation | 7.23 | 3 | 3 | 9 | Moderate |
| Ground-station outage | 22.46 | 5 | 2 | 10 | High |
| Link-window outage | 10.33 | 4 | 3 | 12 | High |
| Random Central Node loss | 11.79 | 4 | 2 | 8 | Moderate |
| Random satellite loss | 8.63 | 3 | 2 | 6 | Moderate |
| Targeted Central Node loss | 13.21 | 4 | 1 | 4 | Low |

Likelihood scores are declared screening assumptions (1 Rare, 2 Unlikely, 3 Possible), not frequencies estimated by the simulator. The consequence score uses the worst mean loss observed across the matched California 60/120, matched India 60/120, and balanced India 180 blocks.

**Table D.10** Recomputed pairwise Pareto fronts in the balanced India 180-satellite block.

| Objective pair | Eligible cases | Pairwise Pareto cases |
|---|---|---|
| Strict completion vs transferred data | 216 | 10 |
| Observation fulfilment vs transferred data | 216 | 6 |
| Strict completion vs P100 $T_{100}$ | 216 | 1 |
| Observation fulfilment vs Central Node count | 216 | 6 |

**Table D.11** India 180-satellite combined-stress strict-completion retention.

| Packet multiplier | Satellite failure | $S_{100}^{\text{useful}}$ retention (%) |
|---|---|---|
| $2\times$ | 10% | 86.44 |
| $2\times$ | 20% | 83.81 |
| $4\times$ | 10% | 60.97 |
| $4\times$ | 20% | 57.49 |

**Table D.12** Priority-stratified strict completion under unlimited forwarding in the India 180-satellite layer.

| Priority | CN30 attainment (%) | CN30 speed | CN100 attainment (%) | CN100 speed |
|---|---|---|---|---|
| High | 50.95 | 0.16 | 65.66 | 0.31 |
| Medium | 80.87 | 0.38 | 95.90 | 0.65 |
| Low | 99.23 | 0.57 | 100.00 | 0.77 |

The speed statistic is the standardized, censor-aware score defined in Chapter 4; a higher value means earlier completion relative to that priority class's deadline. Exact Pareto filtering retains 61 of the 216 balanced cases. Under the declared random-weight distribution, `T180_P10_H0500_I0986_CN050_F1` ranks first in 26.55% of draws and enters the top five in 46.38%. This result is conditional on the India workload and the tested weights; it is not a universal optimum.

## D.3 Included Machine-Readable Result Files

Confirmatory evidence is stored under `data/final_results`. Its principal files cover the completion gate, matched regional deltas, priority-stratified event times, robustness, exact Pareto membership, rank acceptability, and the conditional five-case shortlist.

The 180-satellite campaign evidence is stored under `data/final_corrected_180`; the directory name records its provenance. It includes the verification and case-selection gates, canonical case metrics, Central Node intervals, matched 120-to-180 effects, paired scenario effects, robustness and combined-stress summaries, priority-event tables, exact Pareto membership, rank acceptability, the portfolio extension, and `corrected_180_postprocessing_manifest.json`. These compact artefacts are the numerical source of the final report; the multi-gigabyte raw Parquet campaign is retained separately.

The additional fitted-equation, robustness-dispersion, likelihood–consequence, and pairwise-Pareto tables produced for the final review are stored under `figures/part2_revision/tables`. The folder includes the full pairwise membership table, not only the four counts printed above.

# E Software and Data Availability

## E.1 Software and Evidence Release

The corrected thesis draft and its evidence are maintained in the project repository:

https://github.com/maneeshmanjunath7-lang/Master-Thesis

The immutable simulation-evidence release for this thesis is `v1.1-thesis-final`. It contains the simulator, campaign post-processing code, compact validated result tables, checksums, and environment metadata for the verified 282-case execution. The tag resolves to the exact evidence commit and provides the reproducible boundary for the numerical inputs used in this report. The final signed submission PDF and its presentation-level tables and figures are distributed in the submission package rather than retroactively changing that immutable tag. The earlier `v1.0-thesis` tag predates the verified execution and is retained only as development provenance; it is not the evidence source for the final 180-satellite findings.

## E.2 Repository Layout

```
Master-Thesis/
|-- simulation/          # Full891 simulator and tests
|-- postprocessing/      # Final and reusable analysis code
|-- figure_generation/   # Report-figure generator and map data
|-- results/             # Canonical compact CSV/JSON/Parquet evidence
|-- figures/             # Generated thesis figures
|-- thesis/
|   |-- latex/           # Complete TUM LaTeX source
|   +-- Master_Thesis_Maneesh_Manjunath_2026.pdf
|-- checks/              # Evidence-manifest and release checks
|-- environment/requirements.txt
|-- REPRODUCIBILITY.md
+-- CITATION.cff
```

## E.3 Software Needed for Reproduction

The scientific results do not depend on the editor or operating system used to prepare the thesis. Reproducing the post-processing, tables, and figures requires Python and the version-pinned packages in `environment/requirements.txt`; these include the data-frame and Parquet readers, numerical and statistical routines, and plotting libraries called by the published scripts. Re-running the underlying event campaign additionally requires the Full891 simulator, its TAT-C-based orbit and access layer, the frozen task files, and the campaign configuration. A LaTeX distribution is needed only to rebuild the report from its source.

Git and GitHub provide version control and release distribution, but they do not affect the numerical method. Visual Studio Code, the VM desktop, archive utilities, and ChatGPT/Codex are likewise not reproduction dependencies. The role of ChatGPT/Codex is disclosed in the acknowledgements; all scientific evidence needed to reproduce or audit the reported results is contained in the tagged code, configuration, environment, and result artefacts described here.

## E.4 Reproducing the Analyses

After creating a Python environment and installing `environment/requirements.txt`, the main post-processing command is:

```
python postprocessing/complete/postprocess_complete.py \
  --region /path/to/California.zip \
  --region /path/to/India.zip \
  --output reproduced_results
```

The simulator and each post-processing package include focused tests. The repository workflow repeats the release checks and tests on pushes and pull requests. `REPRODUCIBILITY.md` gives the Windows and Linux command sequences, expected inputs, evidence boundaries, and verification procedure. The `v1.1-thesis-final` verification report confirms that `results/final_corrected_180` and the associated campaign evidence are present and checksummed. The separate submission package contains the final LaTeX source, the additional reviewer-requested fit, uncertainty, risk-screen, and pairwise-Pareto outputs, and the signed PDF.

## E.5 Evidence Included

The machine-readable evidence is under `results`. The California–India 60/120-satellite block is under `results/final_results`. The verified 282-case India 180-satellite extension is under `results/final_corrected_180`; the directory name is retained for traceability. It contains the 773-task verification gate, balanced CN30–CN100 summaries, matched India 120-to-180 effects, robustness and priority evidence, and the large-fleet portfolio candidates. The formatted workbook `results/Thesis_Result_Tables.xlsx` provides a reader-oriented view of the complete campaign while retaining the source CSV path on every sheet.

The compact release allows a reviewer to audit every numerical value used in the report without downloading the multi-gigabyte raw case archive. It does not contain the raw transfer-event partitions required to reconstruct every simulated link event. Those large archives remain external project artefacts and are identified by provenance and checksum records in the compact evidence.

## E.6 Integrity, Security, and Reuse

`checks/build_evidence_manifest.py` creates SHA-256 checksums for the published evidence, and `checks/verify_release.py` checks required files, oversized artefacts, result scope, and common credential patterns. API credentials, local virtual environments, raw downloads, temporary output, and VM-specific files are excluded. Original project code is released under the MIT License; third-party TUM classes, data, and dependencies retain their own notices and terms. The citation metadata in `CITATION.cff` provides the preferred software citation.

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